SmartEP

Evoked Potential System
Manual Version 3.94

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Safety Instructions

Complying with the following warnings and caution statements will ensure safe operation of the equipment as well as the safety of the users and patients. The following labels identify dangerous conditions and their possible effect. These symbols are used throughout the manuals:

WARNING:
This symbol identifies conditions or practices that may present danger to the patient and/or user.

WARNING:
This symbol identifies conditions or practices that may present risk of electrical shock to the patient and/or user.

CAUTION:
This symbol identifies conditions or practices that could result in damage to the equipment or in incorrect results.

WARNING:

Due to safety considerations, the system must be plugged into an Isolation Transformer in order to avoid the risk of electrical shock.

The computer and printer used with the system must be connected to the same Isolation Transformer.

WARNING:

Third party equipment that was not provided or approved by IHS must not be connected to the Isolation Transformer.

The Isolation Transformer must not be placed on the floor in order to prevent the accidental ingress of fluids.

In systems with a power switch, connect the power cable ONLY when the power switch is in the OFF position, failure to do so may damage the equipment.

Power cables shall be connected to electrical outlets approved by international standard IEC-60601. This is necessary for grounding reliability and proper isolation.

In systems with a power switch, connect all cables and verify all connections before turning ON the power.

The System must be tested for leakage current by the biomedical engineering department of your institution.

The biomedical department of your institution must make sure that the leakage current test results comply with the requirements of your region.

The system must be tested thoroughly before connecting any patients.

Computers connected to a network need to be tested for leakage current while connected to the network, a network connection may affect total leakage current.

WARNING:

The System is not explosion proof and must not be used in presence of flammable gases or substances.

The system is not protected against defibrillators. Remove patient connections before defibrillation. If defibrillation occurs, retest the system for leakage current to ensure safety.

Hazardous electrical output may occur. This equipment is for use by qualified personnel only.

> The operator must not touch any of the electrical cables and the patient at the same time.
>
> Patient connections are not intended for direct cardiac contact, although they are electrically isolated.

Read and follow the instructions in the Hardware Installation manual when installing or transferring the system.

In systems that use battery powered parts or accessories, dispose properly of batteries; follow the rules and regulations of your area, or institution, for proper disposal. Batteries must be removed from the equipment when the system will not be used for extended periods.

Only individuals who are trained in the use of the device and its principles should use the SmartEP system. All results should be reviewed by a certified audiologist or licensed physician who has knowledge of Auditory Evoked Potentials.

Do not clean ear canal unless you are a certified audiologist or medical professional trained in this procedure. Irreversible injury and hearing loss may result to patient if an unqualified person cleans the ear canal.

Only cables supplied by IHS shall be used to connect the equipment to other parts or to external components as specified in the manuals.

CAUTION:

Do not operate with damaged cords or plugs. Inspect all connection cords periodically for fraying or other damage.

The power supply provided with the Universal Smart Box is intended for use with that unit only; do not attempt to use this power supply with other devices.

The power supply provided with the Universal Smart Box Lite is intended for use with that unit only; do not attempt to use this power supply with other devices.

When connecting all cables, do not force the connection, damage to the hardware may occur.

Read and follow the instructions provided in the user manual to ensure safe and proper use of the equipment.

Ensure that the patient electrode leads are correctly attached and connected to the appropriate plug. Inverting the electrode connections may cause incorrect results and misdiagnosis.

Test results should only be interpreted by a qualified professional.

Non-medical equipment that will be connected to all IHS devices must comply with the necessary standards for you area.

Turn off and disconnect all equipment and all parts of the equipment when performing maintenance and cleaning.

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Introduction

1

Welcome to the Intelligent Hearing Systems family of customers. Please be sure to read all the enclose documentation. Doing so will allow you to use the equipment safely. It will also allow you to maximize the benefits from this great investment. If you cannot wait to use your equipment, feel free to skip to the tutorials. Tutorials will give you a general idea of how to use SmartEP. If you are new to SmartEP, please try and read the manual in its entirety; doing so will help you understand the power of this great software package.

Indications for Use

使用目的 诊断

The intended use of the SmartEP product is for the Diagnosis of patients of any age using evoked potential techniques. This product is intended for use as an aid for diagnosing hearing related disorders, and as a tool in the estimation of hearing thresholds and other conditions which affect evoked potential responses to auditory, tactile or visual stimuli.

SmartEP can be used for patients of all ages. It is intended for use by trained personnel in a hospital, nursery, clinic, audiologist's office, or other appropriate setting. The SmartEP research and diagnostic components may be used by investigators and students determining the usability and effects of acquired evoked potentials.

Product Features

SmartEP is a full featured auditory evoked potential diagnostic system. It is a multi-environment software package used for complete powerful recording analysis or simple screening. Some of its features are:

- Split-screen displays, 分屏显示
- The ability to drag and place recordings anywhere on the screen,
- On-screen multi-page report generation,
- User-specified testing protocols,
- Digital stimulus generation,
- Latency-intensity plots,
- Easy peak-labeling,
- Intensity series displays,
- Time series plots, FIR and Spectral digital filtering,
- Spectrum estimation

SmartEP works under the Windows© 98SE, ME, 2000, XP and Vista operating systems. This manual assumes that you are familiar with the basic operation of these operating systems and their capabilities. Please refer to the operating system manuals when necessary.

Product Installation 安装

Please refer to the installation manual that should be included in this folder. If you did not receive installation instructions, you can view an electronic version (PDF format) of the manual

in the IHS Programs CD that was enclosed, or contact Intelligent Hearing Systems for a new hardcopy of the manual. Adobe Acrobat Reader© is needed to open PDF files, the Acrobat Reader® can be downloaded from Adobe Systems at http://www.adobe.com.

Hardware Considerations

The functionality of the software will be dependent on the hardware that is being used in combination with it. The Universal Smart Box, Universal Smart Box Jr. and Universal Smart Box Lite will have different capabilities when it comes to the addition of software and hardware modules due to their physical characteristics.

USB Box capabilities within SmartEP:

- Up to 6 or 8 channels of acquisition (Maximum is dependent on the hardware options that are installed). 8APE通道 2OAE通道
- User selectable filters, gain, and notch filter. 可选择
- Electrode impedance measured using the Opti-Amp Transmitter.
- Can run all modules available to SmartEP.

USB Jr. capabilities within SmartEP:

- Up to 2 channels of acquisition. 2AEP通道 1OAE通道
- User selectable filters, gain, and notch filter.
- Electrode impedance measured using the Opti-Amp Transmitter.
- Can run all modules and functionality available to SmartEP that do not require additional specialized hardware. Specialized hardware is required for Ipsilateral Masking, Notched Noise, Somatosensory, and Visual Potentials. 科研类功能包括对侧掩蔽、同侧陷波刺激、体感诱发电位、视觉诱发电位。

USB Lite capabilities within SmartEP:

- 1 channel of acquisition. 1AEP通道 1OAE通道
- Filters and Gain are Hardware-based and may not be changed. 滤波值和增益无法更改
- Notch Filter is always ON. 陷波滤波器无法关闭
- Electrode Impedance is shown in EEG and Amplifier window.
- It can acquire ABR. Limited ECochG and MLR acquisition due to hardware filters.

Customer Responsibility

The equipment, its components, and included software shall perform reliably as long as it is operated and maintained in accordance with the instructions contained in the manuals, 一次 associated labels, and inserts. A defective component must not be used. New IHS-manufactured parts should promptly replace broken, missing, or worn parts. If you suspect a part of the equipment to be defective or if you need additional information, please contact IHS.

The responsibility of IHS is limited by the warranty as stated in the technical reference manual. Should the repair or replacement of the product become necessary after the end of the warranty period, the user must consult IHS before such repair or replacement. If the product is in need of repair, do not use it until all repairs are completed, the unit is functioning

properly and is appropriately tested. The owner of this product has sole responsibility for the following:

- Any malfunction resulting from improper use,
- Maintenance or repairs done by other than IHS,
- Any malfunction caused by parts that are damaged or modified by anyone other than IHS authorized personnel.

Tutorial

2

This chapter will provide you with a basic description and understanding of how SmartEP operates. You will learn the basics about data processing, displaying, peak labeling, printing, report generation, and acquiring new data. A set of sample data was provided at program installation for use with this tutorial. After completing this tutorial, you should be ready to collect data on your own. Make sure to practice by using your system on a few cooperative subjects, so that you become familiar with your new system before recording from patients. Make sure to read the manual as a whole in order to understand SmartEP fully.

Starting SmartEP

Turn on the computer and wait for Windows to finish loading. Once Windows is up and running, you may start SmartEP by using one of the following ways:

- Double click on the SmartEP Icon on your Windows desktop, if available.
- Double click on the Launch Pad icon on your Windows desktop, if it is not open already, and then click the SmartEP button in the Launch Pad program.
- Click on the SmartEP shortcut from the Windows Start Menu.

Once SmartEP is started, the splash initialization screen shown in Figure 1 will appear. If any error messages are displayed during initialization, it is probable that the equipment is not properly connected or has not been turned ON. You will be able to operate SmartEP in demonstration mode when the hardware is not present. Refer to the troubleshooting section of this manual for error details.

Figure 1 – SmartEP splash screen

If a desktop shortcut to SmartEP is not available, you may create one:

1. Find the directory where the IHS programs were installed, typically IHSPROGS in the local disk C, and open that folder.

2. Find the program called sepwin.exe and click on it with the right mouse button.
3. Select “Create shortcut” from the pop-up menu.
4. Click on the newly created shortcut with the left mouse button and drag the shortcut, while pressing the button, to the desktop.
5. Rename the shortcut as desired.

Learning your way around SmartEP

After initialization, the main system screen will be displayed. The system screen is composed of four functional areas: the Control Panel, Page Menu, Display Area and the Main Menu. Functionality of these areas is explained in later chapters.

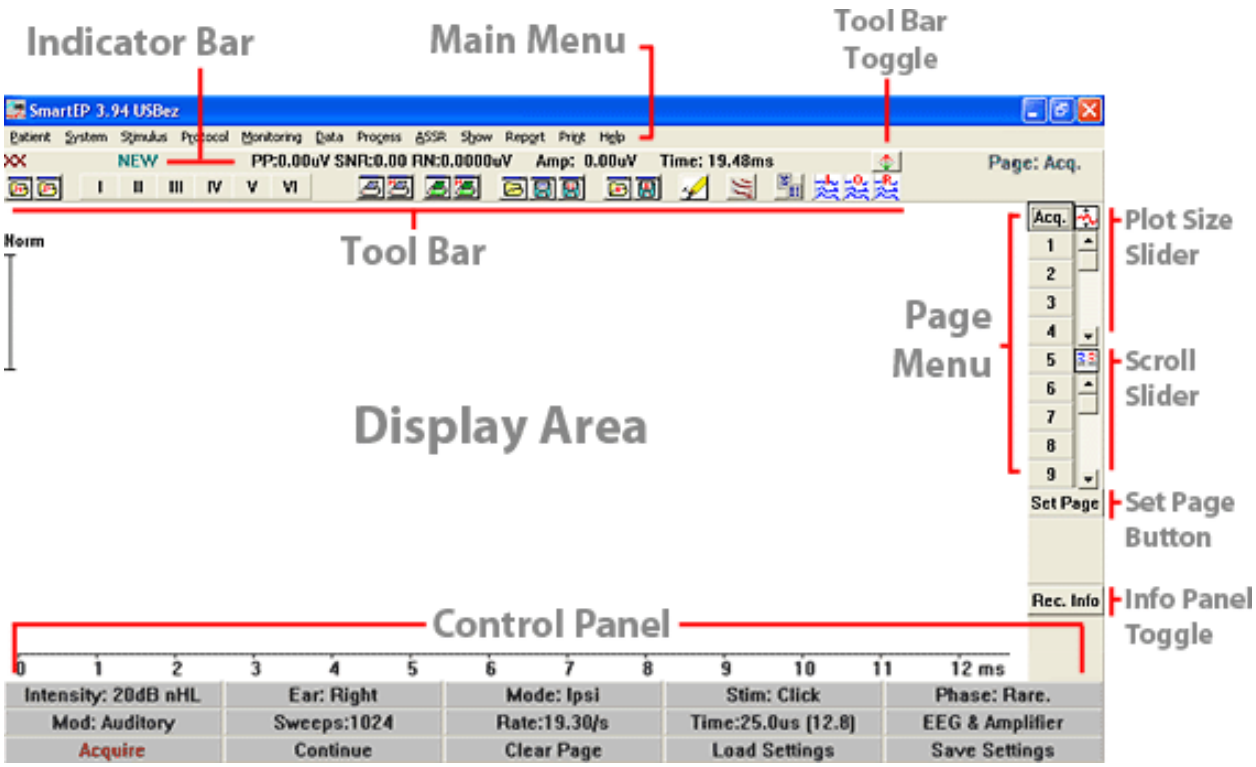

Figure 2 – SmartEP screen

SmartEP has two types of screens, one Acquisition Page screen and nine (9) Display Pages screens. The Acquisition Page screen, as the name implies, is where acquired data are immediately displayed. The Control Panel is found at the bottom of this page and it can be used to set most acquisition parameters. The Control Panel will be hidden while viewing any of the nine display pages. Display pages are mainly used to organize recordings into understandable reports. The acquisition page always has a time scale shown at the bottom of the screen; however, the display pages do not show it unless a recording has been loaded into it. Use the mouse or any other available pointing device to use the Control Panel and the Side Menu. It is possible to navigate to the Main Menu using only the keyboard arrows by pressing the [Alt] key on the keyboard. While in any of the SmartEP dialog boxes, press [Enter] to close the dialog box and confirm any changes, press [Esc] to exit without making any changes. 二次

Tutorial

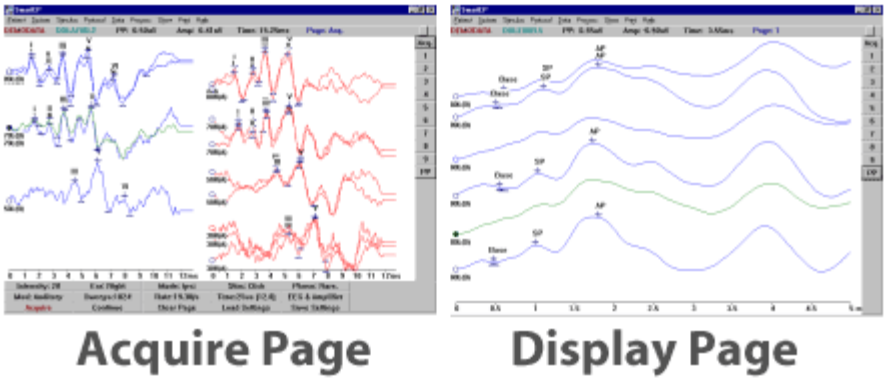

Figure 3 – SmartEP Page Types

Loading a Patient File

1. Click on **Patient** in the Main Menu and select **Open**. A dialog box appears containing a list of saved patient records.

Figure 4 – Patient Menu

2. Select a patient from the list by clicking once on the patient's name to highlight it and then clicking the OK button or pressing [Enter].

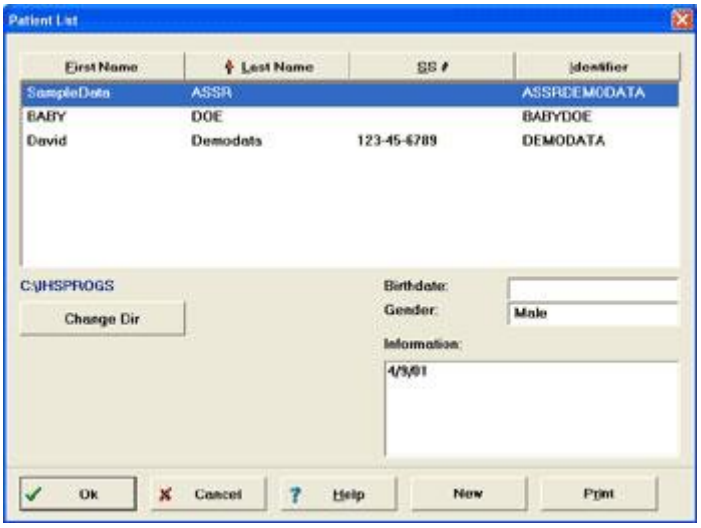

Figure 5 – Patient List window

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3. The patient's information is displayed in the Patient Demographics window. Click the OK button to exit and start recording data for this patient and automatically save any changes made to the patient's data. All subsequent data recorded will be saved and linked to this demographic file until a new patient is selected. If there is missing information, you may be asked to provide it before proceeding.

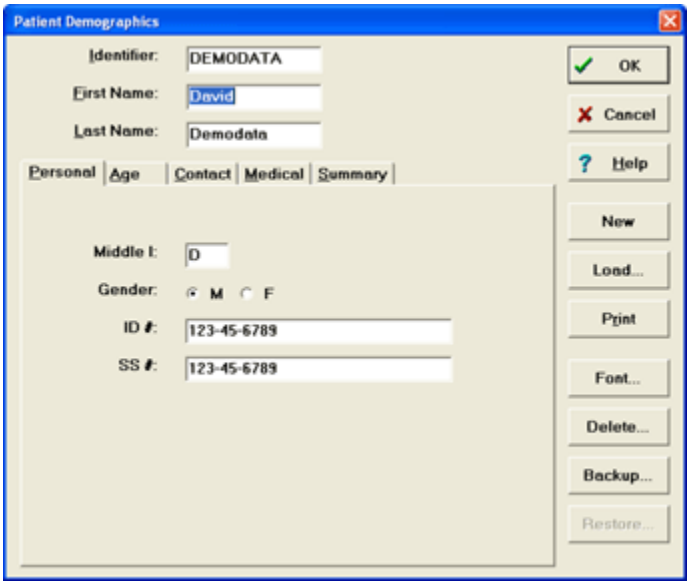


Figure 6 – Patient Demographics window

Loading Recordings

To load previously recorded data for the current patient:

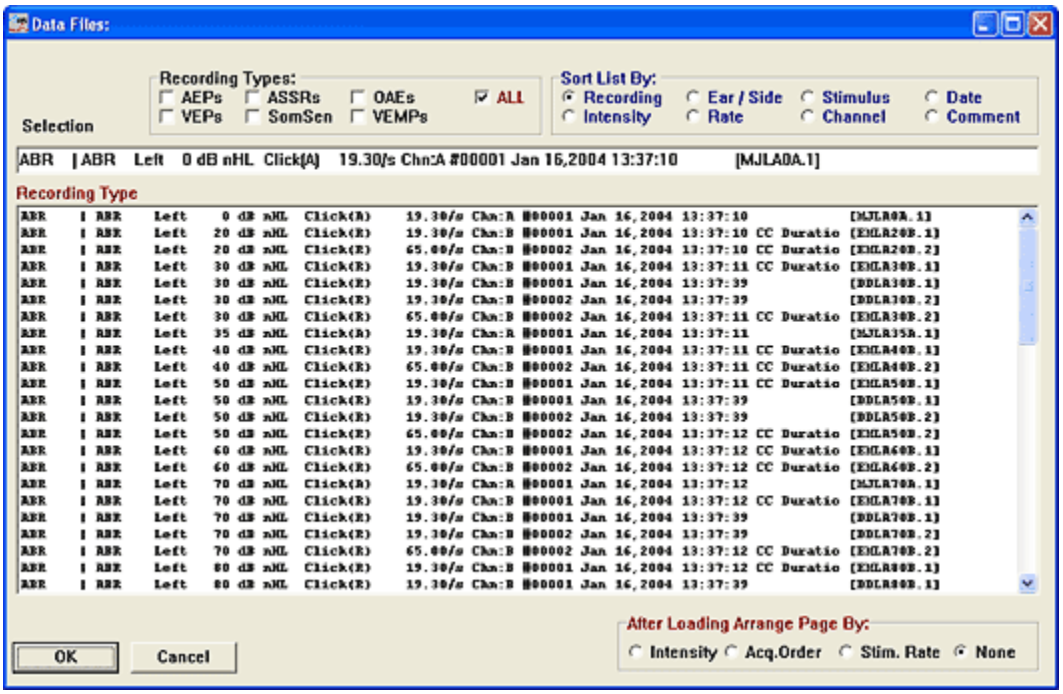


Figure 7 – File List Window

Tutorial

1. From the SmartEP Main menu, click on **Data > Load File,** or click on the Load File button from the quick bar. A listing will appear displaying all the available protocol files for the current patient. See Figure 7.
2. Click on the recording or highlight the recordings that you wish to load (choose any of the ones shown for this tutorial). To select several recordings, drag the mouse over the names of the files while pressing the left mouse button. If the desired recordings are not in consecutive order, you may select one at a time by holding down the [Ctrl] key while clicking on their names.
3. Click **OK** to finish the loading of the recordings to the current page.

The information displayed in the EP files dialog box is (left to right):

- Type of recording,
- Side tested,
- Stimulus intensity, type, rate and channel recorded from,
- Date of acquisition,
- Additional comments,
- and complete filename

Older versions of SmartEP may contain less information in this window. Files acquired with older versions of SmartEP may not include some information. Notice that the recordings can be sorted in different ways using the top right radio buttons; the results may also be filtered by recording type using the type check boxes.

Activating and Moving a Recording

Up to two hundred and fifty (250) objects may be shown on-screen for manipulation, and only one object can be active at a time. The active recording may be moved, filtered, have its peaks labeled, and have many other operations applied to it. Although this limit imposes a restriction on how many recordings are displayed at any time, the total number of recordings that can be acquired for each patient is only limited by the size of your computer's hard drive.

To make a recording active, click on the recording handle, a small circle located to the left of the recording. The active recording is displayed in green and its handle in black. To move a recording:

1. Move the recording around the screen by clicking on its handle and dragging. Notice that the recording snaps back to the correct time axis.
2. Drag the handle and drop it on top of one of the recording page access buttons to the right hand side
3. Click on the access button where the recording was dropped. Notice that the recording was moved to a new page.

The inactive recordings are displayed in red for right ear, and blue for left ear with a clear recording handle. The active recording is always displayed in green with a black filled handle. The intensity and the ear (L or R) are listed below the recording handle, along with the recording channel.

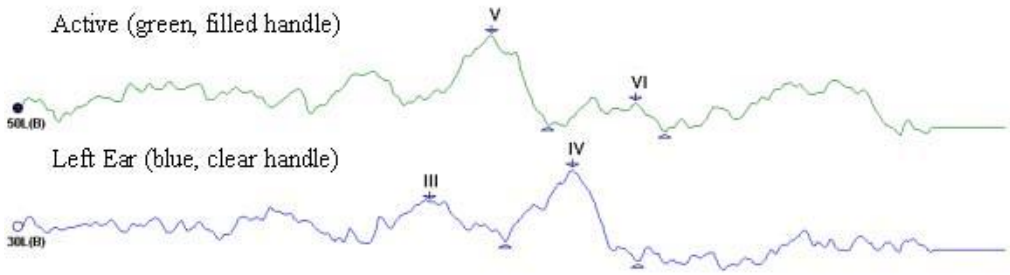


Figure 8 – Active and inactive recordings

1. Click the **Set Page** button (right side of the screen) to show the **Page Parameters** menu.
2. Select **Scale > Normalized**. You can also select one of the standard values from the submenu or enter a different value by choosing **Other**. Remember that the larger the scale value used, the smaller the recordings will appear on the screen. When the normalized option is entered, all the recordings will appear the same size on the screen regardless of actual amplitude.

Multiple recordings may be activated at a time to perform certain operations, however only the last activated recording can be moved by dragging.

Data Manipulation and Processing

To check the parameters used to generate a specific recording:

1. Make the recording active by clicking on its handle
2. Press the right mouse button over the handle.
3. Move the mouse over the first item in the pop-up menu, where the file name is displayed.

A submenu will appear displaying all the acquisition parameters for that recording. Click outside the submenu to close it.

To clear a particular recording from the screen:

1. Make the appropriate recording active by clicking on its handle,
2. Press the right mouse button over the handle.
3. Select the clear option in the submenu.

The recording will be removed from the current display page, but will not be deleted from the hard drive. However, any changes that you have made to the recording will be lost if you have not saved them.

Recordings may be added, subtracted, multiplied and inverted. To add two recordings:

1. Activate the first recording to be added by left clicking on its handle.
2. Activate a second recording to be added by left clicking on its handle while pressing the [Ctrl] key on your keyboard. Repeat if you need to add more recordings.
3. Select **Process > Add Selected** from the SmartEP main menu or press the [+] key on your keyboard.

The addition of the selected recordings will be displayed as a new recording, in a new buffer. Remember to save the new recording for future use by clicking on **Data > Save File As** from the SmartEP main menu.

To subtract two recordings:

1. Activate the first recording to be subtracted by left clicking on its handle
2. Activate the second recording by left clicking on its handle while pressing the [Ctrl] key on your keyboard. Remember that only two recordings may be selected at a time for subtraction operations.
3. Select **Process > Subtract Selected** from the SmartEP main menu or press the [-] key on your keyboard.

The addition of the selected recordings will be displayed as a new recording, in a new buffer. Remember to save the new recording for future use by clicking on **Data > Save File As** from the SmartEP main menu.

Multiplication is an active operation in SmartEP, which means that the operation will be performed on the active recording itself, not on a new buffer as with addition and subtraction. It is preferable to copy the recording data to a new buffer if you plan to save the recording after multiplication. To multiply a recording:

1. Activate the recording to be multiplied by left clicking on its handle.
2. Select **Process > Multiply** from the SmartEP main menu.
3. Enter a value to multiply the recording by in the pop-up window and then press OK.

Inversion is another active operation in SmartEP; make sure to copy the recording to a new buffer if you plan to save the recording after inversion. To invert a recording:

1. Activate the recording to be inverted by left clicking on its handle.
2. Select **Process > Invert** from the SmartEP main menu.

Applying Filters

SmartEP will acquire recordings using the settings that were selected in the EEG and Amplifier dialog box at the time of acquisition. It is possible to filter the recordings even further for a smoother signal. This kind of filtering will not affect the actual recording file that was saved, only the way it is displayed. To filter a recording:

1. Activate the recording to be filtered by left clicking on its handle.
2. Select Process > Filter Type from the SmartEP main menu.
3. On the pop-up window, select the filter setting desired, FIR, finite impulse response, or spectral digital filter and their respective values. Click OK to confirm and close the pop-up window.
4. With the recording still active, select Process > Filter Active from the main menu to filter the active recording.
5. Alternatively, you may filter all the recordings in the current page by selecting Process > Filter ALL Displayed from the main menu.

If a multiple selection has been made, note that only the last activated recording will be filtered when using the Filter Active option.

Using Cursors

SmartEP includes cursors. These are visual aids to help achieve a more accurate analysis of recordings and faster access to the latency information, which is most important to your application. To use cursors:

1. Select Show > Show Cursors from the SmartEP main menu to activate cursors.
2. Move the cursors by dragging the small handle (circle) at the bottom of the display screen using the left or right mouse buttons.

Each of the two available cursors will display the current time position on the screen. The rightmost cursor will also display latency information (i.e. difference in milliseconds between the two cursors) and the frequency equivalent to that time difference. In the same menu, you will find the Show Zero Time option which can be helpful for reference purposes.

Cross Correlation

Cross-Correlation is a method of comparison commonly used for auditory evoked potentials. Two cross-correlation operations are available in SmartEP. The cross-correlation operation may be used to cross correlate a recording that was acquired using alternating modality, which will calculate the correlation between the condensating and rarefaction buffers contained within that recording. To compute the cross-correlation of an alternating recording:

1. Activate the recording by left clicking on its handle.
2. Select Process > Cross Correlate (Split Sweep) from the SmartEP main menu.

A pop-up window will show the result of the cross correlation for the entire recording. The cross-correlation operation may also be used to compare two separate recordings. To compute the cross correlation between two recordings:

1. Activate cursors, if not already activated, using the Show Cursors option in the Show sub-menu.
2. Move the cursors to delimit the region that you wish to cross correlate.
3. Activate one of the recordings by left clicking on its handle.
4. Activate the second recording by left clicking on its handle while pressing the [Ctrl] key on the keyboard. Only two recordings may be selected at a time for cross-correlation operations.
5. Select Process > Cross Correlate Selected.

A pop-up window will show the result of the cross-correlation operation for the defined area. If cursors were not activated before choosing the cross-correlate option, an error message will appear and cursors will be activated automatically.

Peak Labeling

Peak Markers are visual aids used to measure the peak amplitude and latency. There are three ways to mark the peaks of a recording:

- Select the peak marking option from the quick bar at the top of the window so that the label turns red, then left click once over the active recording to place the label where the mouse click occurred. Remember to turn off the peak marking option after applying it.
- Activate the recording you wish to label by left clicking on its handle
- then right-click over the location where the peak marker should be placed, and
- from the pop-up menu select Mark Peak and choose the appropriate label from the list that appears.
- If the peak marker you wish to place is not listed in that menu, select Mark Other Peak from the pop-up menu to choose the label from the label placement window, or to create a customized label.

To remove a peak label:

1. Activate the recording for which you wish to remove the label by left clicking on its handle.
2. Place the mouse pointer over the label you wish to remove and press the right mouse button.
3. Now select the Remove Peak option from the pop-up menu. The label will be removed from the recording.

After a peak has been labeled, its amplitude may be measured by dragging the lower peak marker (upward pointing triangle) to the following or leading valley, from which you would normally measure the peak. To move the marker, simply position the mouse pointer over it and then drag it while pressing the left mouse button. The triangle will leave a trail as it is moved.

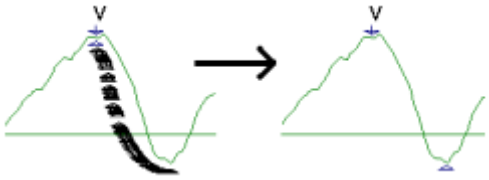


Figure 9 – Peak Positioning

There are two ways to view the peak latency and amplitude values for the labeled recording:

- Right click on the signal and select the Peak Latency option from the menu that appears. A panel will show the peak latency values in milliseconds, followed by amplitudes in μ V. Inter-peak and amplitude ratio values are also displayed for the major peaks (I, III and V) that have been labeled.
- Right click on the signal and select Show Text > Side or Show Text > Below. A text field will show the settings under which the recording was acquired, the peak latency values in milliseconds for the major peaks followed by amplitudes in μ V and Inter-peak and amplitude ratio values for the major peaks (I, III and V) that have been labeled. These options are also accessible on the Display tab of the Recording Information panel.

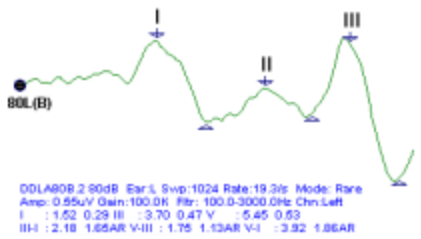


Figure 10 – Recording with peaks and text

Showing and Printing Results

When loading or acquiring many recordings, they may be shown overlapping each other on the display area. The recordings can be organized by using the options in the Show menu or they may be organized automatically at loading time by choosing one of the options at the bottom of the Load Data dialog box.

1. Select **Data > Load File** from the SmartEP main menu.
2. From the Load Data dialog box, select some recordings from the left and right ears. Press [Shift] or [Ctrl] to select multiple recordings.
3. On the main menu, click on **Show > Show by Intensity**.

Notice the recordings are automatically arranged in order of decreasing intensity. If data corresponding to both ears are on the page, the display will be switched into split-screen mode, where the data corresponding to the left ear is placed on the left side and data corresponding to the right ear is placed on the right side.

After you have arranged the recordings, you may generate a hardcopy report by selecting one of the three printing options from the Print menu. The layout of the recordings on the printouts is the same as on the screen. A table containing the acquisition parameters, peak latency, and amplitude information is also printed below the recordings.

Creating a New Patient File

Before you start the acquisition process on a new patient, the patient's information should be entered into the system. This will ensure that the data is stored in an organized manner and will make it easy to retrieve later. If the new patient file was not previously created on the Launch Pad, you may create one inside SmartEP.

1. Select **Patient > New** from the SmartEP main menu to open the Patient Demographics dialog box.
2. Enter all the information in the appropriate spaces. Identifier, First Name, and Last Name are the only required fields.
3. Press [Enter] or click on **OK** to save the patient file and close the dialog box.

If the patient being tested has already been tested, or his/her information already entered into the system, you may retrieve it using the load command before starting a new test. Refer to page 6 for a description on how to load an existing patient file.

Settings for Acquisition

The program defaults to ABR settings when first run. This is convenient since the majority of recordings acquired with SmartEP will be ABRs. However, SmartEP has the capability of acquiring ECochG, MLR, LLR, and P300/MMN (if purchased) among other evoked potentials. Some systems may also have the ability to perform high frequency testing. To differentiate all these types of recordings and to have the proper display, it is necessary to change the acquisition setting in SmartEP. To change the setting, choose the appropriate option under Stimulus > Modality in the SmartEP main menu. When changing modality, the default settings for that mode will be loaded automatically. If you wish to use customized parameters every time, change the mode and the settings, and then save a settings file using the button on the control panel. The current modality is saved along with the settings file, so it is not necessary to change modality when loading a file.

Setting the Artifact Rejection Level

The artifact rejection level allows the system to ignore certain parts of the signal that may cause sections of the EEG to clip, and therefore be displayed as a flat line. The post auricular 耳后肌 muscle is one of the most common sources of artifact. The artifact may be eliminated from the incoming signal.

1. Click on the EEG & Amplifier button from the Control Panel.
2. On the EEG and Amplifier Settings dialog box, adjust the level slider up or down so that the signal is contained within the markers. Zoom in or out as necessary using the Display slider.
3. Adjust the rejection region to fit the type of test that is being performed. Usually, the desired region is between 1 to 10 milliseconds for an ABR recording.

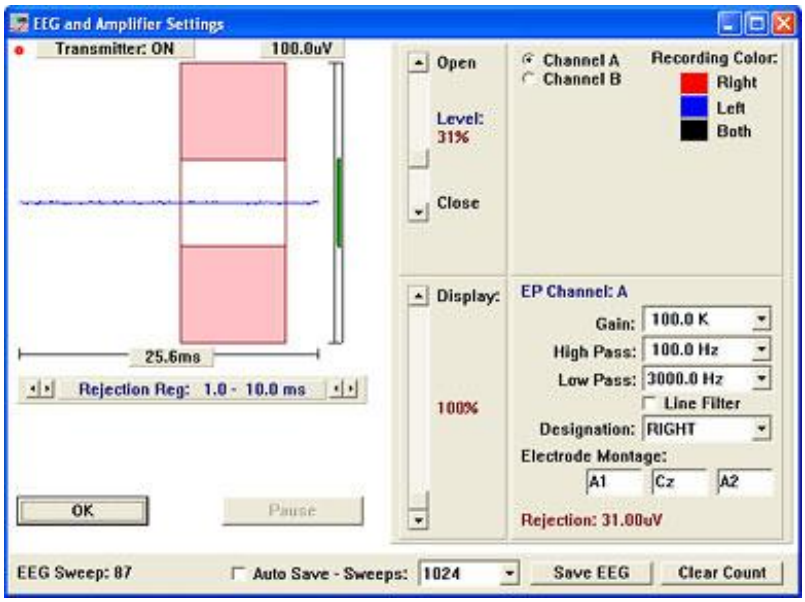


Figure 11 – EEG and Amplifier Settings dialog box

Tutorial

4. Repeat the adjustments for the second, third and fourth channels if available. Switch channels by clicking on the radio button next to the channel name in the upper right hand corner of the dialog box

When using USB Lite hardware, you may also check the impedance level by pressing the "Check Impedance" button at the bottom of the channel selection area. Clicking OK will confirm the adjustments and return to the SmartEP display area. Do not close this window at this time if you are continuing with the tutorial.

Setting Filters and Gain

It may be necessary to adjust the filter and gain settings to acquire data. These changes are usually necessary when testing patients of different ages, when the testing is being done in a noisy environment, or when recording a different kind of auditory evoked potential. To change the gain and filter settings:

1. Open the EEG and Amplifier settings dialog box, if it is not already opened, by clicking on the EEG & Amplifier button in the SmartEP Control Panel.
2. Select the filter or gain setting that you wish to change from the drop-down list that corresponds to each setting.
3. Repeat the procedure for each channel as necessary. Switch channels by clicking on the radio button next to the channel name in the upper right hand corner of the EEG and Amplifier dialog box.

There are certain recommended settings for each acquisition type. Refer to page 101 for more information on recommended filter and gain settings. Note that the high pass filter, low pass filter, and gain of the USB Lite are set to 30Hz, 1500Hz, and 100x respectively and may not be modified.

Changing Stimulus Settings

Before you start acquisition, you must make sure that the system knows which stimulator and stimulus are to be used during the test. This is an important step during testing. Selecting the wrong stimulator will not only load erroneous calibration values; but also, it may cause the recordings to be inaccurate. The stimulus can be changed in the Auditory Stimulus Generation dialog box.

1. Click on the Stim button from the Control Panel or select Stimulus > Select Stimulus from the SmartEP main menu.
2. Select Pure Tone as your stimulus type.
3. Enter a value of 2000 in the frequency field.
4. Change the selected stimulator and notice how the Total calibration changes for each stimulator in the calibration panel (lower right).
5. Return the stimulator choice to ER3A Insert Earphones
6. Make sure that the rectangular envelope is selected and click on the Display button at the bottom of the dialog box to view the current stimulus in the frequency and time domains.
7. Select the Blackman envelope and click on Display again to see the new stimulus.
8. Close the stimulus window and save your changes by pressing the OK button.

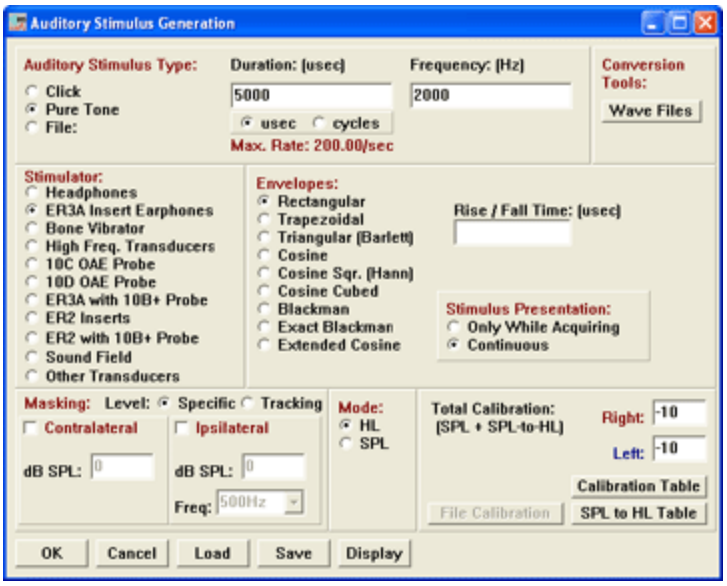


Figure 12 – Auditory Stimulus Generation window

Access to the calibration tables and the SPL to HL tables will require the use of the system password.

Specific就是给固定强度的掩蔽，tracking会给一个跟随刺激强度变化的掩蔽，设置的值为tracking和刺激之间的差值，正负值都可以设置

Masking

In some situations, such as when testing using a bone conductor, you may want to provide stimulation to the contralateral ear to mask the stimulus response from the opposite ear to the one being tested. Contralateral masking is a capability included in the basic functionality of the SmartEP Software. In addition, it may be necessary to use Ipsilateral Masking techniques when performing procedures such as Notched Noise testing. Ipsilateral Masking needs the addition of special hardware in order to be operational. To use masking:

1. Click on the **Stim** button from the Control Panel to bring up the Auditory Stimulus Generation window, if it is not already open.
2. At the bottom left of the window check the masking option that you want to activate.
3. Make sure you select the level type, specific or tracking, best suited for your purpose. Specific will apply the absolute masking level entered in the dB SPL box. Tracking will use the entered value as a difference between the stimulus and the masking levels; negative numbers will apply masking levels lower than the stimulus and positive numbers will apply higher levels respectively.
4. Enter the masking level. If performing ipsilateral masking, select the frequency at which the notch in the noise should occur from the drop-down menu.
5. Click **OK** to confirm your options.

Keep in mind that masking will be applied using the insert earphones or headphones that are connected to the system. Further details about the masking options can be found on page 40 of this manual. 牢记：掩蔽功能仅用于插入式耳机和头戴式耳机。

Tutorial

Starting Acquisition

Once a patient file has been loaded and all settings have been verified, SmartEP is ready to start acquisition. Note that all settings may be saved as a default, in that way, settings may never need to be modified again. To start acquisition: 设置参数文件保存后可永久使用，可设为只读。

1. Select the number of sweeps to be acquired by double clicking on the **Sweeps** button on the Control Panel.
2. Select the ear to be tested by pressing the **Ear** button in the Control Panel.
3. Also, select the intensity, rate, mode, and phase by using the respective buttons in the Control Panel.
4. Click on **Acquire** at the bottom left of the Control Panel to start acquisition.

Once acquisition has started, it may be paused at any time by pressing the **Pause** button, which replaces the **Acquire** button during acquisition.

Post-Acquisition Filtering

Acquired recordings can be filtered digitally after acquisition in order to correct problems with the acquisition, to obtain the results in a certain frequency band, or simply to display a smooth recording. If smoothing is desired, follow these steps:

1. Select **Process > Filter Type** from the SmartEP main menu.
2. Select FIR from the filter type options and move the slider to the right to increase the smoothing of the wave as needed. Choosing 7 will apply very little filtering, while choosing 25 will filter out the fine structure entirely.
3. Click **OK** to close the window and make your selection.
4. Select the wave you wish to filter and from the Process Menu, choose Filter Active. If filtering all recordings on a page, select that option from the Process Menu.
5. Save the wave(s) if needed by choosing the **"Save As..."** option from the Data menu. Choosing the **"Save"** option will overwrite your current data. Saving a report will also save all the unsaved waves on all the pages, without overwriting the previous ones.

Digital filtering may also be useful to remove sinusoidal noise from recordings, such as that introduced by power line noise (60Hz or 50Hz, depending of your area) and its harmonics. It may be useful to select Spectral from the Filter Type window and reapply the acquisition filters (from the EEG and Amplifier window) while keeping the High Pass filter at a value higher than the power line frequency or the harmonic that is causing the contamination of the recording. E.g., set the High Pass filter at 130Hz for a 60Hz harmonic contamination at 120Hz. You must keep in mind however, that applying stringent criteria may cause important response structure to be filtered out. Try different settings to achieve optimal results.

Making a Report

Reports help to keep the patient data organized for future use, especially when acquiring multiple types of evoked potentials from the same patient. To create a report:

1. Move the recordings currently on the screen to the pages and positions that best fit your needs.
2. Load any additional recordings for the patient that needs to be added to the report.

3. Click on **Data > Save Report**, on **Report > Save Report** from the main menu, or click on the Save Report button from the quick bar.
4. Use a descriptive name for the report file for future reference.

Some additional options for advanced report generation are available in the Report Menu. See page 19 for more details on how to use the advanced reporting features.

Reporting a Test Result

Filing a test result into the IHS database will help you automate certain tasks, such as patient referral or letters to the parents. Procedures on how to use and create referral letters can be found in the Launch Pad user's manual. To report test results:

1. Make sure the appropriate data recordings are loaded.
2. Click on Patient > Transfer Results to > IHS Database.
3. Enter the information requested in the appropriate fields.
4. Click on Add Results to post the results and close the dialog box.

To see a tutorial on how to use these results, refer to the Launch Pad manual.

Automating Using Protocols

A protocol is a series of stimulus and acquisition parameters applied in a pre-specified order. Follow this procedure to set up a protocol.

1. Click on **Protocol > Setup Protocol** from the SmartEP main menu. The Protocol Setup dialog box will appear. A protocol is made of one or more Items. Each Item is a group of settings that will be used to acquire a recording. The count is the number of recordings that will be acquired for the settings specified by the Item.

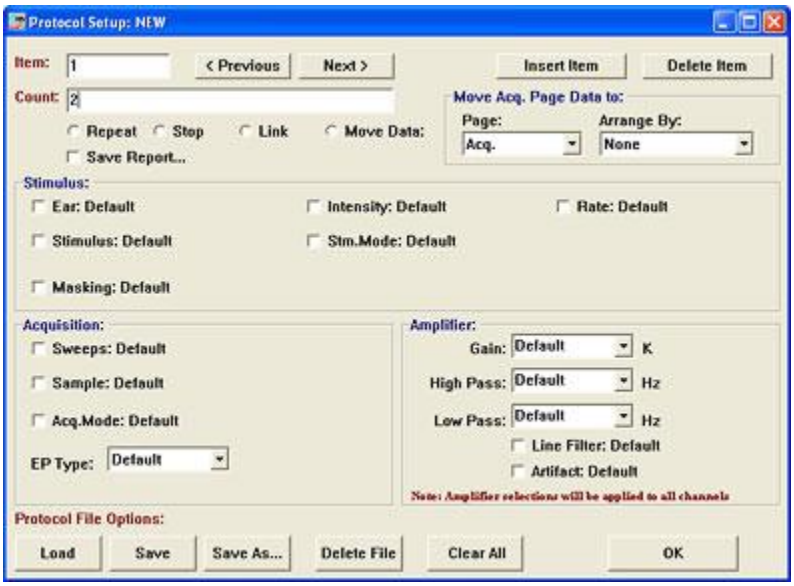


Figure 13 – Protocol Setup window

2. Enter "2" in the count field. The options should now be active. Notice all fields have been set to default, which means that they will use the current control panel values.

Tutorial

3. Click on Intensity and enter 70 in the input box that appears. If a particular parameter is left unchanged, the parameter values from the Control Panel will be used when executing the protocol. Be aware: EEG parameters may change from one patient to the next; therefore, they will not be saved in a protocol file and must be specified before beginning acquisition.
4. Click on Next to enter a new Item. Enter a count and parameters for this second Item.
5. Click on Save or Save as button to store the protocol information to the hard drive. A dialog box will appear requesting a file name. Enter a descriptive name of up to eight characters and do not add an extension (The extension *.pse is automatically inserted).

The Protocol can be now executed by clicking on Protocol > Execute Protocol option from the main menu and then choosing the appropriate filename. Note that protocols that make use of modified amplifier settings will not apply those settings in a USB Lite unit when the file is executed.

Advanced Report Generation

Using the tools provided in the Report Menu, it is possible to create a more advanced report that can contain information as needed.

1. Open the David Demodata patient file and load the report called MainReport.rpt using the Report > Load Report menu item.
2. Notice that the first page contains some patient information, an audiogram graph, and some additional text referring to the tests that were performed.
3. Select the text item by clicking on its handle (similar to the recording handle) and right click to see the available options.
4. Select Edit from the menu and modify the text and the font type and color by clicking on the font button.
5. Click OK to close the edit window and notice the changes applied.
6. Grab the handle of one of the objects and move it around at your leisure.
7. Switch to page 1, by clicking on the number 1 on the right hand side of the screen. The recordings should appear on the page.
8. Select Report > Add Table (Dynamic) from the main menu. Note that the table is placed on top of the recordings and can be moved down the page. If you need to scroll down the page, use the bottom scroll bar on the right hand side of the screen.
9. Switch to page two and notice that it contains a table like the one that was just generated. This table was created while on page 1 and then dropped onto page 2 due to space limitations. Notice that changing the peaks, or adding more recordings to page 1 will modify the data shown on the dynamic table located on page 2.
10. Switch to page 3 and notice the recordings and the included Latency-intensity graph. This graph can be added by opening the Lat-Int Window from the Show menu and pressing the “Copy Image to Page” button.
11. Page 6 contains a couple of transient OAE files that were loaded into SmartEP for reporting purposes.
12. Page 7 contains a couple of images saved from SmartDPOAE showing the distortion product results for the right and left ears.

With the tools in the Report menu, it is possible to add pictures of the patient, text, tables, and data from any other IHS diagnostic program. Refer to page 31 for details about the different items in the Report Menu. Note that when creating advanced reports, it is recommended to turn off the automatic generation of tables in the Print menu by deselecting that option.

Saving Parameter Files

SmartEP allows the user to save the testing parameters for future use. This allows for easy test setting replication. Multiple parameters sets may be saved, one of those sets will be used as the default parameter set when SmartEP starts up. To save your custom acquisition parameters (including amplifier settings and designations),

1. Set the appropriate test parameters in the Control Panel and other options such as the Stimulus Generation and the EEG and Amplifier Settings windows. Left clicking or Right clicking on the parameters will change their value; some option will allow you to enter a value when double clicked.
2. Then click on **Save Settings** in the Control Panel
3. Enter a descriptive filename. Do not type an extension.

This file will automatically be given an extension of *.SET and will be stored in the main SmartEP directory, where the program was installed. Settings can be saved as default parameters to be loaded every time SmartEP is activated. To save the settings as default parameters,

1. Click **Save Settings** from the Control Panel.
2. Press the **OK** button in the Save As dialog box without changing the file name.

When no name is entered, the system automatically saves the current settings in a default file called Sepwin.set. The saved set will determine the settings of the program every time it is started.

Backing Up a Patient's Data

It is strongly recommended that patient data be backed up to disk periodically. This will ensure that data will not be lost in the event of a computer hard disk failure or any other unexpected event. SmartEP contains utilities to backup, delete, and reintroduce data previously backed up. The backup utility is also a convenient way to transfer all data for a particular patient from one computer to another. This procedure will back up one patient at a time. To create backup sets of multiple patients, please refer to the Data Backup Utility described in the Launch Pad Manual.

To save data to a diskette or other removable media:

1. Insert a diskette in disk drive A or B, or other removable media into corresponding port.
2. Select **Patient > Open** or **Patient > Edit Patient** option from the main menu.
3. After selecting the appropriate patient, click the **Backup** button in the demographics dialog box.
4. Select the appropriate Drive letter for your diskette drive or "Other" for other devices and click Start.

The program will copy all the data stored for the selected patient to the designated floppy. If additional disks are required, you will be asked to insert more diskettes. After the data is backed up, you may wish to erase the data that is left on the hard drive by clicking the Delete button, also in the demographics dialog box. Before erasing any data, make sure that you have backed up the data.

To restore patient data to the hard drive: 新版软件已经没有Restore功能，需要借助Backup功能进行反操作，将备份的文件反传输到源文件夹中。

1. Insert the appropriate disk into the diskette drive or other storage device.
2. Click on **Patient > Open** from the main menu.
3. Click on **Change Dir** button and select the correct device.
4. Open the patient file from the list to bring up the Patient Demographics window.
5. Click on the Restore button to bring the data back to the default data directory.

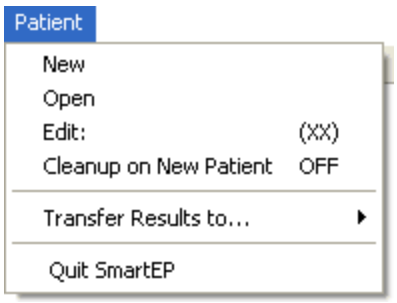
When accessing data from an external source, the Backup and Erase buttons become disabled and the Restore button is enabled. If you plan to work with the recordings you can access them directly from the backup media, or you may restore them to the hard drive in order to speed up the data access time.

The SmartEP Main Menu

3

The following is an itemized description of all the options available on the SmartEP main menu and the software functions they control.

Patient Menu



The Patient menu contains options that allow the loading and creation of new patient files and a utility to edit the demographics for the currently loaded file. The option to quit the program is also in the Patient Menu.

New

Selecting this option opens the Patient Demographics dialog box. All fields of the dialog box will be shown empty, ready to accept information on a new patient. Refer to the Launch Pad Manual for more information on the Patient Demographics dialog box.

Open

Selecting this option opens the Patient List dialog box showing a list of all the patients that have been previously entered in the database. Refer to the Launch Pad Manual for additional information on the Patient List dialog box. Only one patient file may be open at a time.

Edit

This option will open the data of the currently selected patient into the Patient Demographics dialog box. The patient's information may now be modified. Do not modify the Identifier, First Name, and Last Name fields; changing these fields may cause previously acquired data to be lost.

Cleanup on New Patient

When this option is OFF, all the data from the previous patient will remain on the screen when a new patient file is loaded. When this option is ON all data from the screens will be cleared from the pages, when loading another patient file. Although data is removed from the screen, it will not be deleted from the computer's hard drive.

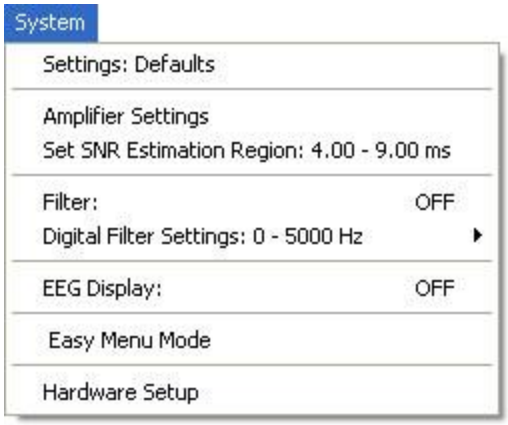
Transfer Results to...

SmartEP supports multiple database formats. Using these options, the information for the current patient may be saved to the appropriate database.

Quit SmartEP

Clicking on this option will exit SmartEP. Acquired recordings will not be lost, processed recordings should be saved before closing the program.

System Menu



Settings

Selecting this option shows the current settings that are loaded. Clicking on this item will open the Load Settings window, allowing the user to load any previously created settings file.

Amplifier Settings

This option will open the EEG and Amplifier Settings window. This window allows the user to set online filters, gain, artifact rejection region, and level; among other settings. It also displays the incoming EEG in real time. See page 35 for detailed information about this dialog box.

Set SNR Estimation Region

This option for advanced users allows setting the region for which the signal-to-noise ratio will be calculated. The SNR result is displayed in the Indicator Line. This option is password protected. Once the correct password is entered, data entry boxes will request new values for the start and ending times to be used in SNR calculation.

Filter

This option will turn the digital filtering option ON or OFF. When the filters are ON, the digital filters will be applied to the incoming responses, in addition to the hardware filters from the EEG and Amplifier window. The digital filter settings option follows.

Digital Filter

This option allows the user to set the low pass and high pass filters to be used when the digital filter is set to ON. The values can be anything from 0 to 5000 in 1 dB increments.

EEG Display

When this option is turned ON, the incoming EEG will be displayed on the acquisition screen. This option is useful when performing monitoring tasks.

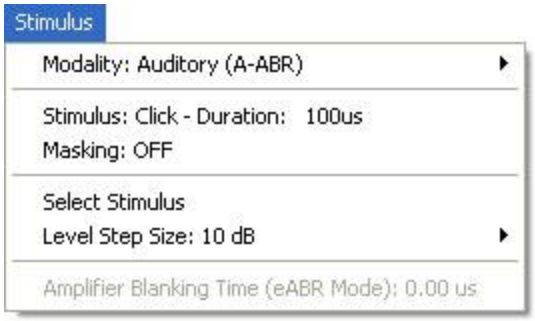
Easy Menu Mode

This option will change the current display to a simpler format with fewer options. It is usually recommended to use this mode when an inexperienced user will perform testing to prevent tampering with certain settings. See page 97 for an extended overview of the Easy Menu Mode.

Hardware Setup

This option will open the Hardware Setup dialog box. The dialog box will allow you to see the serial numbers of your system. It will also allow you to set the software to work offline when processing and analyzing data. This option is protected by the default system password. Additional information about the Hardware Setup dialog box can be found in the Intelligent Hearing Systems Launch Pad manual.

Stimulus Menu



Modality

The modality option should be used when changing the type of recording that will be acquired. After selecting the desired Modality, the software will perform certain changes to the display options and the control panel settings in order to better accommodate the acquisition and display of that particular recording type.

Stimulus

This menu item displays the type of stimulus and its duration, frequency and window as applicable. Clicking on this option will open the Stimulus Generation window, allowing the user to modify the stimulus settings.

Masking

This menu item displays the type of masking that will be applied including type and level. Clicking on this option will open the Stimulus Generation dialog box, allowing the user to change the masking output settings. If no masking is selected, it will show the word OFF.

Select Stimulus

Clicking on this menu item will open the Stimulus Generation window, allowing the user to change any of the stimulation options. For additional information about that window, see page 38.

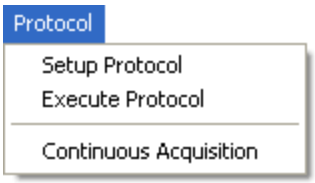
Level Step Size

The step level size is the amount of change that will be applied when the user clicks on the intensity button in the control panel. Single right clicks will increase the intensity by the step size, while a single left click will decrease it by that amount. The available step sizes are 10, 5, 2, 1, or “Other.” By selecting “Other,” the user may define a customized step size.

Amplifier Blanking Time

The amplifier blanking time is an option available only in the eABR modality. It allows the selection of the blanking time range after the trigger from the cochlear implant machine is received. This blanking time prevents the amplifier from saturating due to the electrical stimulation applied by the cochlear implant machine.

Protocol Menu



Setup Protocol

This option will open the Protocol Setup dialog box. In this dialog box, it is possible to set SmartEP to perform a series of test automatically without user intervention. See page 46 for additional information about the Protocol Setup dialog box.

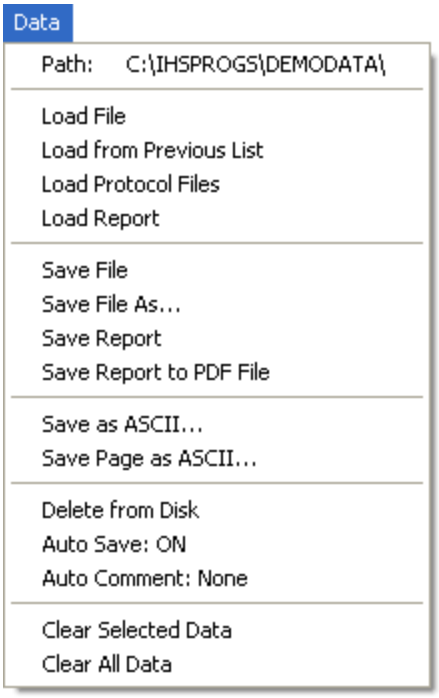
Execute Protocol

Clicking this option will show an open file dialog box, allowing the user to execute any previously saved protocol. Protocol files have the extension PSE.

Continuous Acquisition

Clicking this option will activate a continuous acquisition mode. When starting acquisition, with this option selected, the system will acquire the total number of sweeps specified in the control panel; once it is done, it will start over in a new buffer without any input from the user. The system will continue acquiring until the stop button is pressed in the Control Panel. Note that the system will acquire indefinitely if unattended. Once the maximum number of display buffers is reached, the oldest recording will be cleared from the screen; however, the recording will remain saved in the patient folder.

Data Menu



Path

The current data path for the active patient file is listed in this line. This is the location of the folder containing all the patient data. The path name will change when the current patient file is replaced by another.

Load File

This option will allow the loading of a previously saved recording to the current display page. This option generates a new list from the available files in the patient folder each time it is used.

Load from Previous List

This option allows loading a previously saved recording from a pre-generated list. This option is mostly useful in cases where thousands of files are available for a single patient and generating a new file list will take a long time due to the number of files.

Load Protocol File

This option will load a previously acquired protocol file group. A protocol file group is a group of recordings that were acquired using an automated protocol.

Load Report

This option will load a previously saved report file. A report is a combination of recordings and display objects stored in a single file. It saves the placement of the objects on the pages and may contain data in up to all ten-display pages.

Save File

Clicking on Save File will save the active recording. This option is useful when a recording has been modified by a processing operation and it is desired to save after processing, or then a new file has been created by the execution of a processing operation. Original files will be overwritten with this operation.

Save All Files

This option will save all open files on all pages.

Save File as

When choosing to save as a specific file name, you can choose the name of the file to use by entering it manually in the box provided. It is recommended to maintain naming conventions, as seen in the appendix of this manual, when renaming files. Clicking on any of the other options will allow you to change the type of file to a new recording type; this is especially useful when files have been acquired under the wrong modality, such as acquiring an ECochG while on ABR mode.

Save Report

This option will save into a report the arrangement of all the recordings and screen objects currently loaded in all display pages. All recording files and modifications performed will be also saved when using this option.

Save Report to PDF

This option prints all the currently data-populated pages of SmartEP to a portable document format file (PDF). PDF files can be read using Adobe® Acrobat® Reader

Save Single Sweep Data as

This option will allow exporting the acquired data in a sweep-by-sweep format to a Binary file, a normal ASCII file, or a special ASCII file ready for import into Excel. This option is limited to a maximum of 400 sweeps and is only available in certain modalities.

Save as ASCII

This option saves the information contained in the active recording into an ASCII text file. The text file will provide all available test information and basic patient information, it will also include point-by-point data including time, and intensity in comma separated format.

Save Page as ASCII

This option saves the information from the currently active page into an ASCII text file. The text file will provide all available test information and basic patient information, it will also include point-by-point data including time, and intensity in comma separated format for all the recordings in the active page.

Delete from Disk

This option will delete the currently active recording from the computer's hard drive. This operation is permanent and cannot be reversed. The recording will not be cleared from the screen, allowing the user to re-save the file if necessary.

Auto Save

This option should normally be left ON. When it is ON, all acquired data will be saved immediately after acquisition is completed. When this option is OFF, all acquired data will remain on the screen until cleared, but will not be saved to the computer's hard drive.

Auto Comment

When selecting this option, a dialog box will open to allow the entry of a text. This comment will be automatically entered for all recordings that are acquired after this field has been filled.

Clear Selected Data

This option will clear all selected recordings. Multiple recordings may also be cleared at a time using this option. This option will not delete recordings from the computer's hard drive. Unsaved processed recordings will be lost if cleared.

Clear All Data

This option will clear all recordings from all the display pages of SmartEP. This option will not delete recordings from the computer's hard drive. However, processed recordings that were not saved will be lost by this operation.

Process Menu

Power Spectrum

This option will open the Spectral Analysis window. It will load the currently active recording into the window.

Power Spectrum (Multi Buffer)

This option will load recordings acquired using the Advanced Research Module into the Spectral Analysis window as an average, effectively showing them as a single entity. This option is only available for use with the Advanced Research Module.

Filter Type

This option displays the type of digital filter currently selected. Clicking on it will open the Filter Type dialog box, which allows the user to choose between Finite Impulse Response or Spectral filters and their parameters.

Filter Active

This option will apply the currently selected filter, in the option above, to the Active recording. This option affects one recording at a time and does not modify the original recording file unless it is saved to the same filename.

Filter All Displayed

This option will apply the currently selected filter, in the option above, to all recordings in the current display page. This option will not modify the original recording files unless it is saved to the same filename.

Copy Data to new Buffer 等同于复制波形功能

This option will duplicate the active recording. This is commonly used when multiple operations will be performed on a recording, or when an original and a modified file are required.

Addition/Subtraction Mode

This mode specifies how the operation is done. An addition or subtraction of recordings can be done in two ways: sweep weighted or microvolt weighted. Sweep weighted addition will result in a waveform whose sweep count is the sum of the sweep count contained in each recording that was summed. For example, if waveform xxla80b.1 contained 1024 sweeps and waveform xxla80b.2 contained 512 sweeps and these two recordings were added then the resulting waveform would contain 1536 sweeps, with two thirds of the contribution coming from the first and one third from the second. Microvolt weighted operations will disregard the number of sweeps in the operands and perform the operation as if averaging two sweeps.

Add Selected

This operation will add all the recordings that are currently selected to a new buffer. This buffer must be saved manually for later use if necessary. Multiple recordings may be selected by pressing the [Ctrl] key while clicking on the recording handles one by one; selected recordings will appear with a black or slightly grayed handle.

Add All on Page

This operation will add all the recordings on the displayed page to a new buffer. This buffer must be saved manually for later use if necessary.

Add All on Page (Peak V aligned)

This operation will add all the recordings on the displayed page to a new buffer. The addition will align every pre-marked peak V of all recordings and add them relative to that point. The new recording will have the Peak V aligned to the earliest one of the selected recordings.

Add All on Page (Peak V aligned – Mean Latency)

This operation will add all the recordings on the displayed page to a new buffer. The addition will align every pre-marked peak V of all recordings and add them relative to that point. The new recording will have the Peak V aligned to the mean latency of the selected recordings.

Subtract Selected

This operation will subtract the two currently selected recordings and place the result in a new buffer. This buffer must be saved manually for later use if necessary. Only two recordings may be selected at a time to perform this operation. Multiple recordings may be selected by pressing the [Ctrl] key while clicking on the recording handles one by one; selected recordings will appear with a black or slightly grayed handle.

Multiply

Selecting this option will open an input box; the active recording will be multiplied by the constant entered in the input box, at every point in the recording. Any real value, positive or negative, may be used. Only the amplitude of the recording will be affected. This option may be used before the Add option to add recordings proportionally and to normalize intensities.

Invert

The active recording will be inverted on the screen. It can be used to invert a signal in the event that the electrodes were inverted at the time of acquisition. It is equivalent to multiplying the recording by –1.

Time Shift Data

This operation will add the entered time offset to the active recording. Time must be entered in milliseconds and can be negative or positive depending of the direction of the desired time-shift.

Cross Correlate (Selected)

This calculates the cross-correlation between two recordings. Cursors must be active and two recordings must be selected for this operation to be performed. The resulting number is a quantitative value for repeatability between two recordings. For example, two identical recordings have a cross-correlation value of 1.0. The region over which the cross correlation is calculated corresponds to the distance between the two cursors.

Cross Correlate – Split Sweep

This option will calculate the cross-correlation between the rarefaction and condensation buffers of the active recording. Cursors must be active for this operation to be performed. The region over which the cross-correlation is calculated corresponds to the distance between the two cursors.

Split Buffers

This option can be used to separate the rarefaction and condensation buffers of a recording that was acquired using alternating mode to two new recording buffers, without affecting the original. The new recordings will need to be saved for future use if necessary.

Compare

This option will compare two recordings, one from each ear, with equal acquisition parameters. Selecting this menu item will open a window with recording information for the two selected recordings along with latency differences and asymmetry ratios if applicable. The data from the comparison can be added to the screen as a text object.

2D Spectral Display by Intensity

This option shows a new window with 2D graphical representations of the frequency spectrum analysis of the recordings contained in the currently active page. This option will average the recordings pertaining to each one of the selected intensities. There are five available views combining skewed and elevated options, as well as a side view. The window also allows the user to save the graphic data to an ASCII text file for use in third party analysis programs.

2D Spectral Display by Sequence

This option will show the same graphical representation as the previous one. However, it will show the recordings individually, as opposed to averaged, in the order that they were originally acquired.

ASSR Menu

Analyze Active

This option will show the ASSR analysis window including the polar plot of the response, the frequency domain analysis, and a table of the response analysis. A capture of the information in this window may be sent to the current page as an image object for report generation. Please refer to the ASSR user manual for additional information on how to interpret this information.

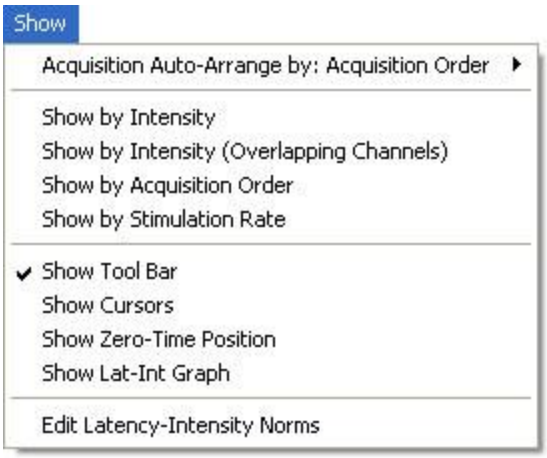
Generate ASSR Response Audiogram

This option will show the ASSR audiogram window with an audiogram reflecting the information gathered from the ASSR responses currently on the page. An image of the audiogram may be copied to the page as an object for report generation. Please refer to the ASSR user manual for additional information about how this graph is generated and for information about correction values.

Extract ABR Components

This option is currently under development.

Show Menu



Acquisition Auto-Arrange

This sub-menu allows the user to select the order in which the incoming recordings are arranged immediately after acquisition. The choices are by acquisition order, by intensity, and by rate. Recordings are not repositioned until after their acquisition has been completed.

Show by Intensity

This option will arrange all of the recordings on the current page in order of intensity; higher intensity recordings will be placed higher, lower intensity recordings will be placed lower. Recordings of the same intensity will overlap. When recordings of the left and right ears are displayed, the page will be split, and it will show right ear recordings on the right and left ear recordings on the left. Recordings will be scaled to accommodate the page.

Show by Intensity (Overlapping Channels)

This option becomes useful when using more than two channels for acquisition. It will perform the same operation as the previous option, but will provide a small shift between channels applicable to the same ear.

Show by Acquisition Order

This option will rearrange all the recordings in the current page according to the order in which they were acquired. Recordings will be ordered in chronological order, starting with the first acquired at the top and ending with the newly acquired at the bottom. If recordings from both ears are present, the page will be split.

Show by Stimulation Rate

This option will arrange the data in order of increasing stimulation rate, from top to bottom. If recordings from both ears are present, the page will be split.

Show Tool Bar

Checking this option activates the tool bar that can be seen at the top of the SmartEP main screen. The Tool bar provides easy access to commonly used functions.

Show Cursors

This option toggles the screen cursors ON and OFF. Screen cursors are used to define the cross correlation area of operation. They may also be used to calculate latencies in a fast and efficient manner. Cursors can be moved by using the "<" and ">" keys; To switch between them, use the "C" key.

Show Zero-Time Position

This option toggles the Zero-Time ruler ON and OFF. This ruler shows on Red and helps when trying to determine the zero-time position on the scale.

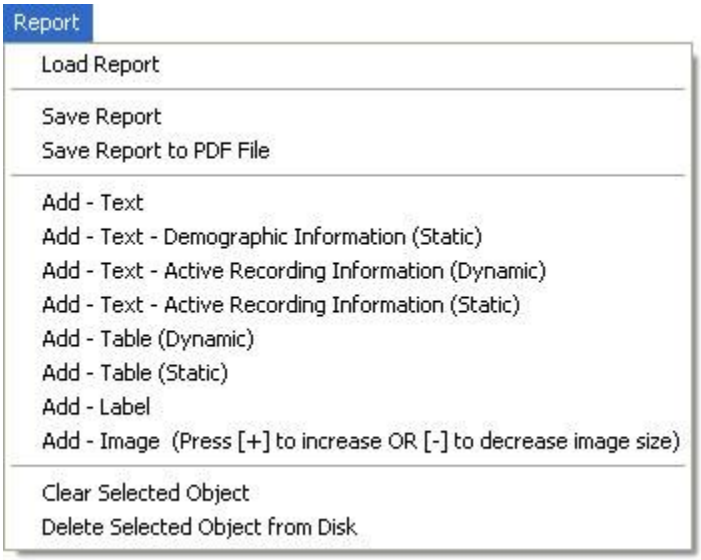
Show Lat-Int Graph

This option will show the Latency-Intensity plot for the currently displayed data. Recordings should have their peaks previously marked for the plot to be displayed. See page 49 for more information about the Latency-Intensity graph.

Edit Latency-Intensity Norms

This option will allow the user to define the normative data that applies to the Latency-Intensity characteristics corresponding to a specific section of the population. This option is password protected to prevent unwanted tampering. See page 49 for information on the Latency-Intensity table dialog box.

Report Menu



Load Report

This option will load a previously saved report file. A report is a group of recordings and objects that are stored in a reference file as they were placed on the pages. A report may contain data in up to all ten-display pages.

Save Report

This option will save the current arrangement of all the recordings and other objects currently loaded in all display pages into a report file. All recordings and modifications performed will be also saved when using this option.

table is that it can be edited manually by the user. Refer to page 44 for additional details about the text-editing window.

Add Label

This option adds a field using a single line of text and can be positioned anywhere on the screen. Regardless of how much is entered in the field, only a single line of text will show and print; any text exceeding the one line limit will be ignored. Refer to page 44 for additional details about the text-editing window.

Add Image

This option allows the addition of an image to the display area. The image must be in bitmap format and it can be resized up or down using the [+] and [-] keys respectively. It is preferable to add images that have been placed in the patient directory. It is recommended to delete images that are not needed to reduce hard drive usage.

Clear Object

Selecting this option will clear the currently active object from the display area. All objects are affected by this option, including recordings. Although the object is no longer shown on the screen, it is not deleted from the computer hard drive.

Delete Selected Object from Disk

This option will clear the currently active object form the screen and delete its source from the computer hard drive. A warning dialog box will ask for confirmation. This action is final; deleted data cannot be recovered or re-loaded later.

Print Menu

Print Page

This will print the current page to the printer selected in the printer setup window. The printed page will include the patient's name, ID, and recordings displayed with acquisition parameters, peaks, and latencies.

Print Page PDF Preview

This will print the current page to a PDF file and open it in the Acrobat Reader automatically. PDF documents contain the same information as pages printed on a standard printer.

Print All Pages

This option will print all pages containing recordings to the printer that was selected in the printer setup window. The printed pages will include the patient's name, ID, and recordings displayed with acquisition parameters, peaks, and latencies.

Print All Pages PDF Preview

This option will print all pages to a PDF document and open it in the Acrobat Reader automatically. PDF documents contain the same information as pages printed on a standard printer.

Save Report to PDF

This option prints all the currently data populated pages of SmartEP to a portable document format file (PDF). PDF files can be read using Adobe® Acrobat® Reader.

Add Text

This option opens a window that allows the user to enter a snippet of text to be displayed in the page. This text field can be of up to 64kB. The edit window also contains options to save and load text as a template file for future use. It is possible to change the font display of the text fields. Note that non-standard font types may not be supported by the program, and will show the default font type. The text field can be moved to any place on the display area using its corresponding handle (note that object handles, although shown in the screen, do not print). Text fields are saved as files to the patient folder. If a text field is going to be cleared and will not be used again, it is recommended to delete the object rather than clear it. Refer to page 44 for additional details about the text-editing window.

Add Text – Demographic Information (Static)

This option opens a window with pre-filled data taken from the current patient information in the patient database. It is possible to edit the information and modify it as needed. Text fields can be positioned anywhere on the display area. Refer to page 44 for additional details about the text-editing window.

Add Text – Recording Information (Dynamic)

This option will add a text field displaying the information of the active recording. This field is a dynamic object, therefore it updates automatically as changes to peak markings are made to the recording. This field can be positioned any place in the display area and can be moved to other pages. If there are multiple instances of the same recording loaded simultaneously, the recording information object will obtain the information from the instance closest to its position in the display pages. Note that the recording must be a saved buffer in order for this information to appear. This text field cannot be edited by the user.

Add Text – Recording Information (Static)

This option will add a text field displaying the information of the active recording. Unlike the previous option, this text is dynamic and can be modified by the user. This information will not be updated automatically if there are changes on the active recording.

Add Table (Dynamic)

This option adds a table containing all the information that pertains to the recordings currently in the page; since the table is dynamic, it will update the information automatically as the peak markings change or as new recordings are added to the page. Although the table reflects the current contents of the active page, it can be moved to a different page for advanced report creating purposes. This table includes acquisition parameters, peak information, and inter-peak latencies. This table cannot be edited by the user.

Add Table (Static)

This option adds a table containing all the information pertaining to the recordings in the current page; however, this table is static and will not update the information if anything changes in the recordings or a new recording is added to the page. The advantage of this

Black & White

When checked, this option will force a black and white printout regardless of object color on the screen. This is especially useful for users with black and white only printers where certain colors may print in lighter gray tones than desired.

Automatic Tables

When checked, this option will enable the printing of recording information tables at the bottom of every page. These tables can be viewed by scrolling down the display page when the option is enabled.

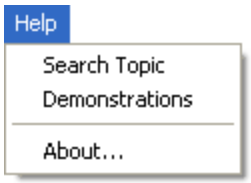
Multi-Page Format

When checked, this option will separate the recordings page from the automatic table when there are more than 16 recordings in one page. It will separate the information into multiple pages as necessary. Not having this option enabled may cause the tables to truncate at the time of printing. This option is only available when the Automatic Tables option is enabled. It is preferable to leave this option ON as default.

Printer Setup

This option will open the Printer dialog box. The dialog box will let you choose the printer that will be used as well as other printing properties. Please consult the printer's documentation for details on printer properties and options.

Help Menu



Search Topic

This option will open the Help dialog box. This dialog box will allow the user to search for specific topics about the SmartEP software.

Demonstrations

Selecting this option will open a dialog box with the names of all the available demonstrations, which show how to use SmartEP.

About

This option displays the system information screen, including the software version, the current detected hardware, type of system, and contact information for IHS.

SmartEP Windows

4

EEG and Amplifier

The EEG and Amplifier dialog box can be accessed from the control panel by pressing the EEG and Amplifier button or in the Main Menu using System > Amplifier Settings.

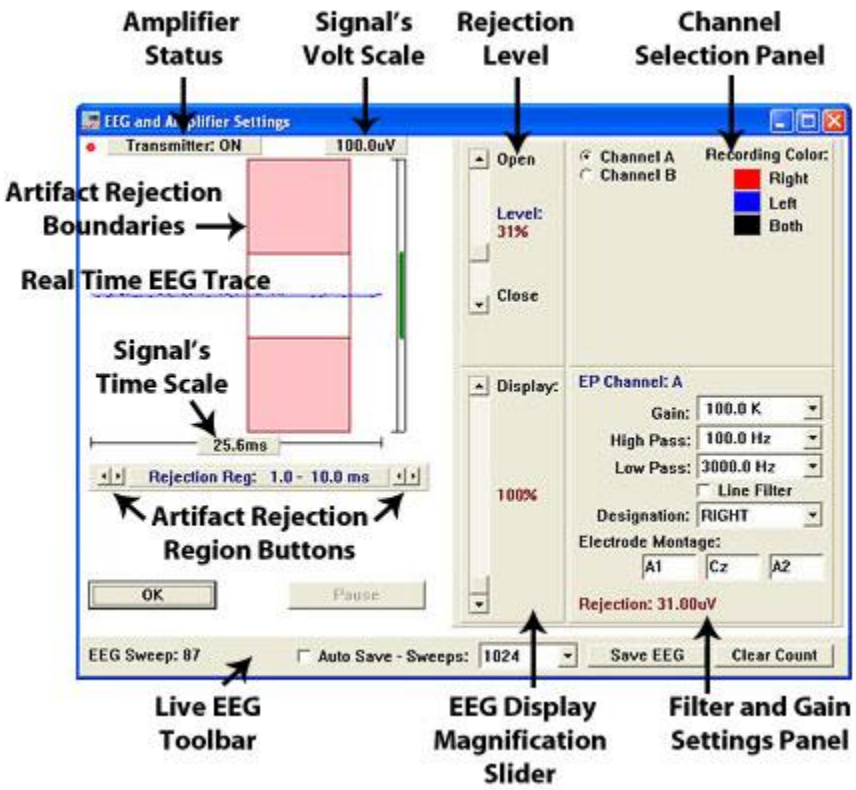


Figure 14 – EEG and Amplifier window

The main window on the left hand side shows the real time EEG trace as it is being sent from the transmitter. The status of the transmitter is displayed above the EEG window. The corresponding scales for the time and voltage axis are displayed on the bottom and right side of the signal's window, respectively.

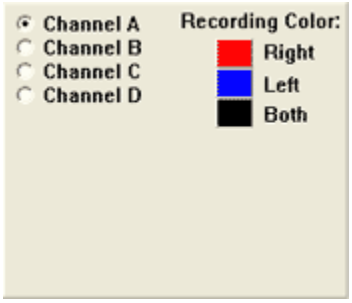
Artifact Rejection

Artifacts are unwanted noise signals that may impair the ability of a device to acquire evoked potentials. Artifacts are present in most environments. To prevent the artifacts from affecting the recording, the software is capable of setting an artifact rejection region and level. The artifact rejection boundaries are graphically shown in the EEG and Amplifier box, see Figure 14. Each type of evoked potential recording (ABR, MLR, etc) will need a specific artifact rejection region that allows the particular type of signal to be acquired; this region can be defined using the Artifact Rejection Region buttons on the lower left of the EEG and Amplifier

window. Most artifact signals will have larger amplitude than a normal auditory evoked potential; setting the artifact rejection level will prevent some types of artifact. The Artifact rejection level slider may be adjusted, until the desired EEG falls inside the boundary lines.

When artifacts are present, parts of the EEG trace may fall on the rejection area. The rejected part of the Real Time EEG will be plotted in red, while the accepted parts will be plotted in blue. An artifact may be produced by certain situations such as patient movement that produces an irregular but brief high signal. The artifact rejection level is defined as a percentage of the total input range of the analog-to-digital converter of the system. This value is converted to a μ V level that depends on the gain of the system.

Channel Selection



The top right panel is the Channel Selection panel. To select a channel, click on the radio button next to the name or click the name of the channel you wish to configure. The amplifier settings displayed on the amplifier settings panel and the recording color settings will change according to the selected channel. Only the channels listed in black lettering are available and can be selected. The availability of the channels depends on your system's configuration. USB Lite systems will also include an impedance-checking button at the bottom of this panel.

Recording Color

The recording color selector allows the user to define the color that will be used to show the recording on the screen. Recordings acquired from the channel-ear combination will inherit the selected colors. This option is helpful for users who wish to acquire easy to recognize ipsilateral and contralateral recordings. When selecting Channel A to have a designation of right, the recordings acquired will show the color selected under right for right-ipsilateral recordings, the color selected under left for left-contralateral recordings, and the one selected under both for recordings acquired with stimulation to both ears. The user may choose any of the standard windows colors or define a custom color from the full RGB color palette. When selecting custom colors, you must take into consideration that certain colors may not show on the screen or print visibly. Once acquired, the color of a recording can be changed using the active recording menu.

Amplifier Settings Panel

Below the Channel Selection panel is the Amplifier Settings panel, displaying the current settings for the selected channel. These settings can be changed by selecting a different value from any of the drop-down menu options. These are the settings and their standard values:

- Gain: 30, 50, 75, 100, 150, 200, 250, and 300 k.
- High Pass: 1, 10, 30, 50, 100, 150, 300, and 500 Hz.
- Low Pass: 30, 100, 300, 500, 1000, 1500, 3000, and 5000 Hz.

- Line Filter: The line filter may be set to 50 or 60 Hz depending on your country's line power frequency (60 Hz in the US). This filter is used to remove power line noise and will filter the signal for frequencies appearing at the appropriate value. It is recommended to keep the notch filter OFF; however, it may be necessary to use it in high noise environments, such as in a NICU (Neonatal Intensive Care Unit).
- Designation: RIGHT, LEFT, ON, OFF. This option designates which channel is to become active during data collection based on the Ear selection and acquisition Mode. If a channel is designated as a Right channel, data will be collected from that channel when the ear is set to right and the Mode is set to Ipsi. Data will also be collected from that channel if the ear is set to left and the acquisition mode is set to Contra. If the designation is set to ON, the channel is always used for data collection regardless of ear and acquisition mode settings. If you have a one-channel system, it is preferred to set channel A's designation to ON. If you have a two-channel system, it is preferred to set the channel A's designation to Right and B's to Left.
- Electrodes: XX-XX-XX. This is a user-specified input used to keep record of the electrode patient connections at the time of acquisition. The electrode configuration gives information about the electrode setup for future reference. This information is stored with each recording file. Changing the electrode configuration information will not change the way the system acquires data. By default, the International 10-20 Electrode System positions are chosen by the system (i.e. A1-Cz-A2).

Note that USB Lite units, designed to work with SmartEP and SmartScreener, will have the Gain set to 100k, high pass filter to 30 Hz, low pass filter to 1500 Hz, and Notch Filter ON. These are the recommended settings for acquisition in infants.

Live EEG Toolbar

The Live EEG toolbar has controls to allow the user to save the incoming EEG signal to a file. When checking the Auto Save feature, the program will save a file with the number of sweeps as defined in the drop-down box to a binary file. This file can be found inside the patient folder. You may also save the EEG that is currently in the buffer by clicking on the Save EEG button which will also clear the EEG count, or you may clear the count and the EEG buffer without saving by clicking on the Clear Count buffer. The buffer count only occurs while the EEG and Amplifier window remains open. This option is mainly intended for debugging purposes.

Recommended Amplifier Settings

The recommended amplifier settings when collecting ABRs from adults are:

- Gain: 100K,
- High Pass: 100 Hz, 新生儿推荐滤波范围30-1500Hz（手术室或重症监护室
- Low Pass: 3000 Hz, 需提高到100-1500Hz，详见P101）
- Line Filter: ON, and
- Artifact Rejection between 15 and 20 μ V.

Remember that the amplifier parameter settings depend on the age of the patient (i.e. infants require different ABR filter settings, typically 30 Hz for high pass and 1500 Hz for low pass). The settings also depend on the type of signals you are acquiring (i.e. ABRs vs. MLRs or

LLRs). Under noisy recording conditions such as an Operating Room or a NICU, it may be necessary to increase the high-pass filter to 100 Hz for infants. Changes made under EEG and Artifact may be stored in a parameter file by using the Save Settings command from the SmartEP Control Panel. Please refer to the appendix section on recommended settings on page 101 for additional details about settings for different modalities.

The Display scroll bar allows the EEG vertical axis to be changed on the EEG display. This bar only affects the display, allowing the artifact rejection level to be easily set and adjusted. If the amplitude of the EEG signal is small on the display, you may expand the EEG by selecting the up button on the display scroll bar.

Auditory Stimulus Generation

The Auditory Stimulus Generation dialog box can be accessed by either clicking on the Stim: Button of the control panel or using the main menu Stimulus > Type. This dialog box contains options necessary to specify the type of stimulus that will be provided to the patient. The dialog box is shown below.

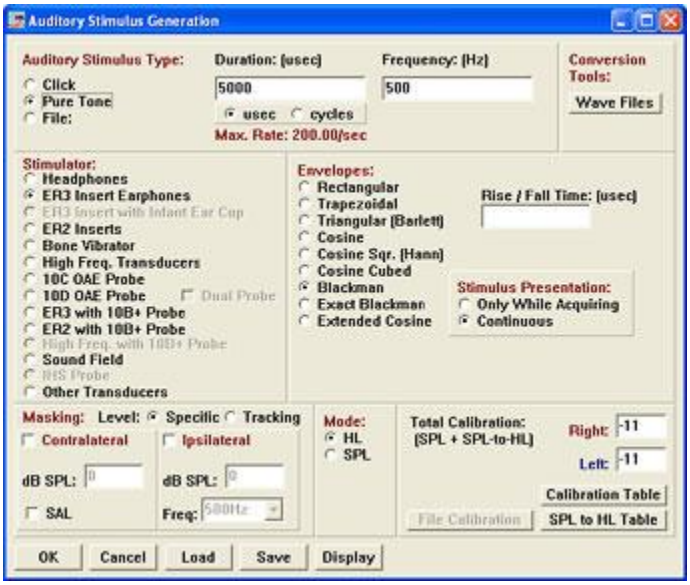


Figure 15 – Auditory Stimulus Generation window

Auditory Stimulus Type

The top panel of the dialog box allows for the selection of the desired stimulus to be applied. The stimulus may be in the form of a click, pure tone, or user defined file.

Figure 16 - Auditory Stimulus Type

A click sound stimulus is generated by presenting a rectangular electrical pulse to the transducer. The length of the electrical pulse is specified by the duration field, given in microseconds. The level is pre-calibrated to 32 dB SPL at 0 dB nHL (when using standard

click duration of 100 microseconds). The calibration value displayed in the calibration panel (bottom right) will be used to calibrate the click signal. The envelope panel and the frequency field are disabled when using click stimulation. The rate of stimulation is defined outside this dialog box, in the SmartEP control panel.

A pure tone is a signal of a given frequency and duration as specified in the frequency and duration fields. Any frequency value between 2 and 16000 Hz can be entered for basic auditory modality; other modalities may have different ranges. The type of envelope used for the signal can be selected from the envelope list panel. The trapezoidal and extended cosine envelopes will require the user to input a rise/fall time in microseconds. The calibration values for the appropriate frequency range, as displayed in the calibration panel, will be used. Calibration values may be changed by accessing the calibration table. In order to store a particular calibration value with a particular stimulus for future use, the new values must be saved as a stimulus file by using the Save button at the bottom of the dialog box. If it is not saved as a file, the system will use the default Calibration Table values every time that stimulus is generated.

The file option under the stimulus type allows the use of a previously saved stimulus file. Clicking this option will call a Load File dialog box, also accessible with the Load button at the bottom of the Auditory Stimulus Generation dialog box. A previously saved stimulus file will contain the stimulus type (click or pure tone), duration, frequency (only for pure tones), and calibration values. The default calibration table and SPL to HL table are disabled when a stimulus file is loaded into the system.

Note: Keep in mind that if the default calibration values are altered, the file calibration values in each previously saved stimulus file will also need to be changed accordingly to maintain recording accuracy and reliability.

- Duration (25-500000 μ sec; integers, no decimals): This option specifies the duration (in microseconds, μ s) of the stimulus signals. Keep in mind that the typical click stimulus has duration of 100 microseconds, while a pure tone stimulus has a duration of 2-5 ms or 2000-5000 microseconds.
- Frequency (2 – 16000 Hz, whole numbers): This option is only active when the stimulus is a pure tone. It specifies the frequency of the stimulus to be applied. A frequency of up to 32000 Hz can be entered if the high frequency modality is installed and high frequency equipment is available. See page 69 for additional information on alternate modalities.

Stimulator

The stimulator panel allows the selection of the type of transducer that will be used. It is important that the correct stimulator be selected, since some of the stimulator types will affect how the data is recorded. Choosing the wrong stimulator may prevent you from acquiring an accurate recording.

Caution: do not change the device selection while headphones or insert phones are connected to the patient.

Envelopes

This option allows the selection of various stimulus windows from rectangular to extended cosine. The Trapezoidal and Extended Cosine envelope options require a rise/fall time. To see how the envelope will affect the way that the stimulus is presented, follow these steps:

1. Select pure tone as the stimulus type.
2. Enter 2000 for the frequency value. The effects of the envelope will be easily identified by using a value of 2000 Hz or higher.
3. Select the Rectangular envelope from the list.
4. Click on the Display button at the bottom of the dialog box. The new window will show a graphical representation of the stimulus, as it will be presented, seen in time and frequency domains.
5. Click OK to close the display window.
6. Select the next envelope type and repeat from step four. Make sure to enter and experiment with the rise/fall time when choosing the Trapezoidal or the Extended Cosine envelopes and see the results on the display window.

Rise/Fall Time (μsec)

This attribute is required for the Trapezoidal and Extended Cosine envelopes. It will only be available when those envelopes are selected.

Stimulus Presentation

This option determines whether the ear is being stimulated only during data acquisition or at all times, while the SmartEP software is running. In Continuous mode, the stimulus is presented uninterrupted. In Only While Acquiring mode, the stimulus is presented only during acquisition and turned off otherwise. Leaving the stimulation on continuously is needed when setting the artifact rejection level.

如果双耳听阈相差较大，则正常耳会影响到测试结果，同时掩蔽值应为刺激声SPL-39dB

Contralateral Masking

Contralateral masking is used to prevent a stimulus signal presented to one ear from stimulating the opposite ear. When there is a large hearing threshold difference between ears, high intensity level stimuli presented to the weaker ear may be picked up by the better ear. The evoked potentials generated by the better ear will make the interpretation of results difficult. The effective masking level is approximately 39 dB below the selected SPL level. This value was obtained using the following correction formula:

Effective Masking = SPL - 10log (Bandwidth = 8000 Hz) = SPL – 39dB

Mechanical masking is that which is provided by the mechanical characteristics of the transducer. Typically, headphones provide a mechanical masking of approximately 40 dB of masking, while insert earphones provide with 60 dB of mechanical masking between the ears. Effective masking is the true masking, that which is actually produced by the

transducer. The effective masking, because of the characteristics of the signal, will be –39 dB from the value entered into the program. Due to these two types of masking, the value entered must be adjusted accordingly in order to obtain the desired value at the sound output. Therefore, when a value of 60 dB is entered in the software, the effective masking (transducer output) will be of about 21 dB; the actual masking will be the effective added to the mechanical, in this case 61dB if using headphones. Therefore, selecting an SPL masking value close to the stimulation level should provide sufficient masking. For example, if you are selecting an absolute masking level and you are testing at 70 dB HL, turn masking ON and select a masking level of 70 dB SPL. 头戴式耳机固有掩蔽值为**40dB**，插入式耳机固有掩蔽值为**60dB**，有效掩蔽值**X**等于测试强度值减去**39dB**，所以掩蔽输入值（最终值）等于**X**加上固有掩蔽值，并直接切换单位为**HL**。

Specific and Tracking Options

The masking Specific option defines a set masking level that will not change unless the masking intensity field is modified. The specific masking level is given in SPL. The masking Tracking option lets the masking level follow the signal level as it changes; the masking level presented will change as the stimulation intensity is changed. If a value of 10 dB is entered as the offset and the stimulation intensity is 70 dB HL, the masking level presented will be 80 dB SPL (70 + 10 = 80). The numeric value of the stimulation intensity is used, disregarding whether or not the value is in SPL or HL.

Ipsilateral Masking

It is possible to provide masking to the same ear that is being stimulated. The provided masking can be white noise or notched noise (at predefined frequencies). This kind of masking provides with frequency specific auditory brainstem responses. Ipsilateral masking is not a standard option of SmartEP; it must be purchased separately.

Sensorineural Acuity Level (SAL)

This option, located in the masking section, switches the masking noise from the insert earphones to the bone conductor. This allows the user to perform a SAL procedure.

Stimulus Mode

This option allows the stimulus intensity to be specified in HL or SPL levels.

Total Calibration

These two fields display the current calibration values used for the selected stimulus and stimulator. When a frequency value is entered, the appropriate calibration values are automatically loaded from the calibration table and displayed as the current calibration values. When the system is set to HL mode, the calibration value displayed is the sum of the SPL calibration value plus the SPL-to-HL conversion value. When the system is set in SPL mode, the value displayed corresponds only to the SPL calibration value. To modify the system calibration, you must access and modify the specific calibration tables by clicking on the Calibration Table button located at the bottom right of the dialog box. Calibration tables are password protected.

Auditory Stimulus Generation dialog box Buttons

OK	Exits the Auditory Stimulation Generation dialog box and applies the changes to the stimulus output.

Cancel	Exits the Auditory Stimulation Generation dialog box and discards all changes made.
Load	Opens the Load file dialog box to load a previously saved stimulus file. Same as choosing file as auditory stimulus type.
Save	Saves the current stimulus as a file along with the current calibration values for future use.
Display	Displays the stimulus signal in the time and frequency domains, which may be sent to a printer.
Calibration Table	Displays the intensity calibration table for the various output channels and frequencies. Password required.
SPL to HL Table	Displays the SPL to HL conversion table for the various stimuli and frequencies. See page 43. Password required.
File Calibration	Displays the current file calibration table for the various output channels. See page 46. Password required.

Stimulus

Accessed through the Auditory Stimulus Generation dialog box, the Stimulus dialog box displays the waveform of the current stimulus both in the time domain and in the frequency domain. The directional buttons at the bottom can be used to scroll through the time domain signal when it is too large for the scale. Clicking Print will send a screen dump of the dialog box to the default printer. Click OK to close.

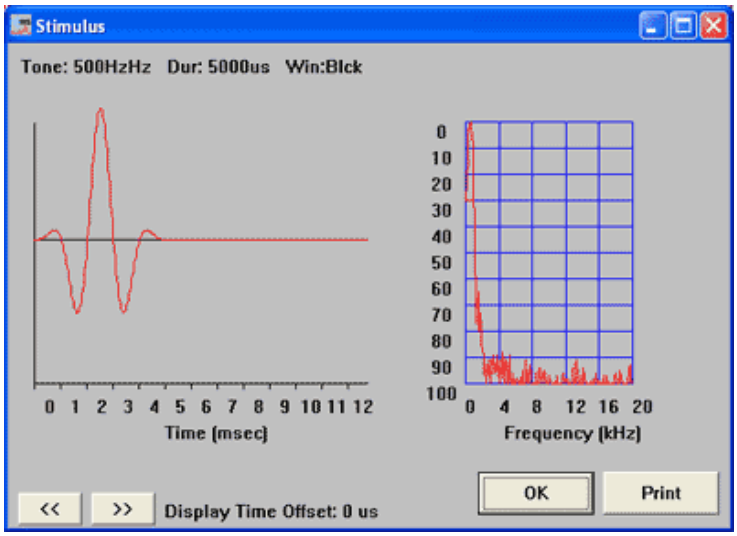


Figure 17 - Stimulus Display dialog box

Calibration Table and Dialog Box

This dialog box allows the user to change the sound level calibration for each stimulus device according to calibration measurements. It is accessed through the Auditory Stimulus

Generation dialog box; it displays a series of correction factors for masking and various frequencies corresponding to the output transducer chosen.

Note: Although the fields are present and editable up to 50 kHz, the system will still present tones up to 16 kHz unless the High Frequency mode is selected and the High frequency equipment is present. For more information about the high frequency modality, see page 69.

The Ear and Stimulator (output device) options are selected by clicking the corresponding radio buttons on the right hand panels. Programmed correction values are displayed. These values can be changed directly in the edit boxes for each frequency and then saved with the stimulus file using the Save button in the Auditory Stimulus Generation dialog box. Remember to change the numbers back to their original values if necessary. The Value options include correction, maximum, and minimum. The correction values are the calibration-offset values. The maximum and minimum values correspond to the upper and lower transducer output levels that can be generated by the stimulator in dB SPL. Maximum and minimum values are limited by the transducer's physical characteristics.

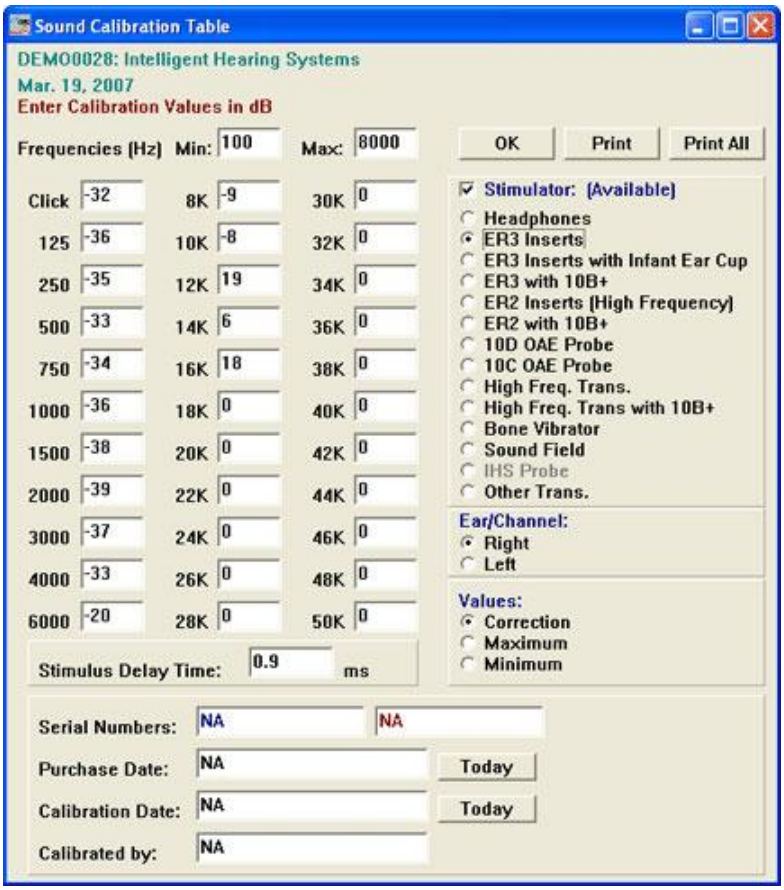


Figure 18 - Sound Calibration Table

SPL-to-HL Table and Dialog Box

This dialog box is selected from the Auditory Stimulus Generation Dialog Box and allows the user to set the dB sound pressure level (SPL) that corresponds to 0 dB hearing level (HL) for

the click and various frequencies. These values are very dependent upon the type of stimulator that is being used on the patient. Note: although the 50 kHz frequency fields are present and can be edited, the system will still only present tones up to 16 kHz unless the high frequency modality is installed and selected. For more information about the high frequency modality, see page 69.

Select the Stimulator by clicking on its radio button on the right hand side of the panel. Programmed SPL to HL values are displayed. These values can be changed directly in the edit boxes for each frequency and then saved with the stimulus file using the Save button in the Auditory Stimulus Generation dialog box.

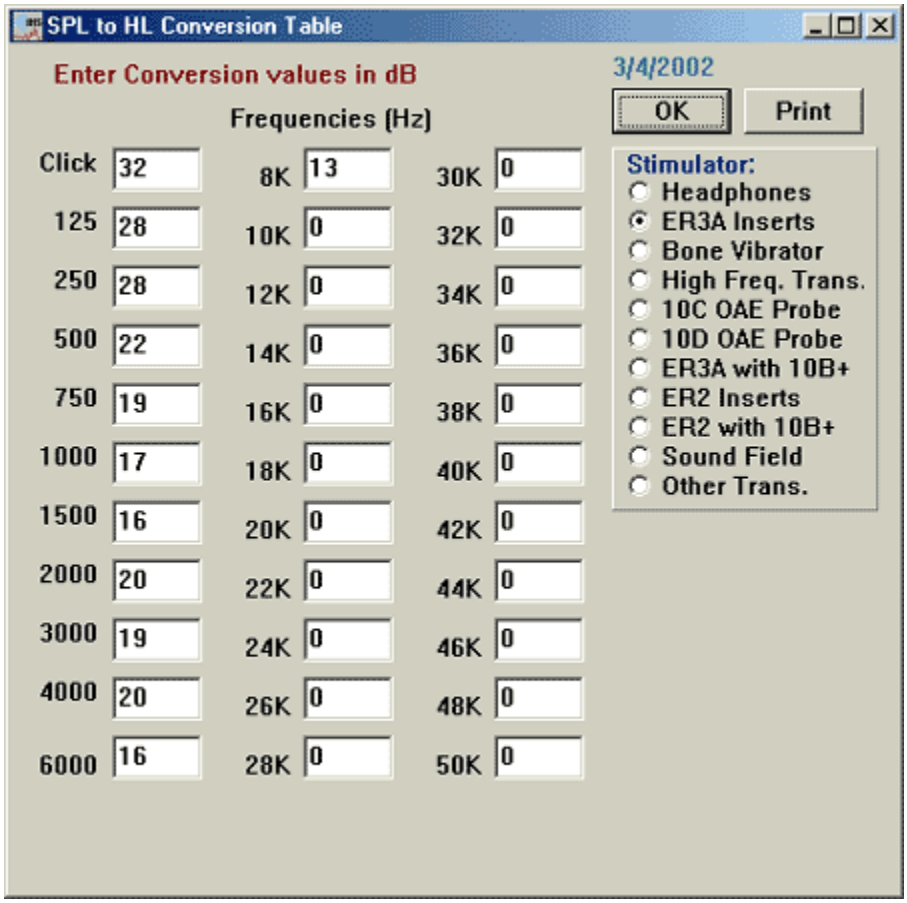


Figure 19 - SPL to HL Conversion Table

Text Edit Window

The text editor window allows the user to insert and edit text to be shown in the display area of the screen. This window will open when selecting the edit menu option of any text object or when creating a new text file; however, it will only allow the editing of text in static objects. Adding or modifying text in dynamic objects will result in the new text being lost once the window is closed. The window can also be used to change the font type, size, and color in static and dynamic text objects.

The size of the text editor window will determine the size of the text object shown on the display area. When reducing the size of the window it may be necessary to check the "Word Wrap" option to allow the inserted text to adjust automatically to the size of the window. It is possible to save and load snippets of text as a template text files to the hard drive for future use. To save the inserted text as a template, click on the "Save as Template Text" button and choose a descriptive name for the contents. For organizational purposes, the software chooses the TEXT_TEMPLATES folder in your software installation directory as the default location for saving template text files.

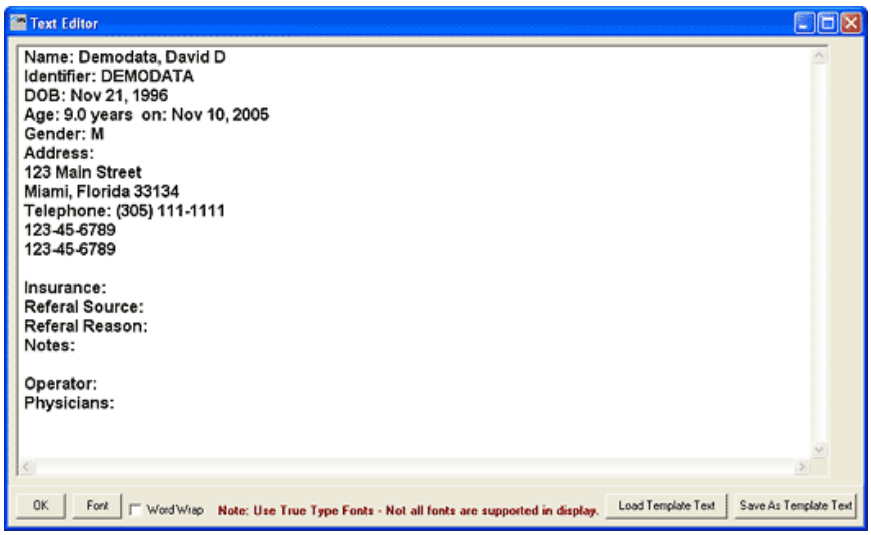


Figure 20 - Text Editor Window

Some special considerations may be needed when changing the font of the text files. Most Open Type and True Type fonts work properly with the system, but other types of fonts may not be supported for display or for saving reports to PDF format. Keep in mind that some font designs use different spacing for each character; using these fonts may result in tables that do not line up correctly.

File Calibration

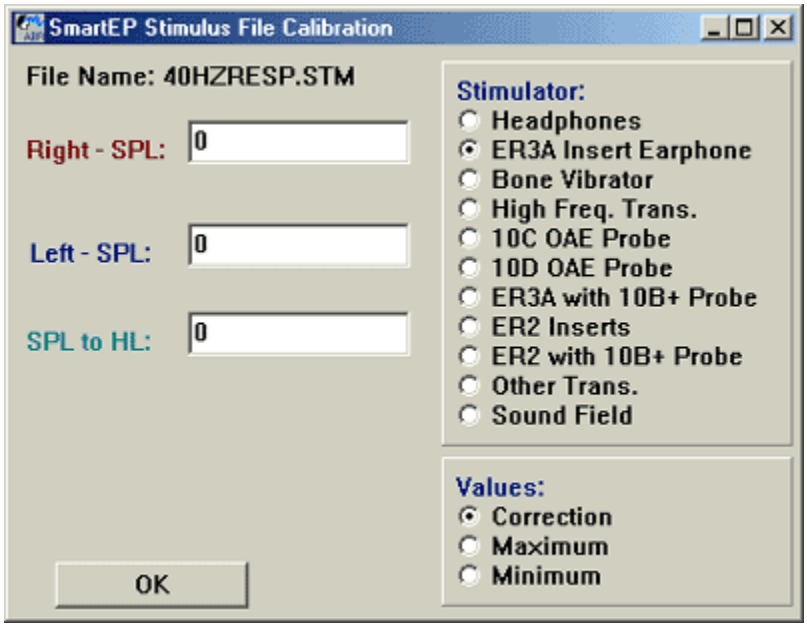


Figure 21 – File Calibration window

This dialog box allows the user to change the stimulus device sound level calibration values for previously saved stimulus files. The File Calibration Dialog Box is only available when a stimulus file is loaded to the SmartEP program and can be accessed from the Auditory Stimulus Generation dialog box. It displays the right and left correction factors for the loaded stimulus data file that corresponds to the output transducer choice. Data files may contain clicks of any duration, pure tones, or other stimulus data such as speech. The SPL to HL conversion level can also be set to the proper value.

The Value options include correction, maximum, and minimum. The correction values are the calibration-offset values. The maximum and minimum values correspond to the upper and lower transducer output levels that can be generated by the stimulator in dB SPL. Maximum and minimum values are limited by the transducer physical characteristics.

Protocol Setup

The Protocol Setup Dialog Box is used to create an automated protocol routine. It consists of up to 64 items; each item may acquire a count of up to 64 recordings using identical parameters.

By default, count is zero (0) and the parameter list (Ear, Intensity, Rate, Stimulus, Mode, etc.) is grayed out or unavailable. To enable the parameter list, enter a non-zero number for count. All of the items set as default will use the current settings from the Control Panel when the protocol is executed. When the Next button is clicked, the protocol Item number is incremented and new settings can be entered for the next item. This procedure is repeated as many times as necessary to build the desired automated protocol.

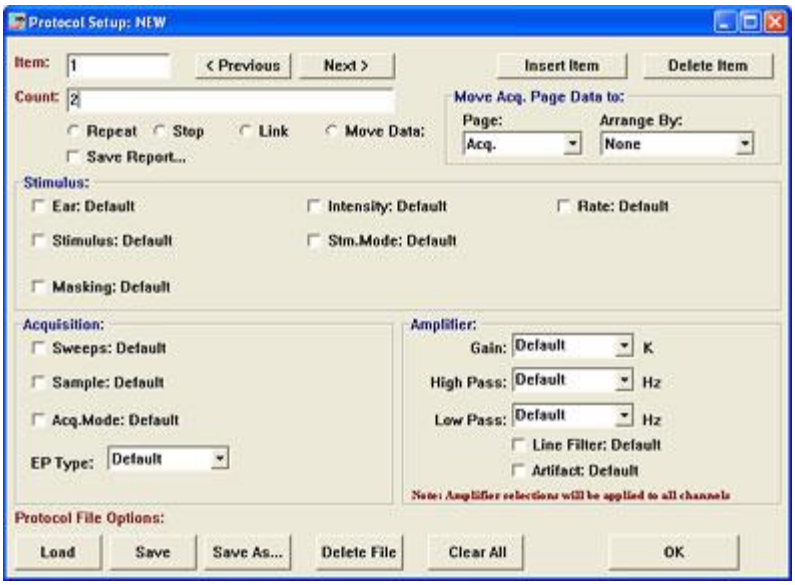


Figure 22 - Protocol Setup dialog box

Protocol Tutorial

This tutorial will walk you through the steps required to create a new protocol file. Follow these steps to make a protocol which tests twice at 70 dB nHL, twice at 50 dB nHL, and twice at 30 dB nHL:

1. Click on Protocol > Setup Protocol to open the Protocol Setup dialog box. The dialog box should open showing a one under Item and zero under Count.
2. Enter "2" under Count. This will make the settings active.
3. Click on the Intensity setting and enter 70 in the input box that appears.
4. Click on Next to move to the second Item.
5. Enter "2" for Count and "50" for the intensity.
6. Create a third Item by clicking on the Next button.
7. Enter "2" for Count and "30" for intensity.
8. Click on Save to save the protocol file to the hard drive.
9. Enter a descriptive filename for easy identification.
10. To run the protocol automatically click on Protocol > Execute Protocol

To change or view Items from a previously saved protocol:

1. Click on Load from the Protocol Setup dialog box,
2. Select the name of the protocol to be edited from the list.
3. Once loaded, navigate the Items using the Previous and Next buttons to find the one that must be changed,
4. Modify the values accordingly,
5. Click on Save to update the parameters to the file or Save As to create a new protocol file.

To delete a previously saved protocol:

1. Click on Load from the Protocol Setup dialog box, and select the name of the protocol to be deleted and click OK
2. Once loaded, click the Delete button.

Protocol Setup Buttons and Options

○ Repeat	Allows the user to repeat the current protocol sequence indefinitely until it is stopped manually by the user. This option requires user assistance during acquisition. No more Items should follow when this option is selected, they will be ignored.
○ Stop	Declares the current item as the end of the protocol sequence. This option can be used to shorten a saved protocol without altering the originally saved protocol list. It is not necessary to provide a Stop at the end of every protocol for it will stop after the last defined Item.
○ Link	This option allows the user to link another protocol file to the current item. The linked Item will be executed after this protocol finishes. A protocol may be linked to itself, however, doing this requires operator supervision since may create an infinite number of files in the patient's folder.
Previous	Displays the previous Item number to allow viewing or editing of that protocol item.
Next	Displays the next Item number to allow viewing or editing of that protocol item. When currently in the last Item, this button will generate a new Item.
Delete Item	Deletes the current Item and automatically renumbers all succeeding items so that no Item number is missing.
Insert Item	Inserts a new item before the currently displayed Item. It will renumber all subsequent items to accommodate the new one.
Clear All	Clears all defined items in the current protocol sequence.
OK	Closes the Protocol Setup dialog box and applies all changes to the protocol. Changes are not saved to file on the hard disk unless Save is clicked.
Load	Opens a Load File dialog box. It replaces the current protocol with the opened file. All changes to the replaced protocol will be lost if not previously saved.
Save	Saves the current protocol file. If the protocol was previously loaded, no filename will be requested, otherwise, an input box

	will request a filename 8 characters or less (no extension).
Save As...	Saves the currently loaded protocol under a new filename. It will open an input box requesting a filename for the protocol.
Delete	Deletes the currently loaded protocol from the hard drive. Deleted files cannot be recovered, they must be recreated.

Latency-Intensity Graph

This dialog box can be accessed from the Show option in the SmartEP main menu. Clicking on Show Latency-Intensity Graph will call the dialog box shown in Figure 23. If data from more than one ear and acquisition channel are present in the current page, the program will automatically generate the graph for the data set with the most samples on that page. If there are 10 right ear recordings and only 2 left ear recordings, the latency-intensity graph for the right ear will be shown. Samples data from other pages are not used in the generation of the graph. Once the dialog box is open, the user may manually select which ear(s) and acquisition channel(s) to use to generate the graph using the choices given at the top-right hand side. Information about the peaks is in table format and displayed on the lower area of the dialog box.

The shown graph or data may be printed or saved using the buttons below the channel selectors. Printing a graph will send a screen-shot to the printer. Printing the Table will print the data as it appears in the lower panel. Saving the table will save the data to a text file in a location specified by the user. Copying the Image to page will generate a graphic object on the report page with the graph. Copying the table to the page will generate a static text object containing the data from the lower panel (refer to page 44 for details about dynamic and static text fields)

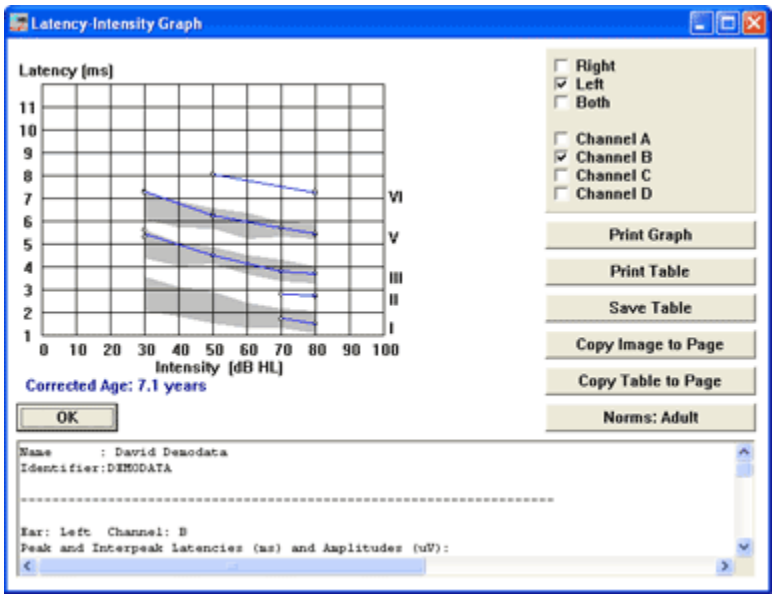


Figure 23 - Latency-Intensity Graph

Different latency-intensity norms may also be displayed along with the data. The Norms button allows the user to select a specific set of norms, which were previously entered and

saved, to be displayed and printed with the current latency-intensity graphs. When the Norms button is pressed, an Open File dialog box will appear displaying a list of all the available norm files. Norms Files can be generated using the Lat-Int program as described later in this manual. Default Norms may open with the Lat-Int graph if a file was defined as the default norms in the Lat-Int Program. When the default norms are used, the age of the patient entered under Demographics will determine the appropriate age norms to be displayed in the latency-intensity plot.

Display Selection:

Multiple combinations of data can be displayed in the Latency-Intensity graph. The Selection panel, in the upper right hand, contains all the necessary options. "Right" and "Left" will select the recordings that were acquired for the respective ears, while the "Both" option will select recordings that were acquired binaurally. The set of channel options is necessary to specify which group of recordings should be displayed. There must be at least one recording matching the options selected for a graph to be generated.

Latency Intensity Norms

The Latency-Intensity Norms Program creates the normative data files that can later be loaded into the Latency-Intensity Graph dialog box (see page 49). It is accessed from the Show option in the SmartEP main menu. To access this program outside of SmartEP, navigate your computer's hard drive and find Sepwlat.exe in the IHSPROGS programs directory. It can also be started using the shortcut found in the Intelligent Hearing Systems Launch pad program. This program is protected by the system password in order to prevent unwanted tampering.

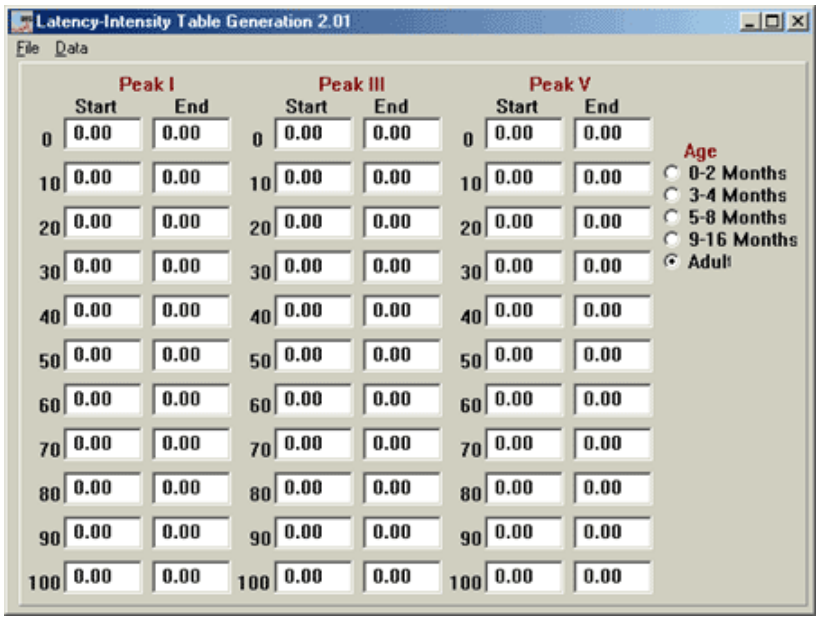
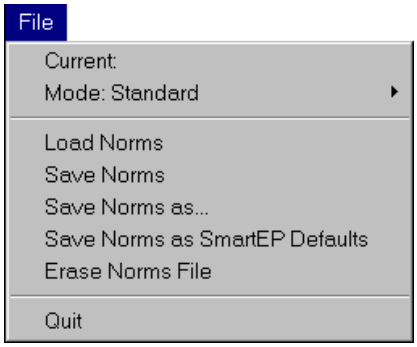


Figure 24 - Latency-Intensity Table Generation window

File Menu

The File menu contains options for loading and saving individual normative data files. There are two types of files: Standard and Special. The Standard file named SmartEP.ltn will

automatically be loaded in SmartEP and displayed along with patient data, if it is present. Other standard files may be saved with different filenames for storage and they can be manually loaded into the graph dialog box, yet they will have the same filename extension. Special files are used to create norm files for specific age groups; they are saved under a different extension (*.lts) and must be manually loaded in the SmartEP program Latency-Intensity dialog box.



Mode (Standard or Special): The standard mode allows the user to enter, load, save, and save as default files containing normative information for all five-age ranges:

- 0-2 months
- 3-4 months
- 5-8 months
- 9-16 months
- Adults

The special mode allows the user to enter, load, and save files containing normative information for any one specific age range or demographic. For example, to build a normative database for the age range 33-34 weeks, select the special mode, and enter the normative information as explained below. Then save this set with a descriptive file name, like, 33-34. This set of norms may now be loaded directly from the Norms button in the Latency Intensity dialog box in SmartEP.

Load Norms: This option allows the user to select any set of norms previously entered and saved. Make sure to select the appropriate mode in order to load the desired file. Select Special when loading a special file and Standard when loading a standard file.

Save Norms: Saves the entered normative information, and adds the file name to the list available for loading.

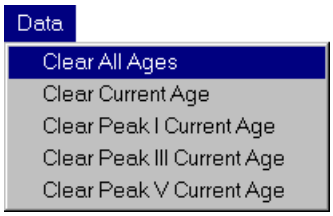
Save Norms as SmartEP Defaults: This option saves the current set of norms as the default, to be used in SmartEP for displaying and printing latency-intensity graphs. When a set of norms has been saved as default, SmartEP will set Norms to Default in the Display menu. This option is available only when in the standard mode.

Erase Norms File: This option is used to remove a norms file that is no longer needed. To remove the file, select the file to be erased, and then press the OK button. Make sure you select the correct file name, for you will not be able to recover the file once it is deleted.

Quit: Will terminate the Latency-Intensity Table Generation program and return the user to the previous screen. This button will not save any changes.

Data Menu

The Data menu contains various options to clear data that have already been entered.



Clear All Ages: Clicking this option will clear all the data in the current file, including all data from all ages if the current file is a Standard file.

Clear Current Age: This option clears all the data in the current age, or all the data in a Special (one age only) file.

Clear Peak N Current Age: This option clears only the data in the current age that corresponds to the named peak.

Age Selection

By selecting the appropriate age range, the corresponding latency-intensity values for that age will be displayed in their corresponding boxes and are available for editing. The start and end latency value ranges are entered directly in the corresponding edit boxes for each intensity and peak. Normative data for intensity values between 0 and 100 dB nHL can be entered. The values entered are in milliseconds (10-3 seconds, milliseconds).

The following tables contain standard ABR latency data for infants and adults. These data were generated with the following acquisition parameters: filters 100-3000 Hz, stimulation rate 21.7 clicks per second. The values are taken from Jacobson, John T., The Auditory Brainstem Response. Copyright ©1985 by Allyn and Bacon. Reprinted by permission. All ages are referenced from conception.

33-34 Weeks AFTER CONCEPTION

Intensity (dB HL)	Peak I (ms)		Peak III (ms)		Peak V (ms)		Inter-peak I–V	
	Mean	σSigma	Mean	σSigma	Mean	σSigma	Mean	σSigma
30	4.17	1.13	6.46	1.03	8.88	0.97	4.71	0.98
40	3.59	0.97	6.17	0.63	8.51	0.77	4.92	0.78
50	3.21	0.88	5.79	0.64	8.19	0.56	4.98	0.73
60	3.07	0.69	5.38	0.84	7.76	0.87	4.69	0.91
70	2.59	0.84	5.22	0.79	7.49	0.56	4.90	0.75
80	2.26	0.81	5.01	0.74	7.38	0.72	5.13	0.67

35-36 Weeks

Intensity (dB HL)	Peak I (ms)		Peak III (ms)		Peak V (ms)		Inter-peak I–V	
	Mean	σSigma	Mean	σSigma	Mean	σSigma	Mean	σSigma
30	4.06	0.95	6.58	0.82	8.77	0.99	4.72	0.87
40	3.57	0.89	6.00	0.71	8.51	0.60	4.94	0.73
50	3.18	0.51	5.72	0.84	8.11	0.56	4.93	0.77
60	2.73	0.71	5.33	0.63	7.68	0.41	4.95	0.52
70	2.33	0.68	4.97	0.55	7.41	0.61	5.08	0.64
80	2.13	0.69	4.47	0.71	7.19	0.48	5.06	0.56

37-38 Weeks

Intensity (dB HL)	Peak I (ms)		Peak III (ms)		Peak V (ms)		Inter-peak I–V	
	Mean	σSigma	Mean	σSigma	Mean	σSigma	Mean	σSigma
30	3.78	1.04	6.31	0.78	8.39	0.88	4.61	0.90
40	3.22	0.78	5.61	0.66	7.95	0.69	4.73	0.71
50	2.94	0.64	5.43	0.73	7.65	0.72	4.71	0.68
60	2.47	0.89	5.06	0.43	7.28	0.55	4.70	0.71
70	2.24	0.67	4.75	0.41	7.17	0.47	4.93	0.54
80	2.02	0.53	4.61	0.47	7.01	0.48	4.99	0.48

39-40 Weeks

Intensity (dB HL)	Peak I (ms)		Peak III (ms)		Peak V (ms)		Inter-peak I–V	
	Mean	σSigma	Mean	σSigma	Mean	σSigma	Mean	σSigma
30	3.59	0.81	5.79	0.84	8.03	0.72	4.44	0.69
40	3.06	0.87	5.36	0.74	7.65	0.55	4.59	0.66
50	2.65	0.72	5.01	0.59	7.31	0.53	4.66	0.60
60	2.38	0.58	4.71	0.62	7.07	0.49	4.69	0.51
70	1.94	0.51	4.57	0.44	6.83	0.48	4.89	0.46
80	1.79	0.59	4.26	0.62	6.72	0.40	4.93	0.59

2 months

Intensity (dB HL)	Peak I (ms)		Peak III (ms)		Peak V (ms)		Inter-peak I–V	
	Mean	σSigma	Mean	σSigma	Mean	σSigma	Mean	σSigma
30	3.11	0.62	5.64	0.39	7.58	0.48	4.47	0.52
40	2.72	0.41	5.18	0.45	7.23	0.31	4.51	0.46
50	2.35	0.44	4.85	0.37	6.91	0.34	4.56	0.37
60	2.14	0.42	4.64	0.35	6.69	0.30	4.55	0.34
70	1.81	0.30	4.27	0.41	6.51	0.38	4.70	0.33
80	1.64	0.43	4.18	0.32	6.39	0.29	4.75	0.36

6 months

Intensity (dB HL)	Peak I (ms)		Peak III (ms)		Peak V (ms)		Inter-peak I–V	
	Mean	σSigma	Mean	σSigma	Mean	σSigma	Mean	σSigma
30	2.86	0.61	5.08	0.50	7.20	0.44	4.34	0.42
40	2.49	0.48	4.93	0.37	6.86	0.33	4.37	0.43
50	2.28	0.41	4.75	0.37	6.68	0.41	4.40	0.36
60	1.93	0.31	4.46	0.40	6.34	0.27	4.41	0.30
70	1.79	0.32	4.26	0.34	6.50	0.31	4.71	0.26
80	1.60	0.26	4.10	0.31	6.27	0.21	4.67	0.28

12 Months

Intensity (dB HL)	Peak I (ms)		Peak III (ms)		Peak V (ms)		Inter-peak I–V	
	Mean	σSigma	Mean	σSigma	Mean	σSigma	Mean	σSigma
30	2.90	0.44	5.11	0.51	7.09	0.39	4.19	0.47
40	2.43	0.40	4.59	0.29	6.62	0.36	4.19	0.37
50	2.18	0.26	4.43	0.32	6.47	0.21	4.29	0.27
60	1.91	0.29	4.01	0.22	6.15	0.24	4.24	0.33
70	1.81	0.23	3.84	0.23	6.14	0.19	4.33	0.23
80	1.62	0.24	3.79	0.18	5.93	0.17	4.31	0.19

Adult

Intensity (dB HL)	Peak I (ms)		Peak III (ms)		Peak V (ms)		Inter-peak I–V	
	Mean	σSigma	Mean	σSigma	Mean	σSigma	Mean	σSigma
30	2.83	0.36	5.00	0.31	6.69	0.29	3.86	0.30
40	2.46	0.31	4.55	0.26	6.29	0.27	3.83	0.15
50	2.23	0.34	4.49	0.18	6.12	0.22	3.89	0.24
60	1.88	0.27	4.11	0.20	5.81	0.27	3.93	0.22
70	1.75	0.21	3.86	0.23	5.67	0.15	3.92	0.22
80	1.59	0.24	3.64	0.17	5.57	0.16	3.98	0.25

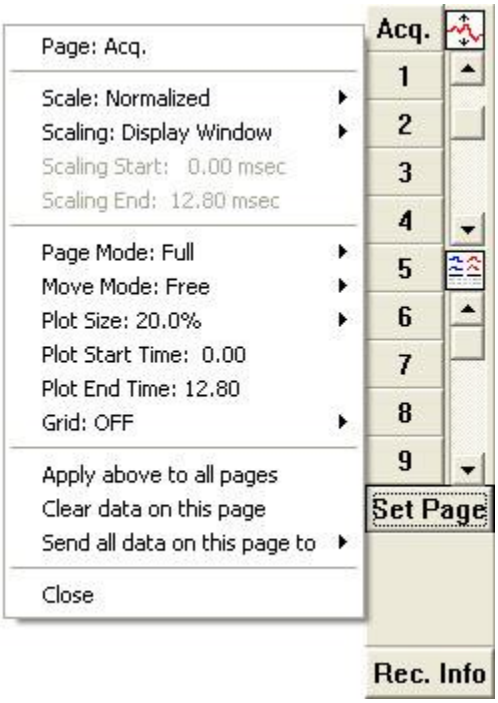
The Display Area

5

The Display Area occupies the center portion of the screen and it is used to view the data acquisition process, review, and label waveforms, and to perform data processing. There are 10 available pages (or screens) in the display area, accessible from the side menu as described below. The display area can hold up to 250 objects at a time, including recordings, text and image objects. Only one of the objects can be active, but multiples may be selected by pressing the [Ctrl] key. Under the Main Menu, time and amplitude values with respect to the current active recording are displayed on the Indicator Line.

The Side Menu

Along the right side of the display area are a series of buttons labeled "Acq," 1 through 9, Set Page, and Rec. Info. The first ten buttons are used to select the current display page. "Acq" corresponds to the acquisition page containing the Control Panel, where any newly acquired recording will be placed. The buttons labeled 1 - 9 correspond to the various report pages where the data can be placed. Report pages do not have control panel access. The Set Page button is used to access the Page Parameters menu, which contains options to change the display settings. The Rec. Info button opens the recording information panel on the right side of the screen. The top and bottom sliders control the magnification of the recordings and the scrolling of the page, respectively.



Page Parameters Menu (Set Page)

This menu contains options to modify the display of all the recordings on the current page. These are descriptions of the available options in the page parameters menu.

Page

This indicates the current page being displayed. This information is also displayed at the top of the screen below the Main Menu, in the Indicator Line.

Scale

This option helps to set the scale of all the recordings on the screen. The following options are available on the sub-menu: Normalized, 1, 0.9, 0.8, 0.7, and "Other." The default scale is normalized. Normalized recordings will be scaled to fit the vertical space allocated to that recording; the vertical space depends on the number of plots per page specified. When a microvolt value is chosen or entered, the current scale is displayed on the top left hand corner of the display window, given in microvolts. All recordings on the current page will be resized

to match the new scale value. Signal display clipping may occur when the scale is smaller than the recording's peak-to-peak value.

Scaling

This option specifies either the use of the full display window or a Special subset range of time display for normalized scaling. When Special is selected, the following two options, Scaling Start and Scaling End, become active. The peak-to-peak Amplitude in the selected range determines the scaling of the recordings. Some clipping (signal cutting off) may occur when other peaks or valleys are larger in amplitude than the scaling region. Set the appropriate time range in the Plot time option to hide unwanted peaks. The actual recording will remain unaffected by changes made to the display using this option.

Scaling Start

Used when scaling is set to Special, to specify the starting time of the scaling range.

Scaling End

Used when scaling is set to Special, to specify the ending time of the scaling range.

Page Mode

This option is used to change the page display between Full and Split page modes. The split page mode displays the recordings on two halves of the screen, the left ear on the left side and the right ear on the right side.

Move Mode

The selection on this option determines whether the recordings can be dropped freely on the axis or snapped to only specific locations on the screen. This location is determined by the number of plots per page.

Plot Size

This option determines the relative size of each recording by specifying the percentage of page area that each recording takes on the screen. Select a value from the menu or a custom value by clicking on "other". Plot size can also be changed using the zoom slider located on the top right of the page buttons.

Plot Start Time

Specifies the starting time (relative to the beginning of stimulation) from which the recordings on this page are displayed. Note: This feature is independent of the actual recording time range; the recording time range is not altered and nothing is changed in the data recording stored on the hard drive.

Plot End Time

Specifies the ending time (relative to the beginning of stimulation) up to which the recordings on the page are displayed.

Grid

This option specifies if a vertical time grid is to be displayed on this page. Each grid line represents one millisecond.

Apply above to all pages

This option applies the current page parameters to all the other pages.

Clear data on this page

This option clears all the data on the currently displayed page. This option does not delete recordings from the hard drive; it only clears them from the screen.

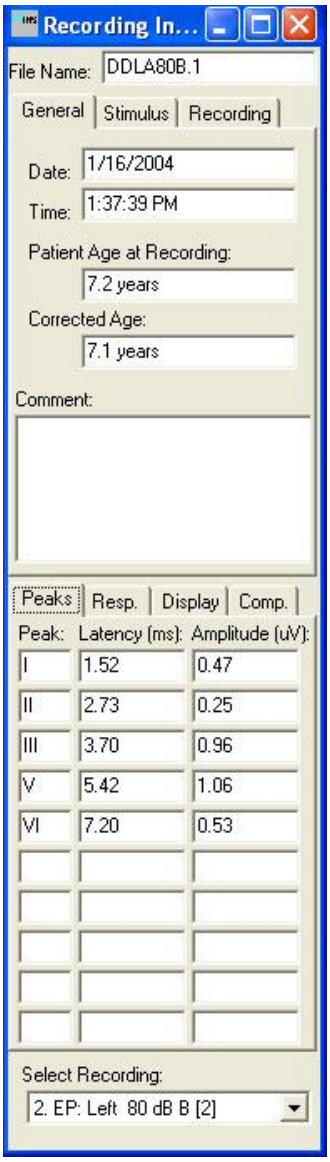
Send all data on this page to...

Transfers all the data displayed in the current page to the user-specified page.

Close

Use this to close the page parameters menu. Clicking outside the menu will also close it.

Recording information



The recording information panel can be opened by clicking on the Rec. Info button, at the bottom of the side menu. When clicked, it will display the information pertaining to the currently active recording.

General Tab

This tab shows the date and time the recording was acquired, the age of the patient at the time of the recording, and the corrected age if available. It will also show any comments added to the recording, which may also be edited from this field.

Stimulus Tab

This tab shows information about the stimulus used for the acquisition of the response. This information includes Ear, Intensity, Mode, Rate, Stimulus, and the use of Masking.

Recording Tab

This tab shows additional information about the recording, including sampling rate, number of sweeps, number of artifacts, and the amplifier settings.

Peaks Tab

The peaks tab at the bottom of the panel shows the marked peaks along with the latency and the amplitude of each one. This panel will show up to 10 peaks.

Response Tab

The response tab shows the Signal-to-Noise ratio and the residual noise for the response. It also includes a field to show the region that was used to calculate the noise measurements.

Display Tab

This tab contains controls to modify the appearance of the recording on the screen and to show and hide peak labels and recording information. The “Show Peak Labels” check box

will show or hide the marked peaks on a recording and the print out. The "Print Parameters" check box will turn on and off the printing of the recording information at the bottom of the page. The Plot Type radio button group allows you to change the way the recording is displayed: as an average of even and odd sweeps, as a split of even and odd sweeps, as the subtraction of even and odd sweeps, or a spectral view of response versus noise. The Show Text radio group allows you to show peak and recording information at the position specified.

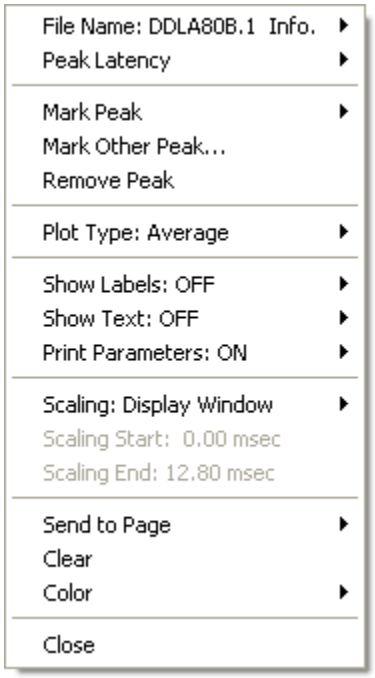
Compare Tab

The compare tab displays comparison information between two recordings of the same type that have been acquired using the same parameters on opposite sides. This is equivalent to selecting the compare option from the process menu.

Select Recording

This drop-down menu shows a list of all the recordings on the current page. Choosing an item from this list will change the active recording to the selection and update the panels above with the corresponding information.

Active Recording Menu



Information about the active recording may be obtained using the Active Recording Menu. This menu is displayed by right clicking the mouse anywhere in the display area of the screen. This menu is used to change some of the recording options, label the active recording, change the scaling, or change the color of the recording. Many of the options on this menu have secondary menus, indicated by a right-pointing arrow, as shown in the image to the right. A thorough description of the contents of the Active Recording Menu follows.

File Name Info

Displays the file name (DDLA70B.2 in this example) and shows the acquisition parameters information as a side panel when the mouse pointer moves over the selection, including the following items:

- Stimulus Side (Left or Right),
- Stimulator (Insert, Headphone, etc.),
- Stimulus Type (Click, Tone, etc.),
- Intensity (dB nHL),
- Stimulus Mode (Rarefaction, Condensation, Alternating),
- Stimulus Rate (presentations/second),
- Masking (dB SPL),
- Number of Sweeps,
- Artifacts (sweeps rejected),
- Sampling Period (in microseconds, μ s),
- Amplifier (gain high pass – low pass),

- Electrodes (Configuration of electrodes),
- Comment (optional),
- Date and time of recording.

Peak Latency

When the mouse pointer is run over this option, a side menu appears that shows the following information for the active recording:

- Time latency (i.e. 1.73, 2.78, 3.83, 5.68)
- Inter-peak latencies (2.10, 1.85, 3.95)
- Amplitudes of the peaks (0.07, 0.16, 0.25, 0.32)
- Amplitude ratios (AR) (3.79, 1.29, 4.90)

Mark Peak

This option is used to mark ABR peak labels for the current recording. The sub-menu shows options to mark peaks one through six of an ABR recording.

Mark Other Peak

This option is used for marking peaks other than ABR peaks. It will open a dialog box with a list of standard labels for all possible recording types. This dialog box will also allow the definition of non-standard labels.

Remove Peak

This option is used to remove a previously marked peak. The peak closest to the place where the right click was done will be removed with this option.

Plot Type

This option changes the type between the four possible types of plot displays: Average, Split sweep, Plus-Minus, or Spectral. See page 63 for further details about the different plot types.

Show Labels

This is used to enable or disable the displayed peak labels. Labels can be disabled to relieve a cluttered look on the screen while arranging recordings.

Show Text

It is used to select if and where the parameter and peak information is to be displayed next to the active recording. ASSR and TrOAE recordings will display their respective parameters, frequency analysis, and response detection information. The information may be displayed next to the recording, above it, or turned off.

Print Parameters

This option determines whether the recording acquisition parameters are printed along with the recording when the page is printed.

Scaling

This option specifies either the use of the full display window or a Special subset range of time display for normalized scaling. When Special is selected the following two options, Scaling Start and Scaling End, become active. The peak-to-peak Amplitude in the selected

range determines the scaling of the recordings. Some clipping (signal cutting off) may occur when other peaks or valleys are larger in amplitude than the scaling region. Set the appropriate time range in the Plot time option to hide unwanted peaks. The actual recording will remain unaffected by changes made to the display using this option.

Scaling Start

Used when scaling is set to Special, to specify the starting time of the scaling range.

Scaling End

Used when scaling is set to Special, to specify the ending time of the scaling range.

Send to Page

This submenu is used to move the active recording to the selected page from the list.

Clear

Selecting this option will clear the active recording from the page. Note: This option does not delete any recordings from the hard drive.

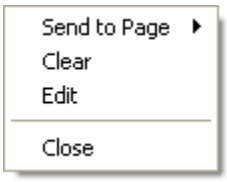
Color

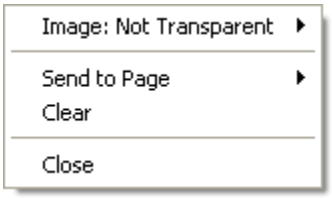
This can be used to change the display color of the selected recording while inactive. Note: While the recording is active, it will always be displayed in green.

Close

This option will close the Active Recording Menu. This can also be accomplished by clicking the mouse anywhere in the display screen outside the Active Recording Menu.

Active Object Menu





Send to Page

This submenu is used to move the active object to the selected page from the list.

Clear

Selecting this option will clear the active object from the page. Note: This option does not delete the object from the hard drive.

Close

This option will close the Active Object Menu. This can also be accomplished by clicking the mouse anywhere in the display screen outside the Menu.

Edit (Text Objects Only)

This can be used to edit the contents of non-dynamic text objects. Selecting this option will open the Text Edit Window; refer to page 44 for additional information.

Image (Image Objects Only)

This submenu provides the option of changing the transparency of a graphic object. Transparent items allow the user to see other objects that may be hidden behind the image object. The transparency property only affects the display and will not be reflected when the page is printed.

Indicator Line

The Indicator Line is just below the Main Menu. It includes basic informational items such as the active recording file name, time, amplitude, peak-to-peak amplitude, and display page number. Placing the mouse cursor over an item will display the name of the item.

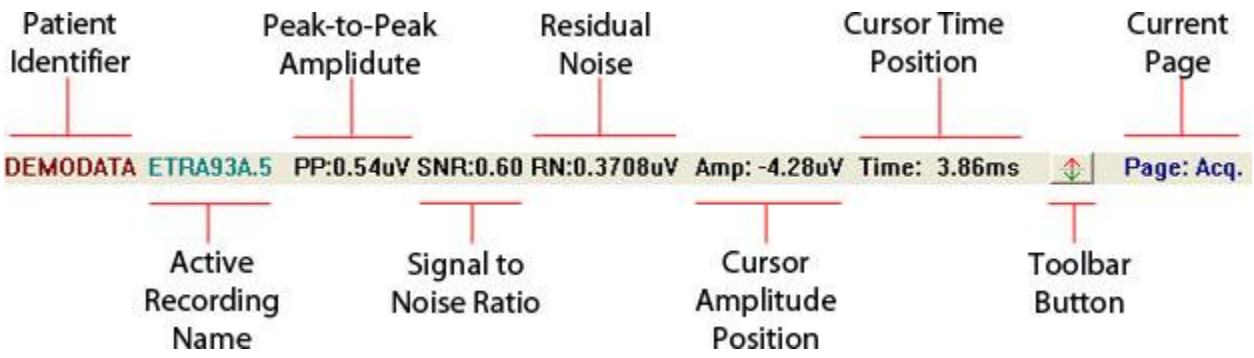


Figure 25 - Indicator Line

Patient Identifier

The Identifier of the currently loaded patient is displayed in the upper left corner of the Indicator Line.

Active Recording

The filename is displayed under the Main Menu on the Indicator Line along with amplitude and latency information for the active recording. Refer to page 109 for additional information about automatically generated file names.

Peak-to-Peak Amplitude

PP值代表了什么，表达了什么意义。

The number displayed after PP is the maximum absolute peak-to-peak amplitude for the active recording.

Signal to Noise Ratio

This number is indicated next to the initials SNR on the information bar. It indicates the Signal to Noise ratio as calculated by the program's internal algorithms.

Cursor Amplitude Position

The number after Amp is the current amplitude with respect to the origin of the active recording. In both cases, the amplitude is measured in microvolts (μ V).

Cursor Time Position

This displays the current time position of the mouse cursor in the display area. Time is referenced to the stimulus onset depending on the stimulus device used (corrections are made for insert earphones).

Toolbar Button

This button will toggle the toolbar ON and OFF.

Page

This displays the name of the current page (Acq. 1, 2, 3, 4, 5, 6, 7, 8 or 9).

The Toolbar

The toolbar contains buttons that will help you achieve common tasks faster. It is equivalent to using the menu items on the SmartEP main menu.

Patient File Buttons

The first of these buttons, (N), will open the Patient Demographics window with all its fields empty, ready to accept information for a new patient. The second button, (P), will open the Patient Demographics window with the fields containing the information for the currently loaded patient.

Label Marking Buttons

These buttons allow you to mark features of the active recording. To activate this option, click once on the label you want to use; once active, click at the location of the feature to place the label. Click on the toolbar once more to deactivate the option, or to activate another label. The labels shown in these buttons will change for the current active recording based on its type.

Print Buttons

The left print button will print the current page to the default printer. The right print button (M) will print all pages containing information to the default printer.

Print PDF Buttons

The left print to PDF button will print the current page to a PDF document. The right print to PDF button (M) will print all pages containing information to PDF.

Data Buttons

The left data button will open the load data window with the list of recordings for the currently loaded patient. The center data button saves the currently active recording. The right data button will save all unsaved recordings. Note that unsaved recordings show the name “NEW” in the indicator bar.

Report Buttons

The load report button will open an open file window to allow the loading of a previously created report. The Save report button will open a save file window to allow the creation of a report file.

Add Text Button

This button will open the Text Editor window. This can be used to add informational text to the page to be saved in a report.

Latency Intensity Button

This button will open the Latency-Intensity Graph window. From that window it is possible to place a screen capture of the graph into the page to be saved in a report.

Page Mode Toggle Button

This button will toggle the current page display between split mode and full mode. It is equivalent to using the same option in the Set Page menu.

Arrange By... Buttons

These buttons will rearrange the recordings on the page by Intensity (I), by acquisition order (O), or by Rate (R). These options will also split the page if the recordings on the page come from the left and the right.

Sweep and Artifact

The sweep and artifact count is only displayed during acquisition. They are displayed in a line just below the Indicator Line that appears only when acquiring data. They are displayed on all pages in order to inform the user as to the status of the current acquisition process.

Sweeps shows the number of EEG sweeps averaged. Data acquisition stops when the number of averaged sweeps equals the number of desired sweeps displayed in the Control Panel or until interrupted with the Pause/Stop button in the Control Panel.

Artifact shows the number of EEG sweeps that have been rejected during acquisition. Artifact rejection is based on the EEG level set by the EEG & Amplifier command in the Control Panel.

Recording Identifier

Next to each of the recordings on the screen, a recording identifier is displayed. The identifier consists of the stimulation intensity, the ear, and the acquisition channel. The recording identifier is used to distinguish recordings on the screen quickly.

Plot Types

Recordings can be displayed in several formats. The format can be changed from the Active Recording Menu options (see page 58). The following formats are available:

Average

The average of all the sweeps acquired post-stimulus. If 1024 sweeps are acquired, choosing this option will display the average of those 1024 sweeps. This is the default plot type and the plot type displayed during acquisition.

Split Sweep

As alternating data is acquired, the rarefaction and condensation sweeps are averaged and stored separately in two different buffers. They are continuously averaged together and the total average is displayed on the screen as the plot type average. The Split Sweep plot type allows comparisons between the two acquiring buffers. In some instances, when the recording has not been acquired in alternating mode, it may display one waveform identical to the Average waveform and one straight line when this format is selected.

Plus-Minus

This plot type subtracts one of the two recording buffers (see above, Split Sweep) from the other, resulting in the EEG alone with (ideally) no stimulus. This may be used to see an unwanted signal acquired with the data, as may occur if the sound output cables rest against the electrode leads.

Spectral

This plot type shows the waveform in the frequency domain (Hz). Again, this is useful to see a continuous interfering signal as explained above in Plus-Minus.

The following displays the same data recording in the four different plot types:

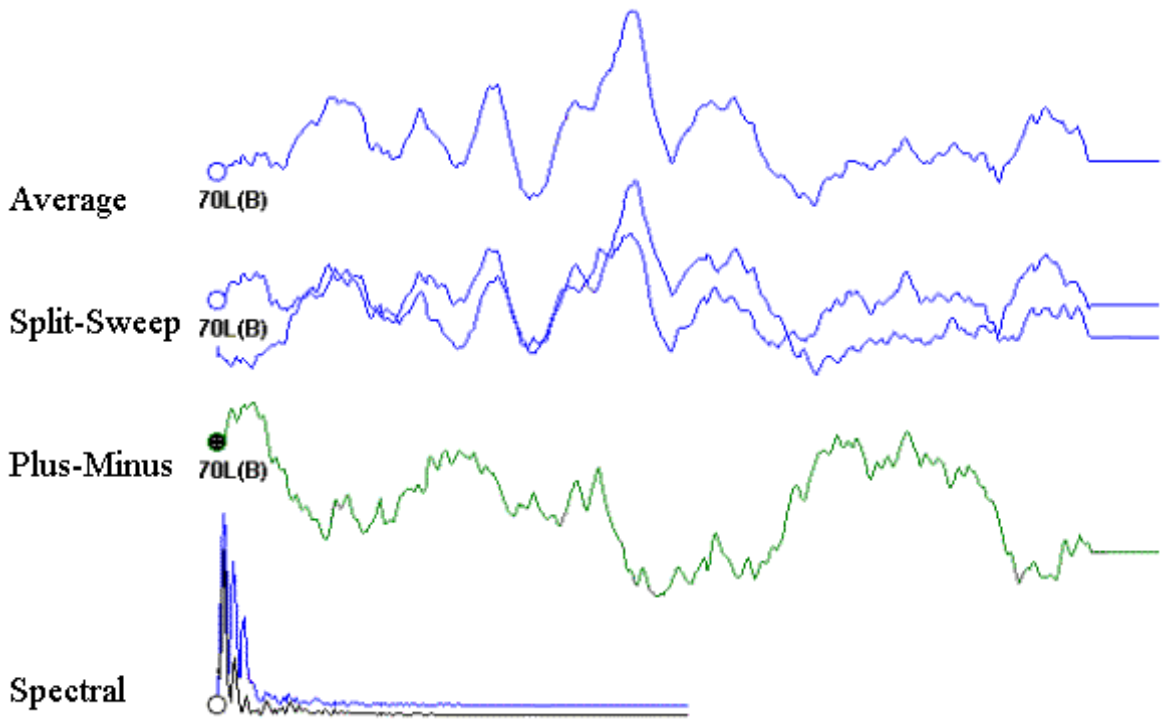


Figure 26 - Plot Types

Keyboard Shortcuts

The following keys may be used for faster system operation on the display page.

SmartEP Keyboard Commands

[Esc]	Used to Pause or Stop recording
[Space]	Used to Pause or Stop recording
[A], [0]	Display Acquisition Page
[1]...[9]	Display corresponding page 1...9
[P]	Display Page Parameters
[+]	Used to add all the currently selected recordings to a new buffer
[-]	Used to subtract the two currently selected recordings to a new buffer

The Control Panel

6

Control Panel

The Control Panel is located along the bottom of the acquisition page. It contains fifteen buttons that control various features of the SmartEP acquisition procedures. Seven of them display the value of the parameter that they represent. Clicking on these buttons directly controls the value of the parameter. The other buttons bring up dialog boxes when pressed or begin and stop the acquisition process. Details follow.

Intensity: 11	Ear: Right	Mode: Ipsi	Stim: Click	Phase: Rare.
Mod: Auditory	Sweeps:64	Rate:19.30/s	Time:25us (12.8)	EEG & Amplifier
Acquire	Continue	Clear Page	Load Settings	Save Settings

Figure 27 – SmartEP Control Panel

Note: The Control Panel, as displayed and described here, is for the Auditory Modality only. If you have other modalities installed, such as High Frequency, Chained-Stimuli, or P300/MMN some of the options will change.

Intensity

Intensity: 11

Increase the intensity of the stimulus by right clicking on this button or decrease it by left clicking. The intensity will be increased or decreased by the amount of the step size given under Stimulus > Level Step size in the main menu. Double click to enter a specific intensity value. The range of the intensity values is 0 - 100 dB nHL in 1 dB steps or 7 - 132 dB SPL in 1 dB steps, depending on the Stimulus Mode setting specified in the Auditory Stimulus Generation Dialog Box.

WARNING: Do not PRESENT continuous stimulus if:

- Using click stimulation intensities above 70 dB HL (~100 dB SPL)
- Using other stimuli at or above a level equivalent to 100 dB SPL

Only stimulate at those intensities while acquiring data. Prolonged presentation of stimuli at levels greater than 100 dB SPL is uncomfortable to patients and in some circumstances may actually cause temporary or permanent hearing loss.

Ear

Ear: Right

Use this button select the ear to which the sound will be delivered. Click the mouse to cycle between Right, Left, or Both. Either mouse button may be used for this function. When using a one-channel system, remember to set the right/left switch to the appropriate ear. When using a two-channel system, remember to connect the leads correctly to the transmitter box.

Mode

Mode: Ipsi

Use Mode to select whether to record Contra (contralaterally), Ipsi (ipsilaterally), or Both (simultaneous contralateral and ipsilateral; available in dual and quad channel systems only) by clicking the mouse button to toggle between the modes. When Contra is selected in a single channel system, the user must set the left/right switch on the transmitter to the setting

opposite the ear being stimulated. When both ears are selected in a two-channel system, both ipsilateral and contralateral recordings will be displayed. Whether a particular recording is ipsilateral or contralateral depends on the channel and the ear being stimulated. When the Both option is selected in a four-channel system, the configuration depends on the setting in the EEG and Amplifier dialog box. Stimulus selections may be stored in parameter files by using the Save Settings command.

Stimulus

Stim: Click

This button displays the stimulus type currently used by the system, such as click, pure tone, or stimulus file. When clicked, it opens the Auditory Stimulus Generation dialog box, see page 38. This dialog box allows for setting the type of stimulation (click, pure tone, or file), the stimulator (bone vibrator, ear insert, etc.), and the ability to enter calibration values and masking intensities. Stimulus selections may be stored in parameter files by using the Save Settings command.

Phase

Phase: Rare.

Select the phase, polarity, of the stimulus by pressing the right or left mouse button. The phase will toggle between Rarefaction, Alternating, or Condensation options. Phase dictates the direction in which the sound pressure is applied. Condensation will produce positive clicks, rarefaction will produce negative clicks, and alternating will alternate between the two. Stimulus phase selections may be stored in parameter files by using the Save Settings command.

Modality

Mod: Auditory

The modality indicator shows the current stimulus modality being used. The default modality for your system is Auditory. Other EP stimulation modalities may include high frequency, chained-stimuli, P300/MMN, visual, and somatosensory depending on your hardware configuration. Refer to page 69 for more information about how to use these testing modalities. For information regarding the availability of other modalities for your system, contact IHS.

Sweeps

Sweeps:64

Sweeps correspond to the number of stimuli that will be presented to the patient and will be averaged into a recording file. Click the right (increase) or left (decrease) mouse buttons to increase or decrease the number of sweeps used in averaging, or double click on the left mouse button to enter a specific value manually. The range of possible values is 1 - 32000 sweeps, in 1-sweep increments. Sweep selections may be stored in parameter files by using the Save Settings command.

Rate

Rate:19.30/s

Increase or decrease the stimulus presentation rate (in presentations per second) by using the right (increase) or left (decrease) mouse button or double click on the left mouse button to enter a specific value manually. The range in the rate parameter is 1.0 - 1000.0 per second, in 0.1 increments.

Caution:
The maximum presentation rate that should be used for acquisition at high frequencies is 100 PRESENTATIONS per second.

Note that the presentation rate will affect the size of the acquisition window (how long the data will be acquired for each stimulus). This makes a difference when acquiring at high rates, for example 100 stimulus presentations per second. See page 98.

Time

Time:25us (12.8)

Time corresponds to the sampling period in microseconds (μ s). The post stimulus acquisition time or analysis time window is displayed in milliseconds, in parentheses, next to the sampling period. Changing the Time option will change the time axis displayed in the acquisition page. Note that the time axis corresponds only to the post-stimulus acquisition window. The acquisition time window may also be affected by the Rate setting. For more information on how the time button alters the functioning of SmartEP, see page 98.

In order to change the acquisition time, press the right (increase) mouse button, the left (decrease) mouse button, or double click on the left mouse button to enter a specific value manually. The time scale will be modified accordingly. The Time selection may be stored in parameter files by using the Save Settings command.

EEG & Amplifier

EEG & Amplifier

When selected, the EEG and Amplifier Settings dialog box will open (see page 35 for details). This dialog box allows the user to set the artifact rejection level and to select filter and gain settings for a channel while viewing the current EEG. To exit the EEG dialog box and return to the main screen click OK.

Acquire (and Pause/Stop)

Acquire

Use Acquire to begin acquiring data. The Acquire button will change to Pause/Stop. Press this button again to pause or stop acquiring data. The [Esc] or [Spacebar] keys may also be used to pause or stop the acquisition process. When Pause/Stop is clicked, the system will stop collecting data and a small dialog box will appear asking if you want to continue. If you want to resume data collection afterwards, press the YES button. The Acquire button will change to Continue. If you do not want to continue collecting data, you can press the NO button to stop. If you select to stop, you will be asked if you want to save the collected data.

Continue

Continue

Use Continue to resume the averaging process after the Pause/Stop button has been clicked. When this option is selected, the data acquisition and averaging begins where it left off and runs until the specified number of sweeps has been collected. All of the data acquired, before and after the pause, are used to produce the final, averaged, ABR recording. If you need additional sweeps to average in to the current recording, you may increment the sweep value before clicking Continue and the system will continue until the new value is reached. The new aggregate average will appear in a new display buffer, and it will be saved to a new

data file. This allows you to determine the effect of the new sweeps without losing the previous results. Unneeded data files or display buffers may be discarded manually.

Clear Page

Clear Page

Use Clear Page to remove all recordings from the acquisition page. This choice does not remove recordings from the hard drive or the nine display pages. Saved data may be loaded again at any time. Recordings that were not saved will be lost.

Load Settings

Load Settings

Parameter files contain all the acquisition parameters that are set in the Control Panel. Select the Load Settings button to list the available files previously saved. Then, select the desired file and press the OK button or simply double click on the file name to load the settings. All the system parameters will be replaced with the parameters stored in the selected file. These include stimulation intensity, stimulus type, rate, sweeps, and all amplifier settings. If you are unsure of the parameters stored in the file you have loaded, check all the acquisition parameters before recording data; the acquisition parameters are critical to the proper interpretation of your results.

Caution:
Make sure that all parameters are correct before acquiring. Different parameters may lead to misinterpretation of the recorded data.

Save Settings

To save your custom acquisition parameters, select the Save Settings option. Type a file name in the box that appears. Make sure the name of the file you enter is easy to identify. All the current acquisition parameters, including all options entered in the Control Panel, Main Menu, Auditory Stimulus Generation dialog box, and the EEG & Amplifier Settings dialog box, will be stored in that file. If the OK button is pressed without typing a filename, the current parameters will automatically be saved as Sepwin.set, and it will serve as the default settings every time SmartEP is loaded.

Control Setting Limits

The following table describes the limit values that can be assigned to several parameters accessed through some of the control panel buttons.

Table of Limits

Button	Minimum	Maximum	Units
Intensity	0	100	dB nHL
Intensity	7	132	dB SPL
Sweeps	1	32000	Sweeps
Rate	1	1000	Cycles/sec
Time	25	2500	Microseconds

Additional Modalities

7

The basic SmartEP system includes only the basic auditory modality, which tests in the frequency range 0 to 16,000 Hz depending on the transducer used. Other modalities are available through IHS, including:

- High Frequency (up to 32,000 Hz)
- Chained Stimuli
- P300/MMN
- Simultaneous ABR-OAE
- Electrical ABR
- Vestibular Evoked Myogenic Potentials VEMP
- Visual
- Somatosensory

You can view the available modalities for your system under the Modality option of the Stimulus Menu. The text of the available modalities is shown in black, while the unavailable modalities are grayed out or not listed. Since some of these modalities result in a different testing scheme, the options of the Control Panel are altered when the modality is modified. This section deals with the different parameters specified using these modalities. Make sure to change the modality under the Stimulus option in the Main Menu before using other modalities. Some of these modalities are intended for research use only; contact IHS for details.

High Frequency

The high frequency modality involves testing using frequencies of up to 32 kHz. When this modality is chosen from the System Menu, the system will check that the hardware can support the modality and will display an error message box if an error is encountered. The Modality (Mod.) button, in the Control Panel (left column middle button), will display the text HF Auditory. The functional change in the software is that the Auditory Stimulus Generation dialog box will accept frequency settings up to 32 kHz, whereas the auditory modality only accepts up to 16 kHz. Note that the high frequency transducers are not allowed to output click stimuli; to output click stimuli, the insert earphones should be used.

Caution:
The maximum presentation rate that should be used for acquisition when using the high frequency modality is 100 PRESENTATIONS per second. Follow all of the instructions on the high frequency transducer release sheet.

Chained Stimuli 链式刺激和Protocol测试的区别是什么北

The SmartEP system can also be set up for the chained stimuli modality. This involves a protocol that presents chains of stimuli of different intensity levels, which are an interval apart. Therefore, several ABR recordings (one for each stimulus) are acquired very rapidly during testing. Each averaged recording will contain the auditory response to one of the stimuli in

the chain. This results in a more time-efficient testing scheme. For more information on the chained-stimuli technique, see Hamill et al, 1991.

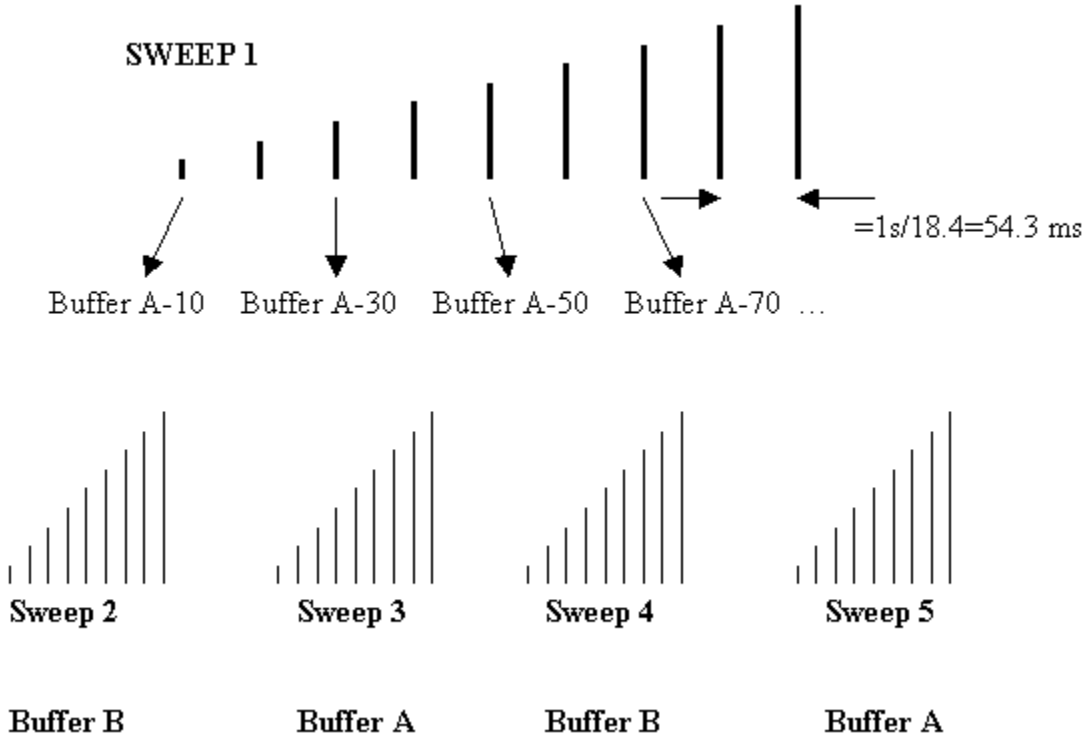


Figure 28 - Chain Stimuli Characteristics

The stimulus used in the chain can be set as in the auditory modality, by accessing the Stimulus Generation dialog box using the Stim button in the Control Panel. The Rate and Time buttons become unavailable when the chained-stimuli modality is selected.

The chain is set up with the Chain Sequence dialog box (below). Up to ten user-defined intensities may be selected to form the chain, but all must be the same stimuli as set within the Auditory Stimulus Generation dialog box via the Stim button. The response to each intensity value is stored into a separate buffer. Ten buffers may be used during each chain stimuli at once to collect ten different responses, all averaged, and displayed on screen during the testing.

The stimulation rate is the rate of repetition of the pulses of the chain. This interval can be chosen as either static (the same interval during and between each chain) or it may follow a bell-shaped curve to produce a “random” distribution of intervals. To choose the interval at one set number:

1. Access the Chain Sequence Setup dialog box by clicking in Setup Chain in the control panel.
2. Set the Standard Deviation to zero.
3. Enter the desired Mean Rate to the interval (in stimulations per second).
4. For a variable interval, choose the number of Standard Deviation to be greater than zero and chose the Maximum and Minimum rates for the time interval distribution.

5. Enter the desired stimulation levels in each of the boxes.
6. Save the changes to a file if this setup will be used repeatedly.

The maximum stimulation rate within each chain is 66/s. In addition, the chain may be presented in increasing sequential order or in a random order. Only ipsilateral tests may be used and only one ear at a time may be tested.

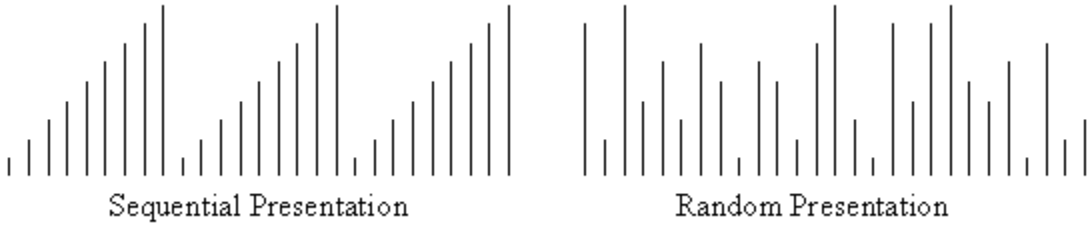


Figure 29 - Presentation Types

P300/MMN

The two techniques P300 and MMN involve investigation of the auditory response that occurs around 300 ms after the onset of the stimulus. In this test, several intensities of varying stimuli are presented and the test is centered on the ability of the patient (or the patient's ears) to discriminate between the signals. To the software, the two tests are conducted identically; however, since the two tests are conducted for different reasons, it is important to specify which test is used for consideration of the results. The P300 test requires the patient to, consciously, count the stimuli as it is presented to them. In the MMN test, patient concentration is not required.

All the options available in the auditory modality are available in the P300/MMN tests including testing both ears at once and testing contralateral or ipsilateral side. The only change in the Control Panel is that the left top button becomes the Setup P300 button. In the dialog box opened from this button, one background (frequent) intensity and up to three (rare) stimuli to be discriminated, are set. The stimuli must already be set and saved in stimulus files using the Auditory Stimulus Generation dialog box accessible through the Stim button. The values in the P300/MMN Configuration dialog box may either be set for the current session only or may be saved as the default settings and will be loaded every time the program is run.

Suggested Parameters

Type of Stimulus	Tone burst
Frequent Stimulus	1000 Hz
Rare Stimulus	2000 Hz
Duration	50,000 μ s (10,000 μ s rise and fall)
Rate	1.1 /second
Polarity	Alternating
Intensity	70 dB
Number of Stimulations	250
Probability of rare stimulus	15%
Presentation ear	Monaural

Transducer	Insert earphones
Masking	Usually not needed
Acquisition	
Amplification (Gain)	50,000
Artifact Rejection level	100 µ volts
Band Pass Filter	1-30 Hz
Notch Filter	None
Electrodes	
Channel 1	Fz-Ai/Ac
Ground	Fpz

Simultaneous ABR-OAE

The simultaneous ABR-OAE modality is currently under development. For more information about this product and its availability, contact Intelligent Hearing Systems.

Cochlear Implant eABR modality

The Electrical ABR modality involves the stimulation of the auditory system using electrical stimulation while bypassing the tympanic membrane and the outer ear. Cochlear Implant eABR is performed by providing electrical stimulation to a patient who has already been implanted. This procedure is performed in combination with a computer provided by the cochlear implant manufacturer and it is mainly used to acquire data that can be used to fit the implant properly. Testing procedure:

1. Connect the Cochlear Implant computer as specified by the manufacturer's instructions.
2. Connect the Cochlear Implant computer's trigger output to the trigger input of the Universal Smart Box.
3. Set up the electrodes on the patient using a vertex positive array.
4. Set up the Cochlear Implant computer for stimulation according to the manufacturer's recommendations and the tests you wish to perform.
5. Start SmartEP and set the modality to Auditory eABR on the Main Menu under the stimulus option.
6. Set the parameters on SmartEP to match the stimulation provided by the Cochlear Implant computer. This will keep the recordings labeled correctly.
7. Set the gain and filters on the EEG and Amplifier window as necessary.
8. Start the stimulation on the Cochlear Implant computer.

When the modality on SmartEP is set to eABR, the software will wait for the trigger signal from the cochlear implant computer to acquire; acquisition will stop once the maximum number of sweeps is recorded or the stop button is pressed. For additional information and specifics about particular brands of cochlear implants, consult the Smart Notes available in your software CD and on the IHS website.

Vestibular Evoked Myogenic Potentials

The VEMP modality refers to the acquisition of Myogenic Potentials related to the Vestibular system evoked by using high intensity stimulus. Due to the nature of the test, in which the user needs to tense the SCM Muscle, it is also necessary to use a method for reducing the amount of sweeps when the patient is not cooperating by tensing the muscle properly. For

this purpose, the system uses a VEMP LED box and an acceptance region, which combined allow for recordings that are more consistent. To use the VEMP Modality:

1. From the SmartEP main menu, select Stimulus > Modality > VEMP.
2. Position the VEMP feedback window in a convenient area of your screen or minimize it if you do not wish to use it. To use video feedback, choose "Video > Open Video Player" from the SmartEP main menu. Select the video to be played using the corresponding option from the Video menu.
3. Click on the PP menu, and change the plot start time to -20 if necessary (this may only be necessary in older versions of the software).
4. Click on the EEG and Amplifier button in the control panel and set the filters and notch filter as desired for each channel.
5. Edit the Artifact Rejection Region settings (for each channel) to prevent artifact while allowing normal EEG activity. When the EEG activity falls within the red area, the sweeps will be rejected; when the EEG activity falls below the red area, the sweeps will be accepted.
6. Edit the VEMP Acceptance area for each channel to prevent the system from averaging when the patient relaxes the SCM muscle, or when there is excessive EEG activity. In the following image, when the EEG activity falls within the green area the sweeps will be accepted; if the EEG activity falls outside the green, the sweeps will be rejected. The minimum and maximum values of the acceptance region are requested as RMS values.

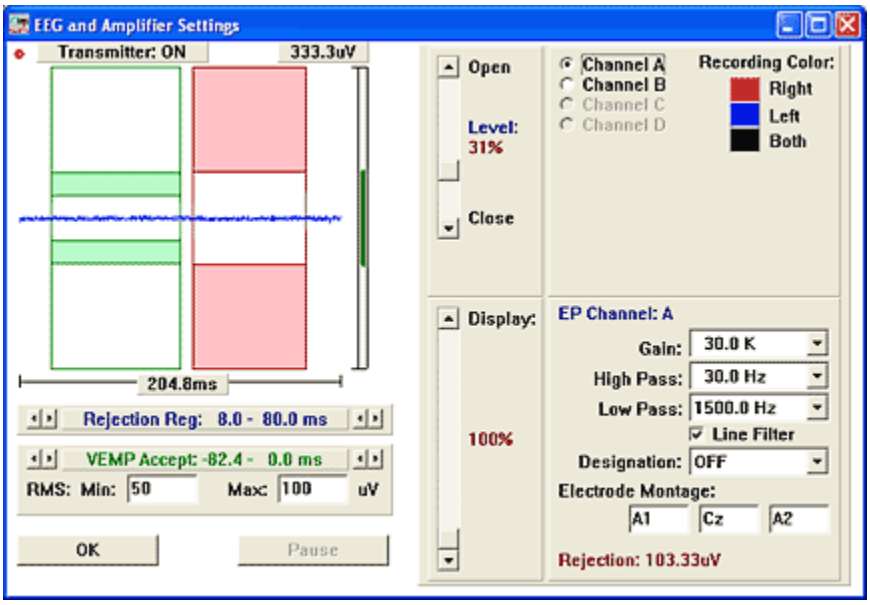


Figure 30 – EEG and Amplifier window

7. In order to select the correct stimulus to be applied, click on the Stim button in the control panel.
8. Select the correct stimulator from the list and the stimulation options desired. Close the window.
9. Choose your intensity, rate, and number of sweeps.

10. Have the patient turn his/her head to the opposite side from the stimulus while tensing the SMC muscle. If the proper tension is being achieved, The VEMP EMG Level Feedback window will show a green level and a happy face. If the video window has been activated, it will be playing video continuously. If the tension is not being achieved, the level will drop, a sad face will show, and the video, if selected, will stop playing.
11. Click Acquire to start testing.

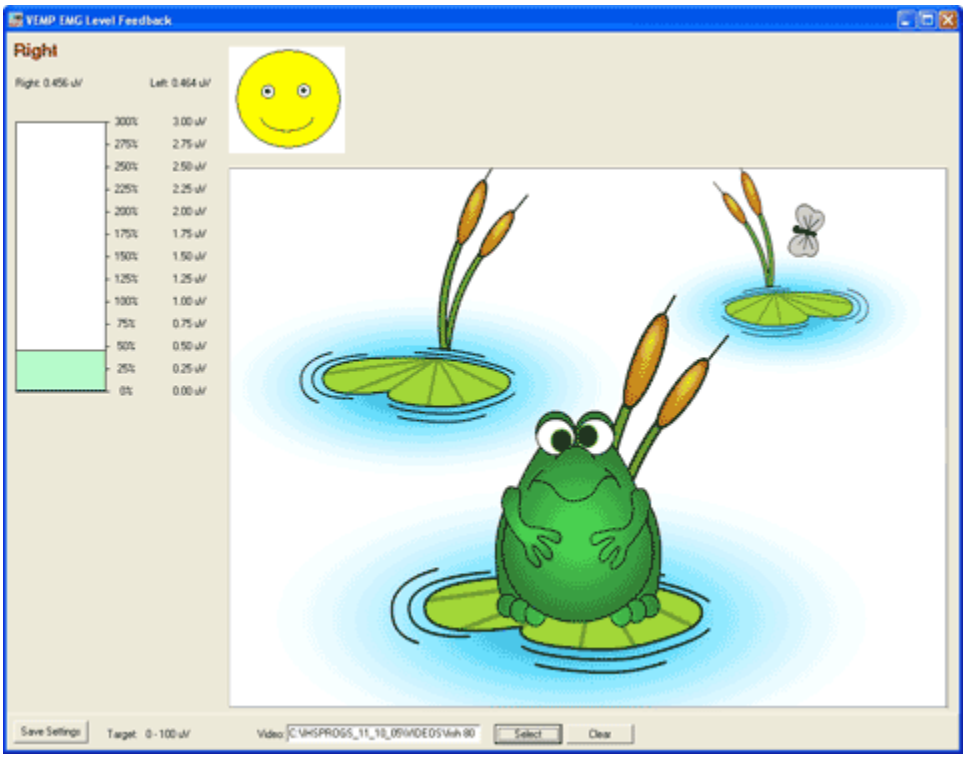


Figure 31 - VEMP EMG Level Feedback window

Once acquisition starts, the software will reject all sweeps that surpass the amplitude allowed by the Artifact Rejection settings and the ones that fall below the amplitude allowed by the VEMP acceptance area. Whenever the sweeps are falling short of the required amplitude for acceptance, the handheld VEMP LED box will present a warning light to indicate that the muscle is not tense; therefore, the sweeps are being rejected. Repeat the acquisition to ensure that the responses are repeatable.

Suggested Parameters

Type of Stimulus	Clicks, 500 Hz or 750 Hz Blackman tones
Duration	0.1 msec click or 2 cycle rise/fall tones
Rate	3.5 to 5 per second
Polarity	Rarefaction
Intensity	95dB nHL Clicks or 120dB SPL Tones
Number of Sweeps	75 – 250
Presentation ear	Monaural
Transducer	Insert Earphones or Headphones,
Masking	None
Acquisition	

Amplification (Gain)	5,000
Band Pass Filter	30 - 1500 Hz
Notch Filter	OFF, ON if excessive electrical noise
Electrodes	
Non-Inverting	Middle of testing side SMC muscle
Inverting	Sternoclavicular Junction
Ground	Forehead

Once acquired, it is possible to add peaks to the wave using the active recording menu, by choosing "Mark Other Peak" and selecting P1 from the peak label window. The Base for VEMP recordings is automatically selected based on the acceptance region that was defined in the EEG and Amplifier window. The base markers may not be seen if they fall outside the current display window; in order to make them visible, use the Page Parameters menu to extend the Plot Start Time. The position of the base markers will affect the calculation of the RMS for comparison calculation purposes. The VEMP recordings can be compared using the Process > Compare menu option. The corrected amplitude is equal to the peak value divided by the RMS value of the wave. The Asymmetry Ratio is equal to the absolute value of the difference over the sum of the corrected amplitudes.

Visual Modality

The Visual modality is used to acquire Visual Evoked Potentials (VEPs) in collaboration with a special visual stimulator unit. This allows the software to present a certain visual stimulus that triggers a response from the patient that correlates to their visual capabilities.

Somatosensory Modality

Somatosensory testing is performed by providing skin surface (cutaneous) electrical stimulation to a patient's upper or lower limbs, and recording the elicited responses using secondary surface electrodes along the neural pathway or cortical somatosensory region. This procedure is useful in determining the integrity of the somatosensory pathways as well as being helpful in diagnosing the nature of any possible sensory impairment. A stimulator probe delivers the constant current electrical stimulus through two metal prongs or electrode connections. Stimulus electrical connections are also available at the Electrical Stimulator Box; however, only one output option should be used at a given time.

WARNING: When using the somatosensory equipment, DISCONNECT THE SUBJECT FROM THE STIMULATOR BEFORE POWERING UP. Electrical shock may occur. When not in use, maintain the Dial on the stimulator to the lowest position.

WARNING: Patients with an implanted electronic device, such as a pacemaker, should not be subjected to any type of electrical stimulation unless indicated and approved by a specialist.

WARNING: Somatosensory testing in humans may be pending clearance from the pertaining governmental entities in some areas. Please consult with your representative or with the IHS offices for updated information.

Connecting the equipment

The Electrical Stimulator Box and Probe must be connected to the USB box as specified by the following instructions to ensure safe use.

1. Plug in the power supply for the USB box to the appropriate slot as shown in the hardware installation manual, and connect it to your isolation transformer using a medical grade power cord.

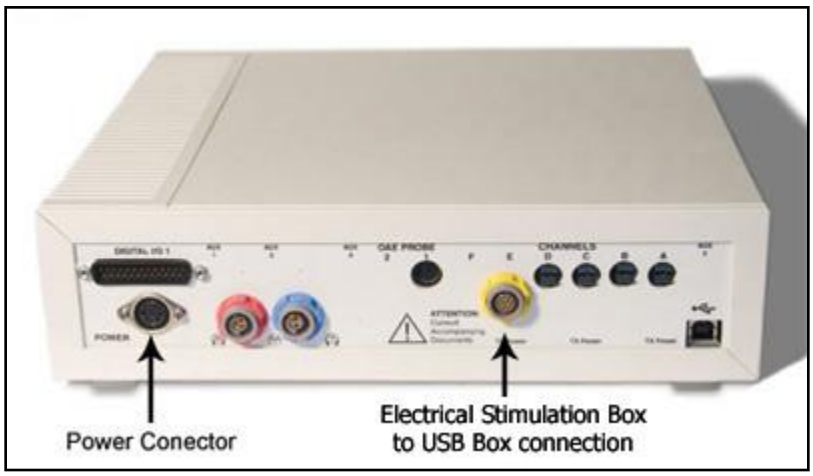


Figure 32 - USB Box Connectors

2. Connect the Electrical Stimulation Box to the isolation transformer using a medical grade power cord.
3. Connect the Yellow DIN connector of the USB box to the Yellow DIN connector of the Electrical Stimulation Box using the cable provided and aligning the arrows on the connectors and cable. This cable is also color coded to avoid confusion.

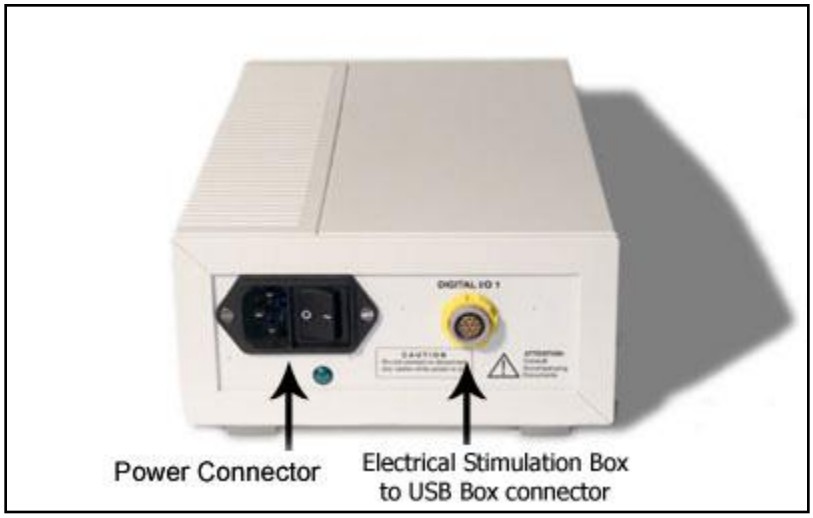


Figure 33 - Electrical Stimulation Box Connectors

4. Plug the Electrical Stimulator Probe to the Electrical Stimulation box by plugging the cable into the black connector located at the front of the box. NOTE: Never use the system without the probe being connected

Figure 34 - Electrical Stimulation Box and Stimulator Probe connections

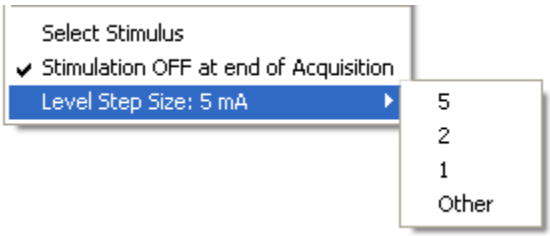
5. Turn the dial of the Electrical Stimulator Probe all the way to the left (lowest setting).
6. Turn ON the USB box and then turn ON the Electrical Stimulation box, in that order. Never power or reboot the system while the patient is connected to any of the electrode stimulation outputs.

The Control Panel

Once the software is set to somatosensory mode (Click on Stimulus > Modality > Somatosensory > Standard from the main menu), the control panel will display a different set of options. The modified options are described below.

Intensity: 20mA	Stim: OFF	Limit: 40mA	Side: Right	Phase: Neg.
Mod: Somato	Sweeps:1024	Rate: 5.10/s	Time:100.0us (51.2)	EEG & Amplifier
Acquire	Continue	Clear Page	Load Settings	Save Settings

Figure 35 - Somatosensory Modality Control Panel



- Intensity is now shown in milliamperes. This amount can be incremented and decremented, by the given step size, by left or right clicking the button respectively. The step size is defined under Stimulus in the main menu; minimum step size is 1 milliamperes. The intensity field will change color to yellow and red as it approaches the maximum.
- Stimulus ON and OFF determines whether the stimulation is ON or OFF. The indicator light on the Electrical Stimulator box will reflect this state.
- Limit is the maximum current value that can be selected in the intensity option. This value can be entered by double clicking on the button. The limit's maximum value is 100 milliamperes.
- Side is a button used for file naming purposes when testing the right or left sides; it has no effect on how stimulation is delivered.
- Phase is a button that determines how the electrical stimulus is delivered. It can be set to Positive (POS), Negative (NEG), or Alternating (ALT); where setting it to positive will

deliver the stimulation with the polarity as designed, negative will invert the polarity of the stimulus and alternating will alternate between the previous two options.

- All other buttons and settings are kept the same as previously described (see page 65).

The stimulus menu at the top of the window also contains an option to turn the stimulation OFF at the end of acquisition. It is recommended to keep this option selected at all times.

The Somatosensory Stimulus Generation Window

This window is used to generate stimulation files for the somatosensory programs. It is accessible through the stimulus option of the main menu (Stimulus > Select Stimulus). The utility can be used to build stimulus files with up to four continuous segments, each with different stimulation parameters. The following figure shows the Stimulus Generation window with some defined fields:

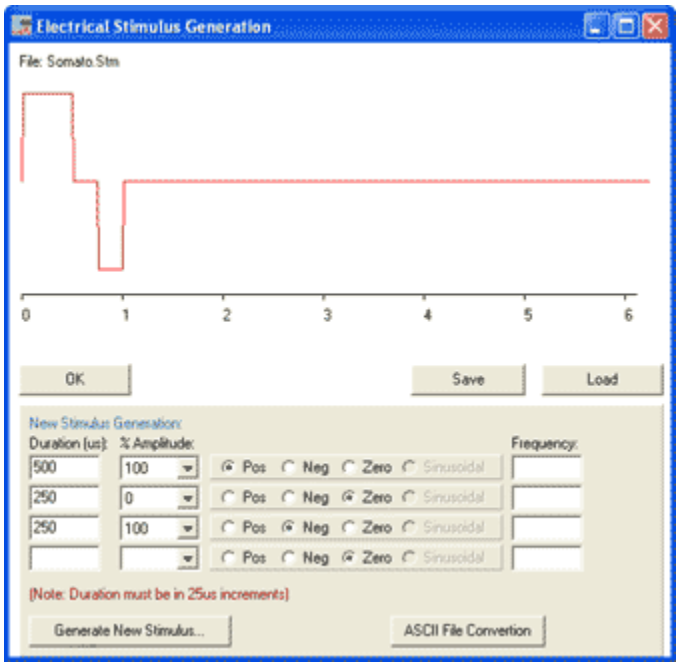


Figure 36 - Somatosensory Stimulus Generation Utility

- Duration: This option defines the duration of the stimulus component; this quantity needs to be entered in microseconds. Do not exceed 600 microseconds for human testing. The total duration of the components may only add up to 1000 microseconds. Stimulus duration must be specified in 25 microsecond increments.
- % Amplitude: This option defines the amplitude of the component when related to the Intensity provided. Example: For a component defined at 40%, if the test intensity is set to 40 milliamperes in the control panel, the intensity of the component will be 16 milliamperes.
- Component Type: These radio buttons define the polarity of the pulse or a region without a signal.
- Sinusoidal and Frequency options: These options are currently pending implementation.

It is necessary to save the created file before using it. To save the file, click on the Save button to load and/or edit an existing stimulus file, click on the Load button and select the

desired file. To close the window and use the selected stimulus as your stimulation file, click on the OK button. NOTE: when using pulses, a bipolar pulse is recommended to prevent skin damage. No D.C. components should be used for stimulation.

The Current Indicator can be found on the right hand side of the screen, right below the side menu. This item displays the percentage of current that is being returned (i.e. flowing) as a factor of the selected level. The indicator is yellow when the stimulation is OFF, and will be red when the percentage is less than 80%, indicating that the actual amount of current being delivered is lower than the selected value in the intensity option. This could be caused by either a high skin stimulation electrode impedance (>4kOms) or a low setting on the probe potentiometer. The indicator will turn green between 80% and 110% meaning that the output level is close to the selected level. If the percentage goes over 110%, the display will turn red again as a warning. The value shown by this indicator will be included in the recording file when it is saved.

Performing a Test

1. Connect the Electrical Stimulation Box and the USB Box as previously outlined. Make sure the patient is not connected to the system at this time.
2. Start the system by turning ON the USB box, turning ON the Electrical Stimulation Box and starting SmartEP.
3. Choose **Stimulus > Modality > Somatosensory > Standard** from the SmartEP main menu.
4. On the control panel, make sure the stimulus is OFF (Stim: OFF).
5. Set the desired maximum limit level on the control panel. The default maximum limit is set to 20 milliamps.
6. Place surface electrodes as needed and connect them to the transmitter box.
7. Select the appropriate side, rate, phase, and number of sweeps as needed for testing. Refer to the following table for recommended settings.

Setting	Value
Rate	Between 3 and 7 Hz, 5.1 and 5.3 Hz are the recommended values
Phase	Pos
Number of sweeps	800 to 500
Analysis Time Window	50 to 100 milliseconds
Amplifier Gain	30 to 50K
Filters	30 to 1000 Hz
Artifact Rejection	25 microvolts

8. If you do not want to use the preloaded stimulus (bipolar pulse of three-200 microseconds components, one positive, one zero, and one negative in that order), define your stimulus as necessary using the Somatosensory Stimulus Generation window as explained above. When testing humans, a pulse should not exceed 600 microseconds for optimal results. Click the Save button when done, then click OK to close the window.
9. Turn the stimulation ON.
10. Apply the two metal prongs of the stimulator probe to the testing area and press the side button to apply the stimulus. Graduate the potentiometer dial as necessary to find the muscular reflex threshold; an LED bar located under the potentiometer dial will

indicate the approximate percentage applied to the skin from the intensity value set in the control panel. Mark the electrode sites, set the intensity dial back to 0, and turn the stimulation OFF in the control panel.

11. Connect the stimulation electrodes to the previously marked sites and to the outlets of the electrical stimulator probe. Stimulation electrodes may also be connected to the outlets on the Electrical Stimulation Box.

WARNING: DO NOT CONNECT THE STIMULATION ELECTRODES TO THE TEST SUBJECT'S HEAD. CONNECT ONLY TO THE TEST SUBJECT'S LIMBS.

12. Repeat step 10 to find the muscular reflex threshold.
13. Click on the Acquire button on the control panel to begin acquisition.
14. Set the stimulation OFF and the intensity dial back to the lowest level.
15. Once done, label, analyze, and organize your recordings as needed.

If the impedance on the testing site is too high (less than 80% current return) the Impedance LED will light up in the box and in the probe.

Software limits

Due to government regulations and safety considerations, certain limits have been implemented in the software and hardware. Other limitations have been implemented by regulatory entities for human subjects. Limitations are given in the following table:

Units	Limits
Voltage (Hardware)	400V
Current (Software)	100mA
Stimulation Rate (Software)	100/s
Pulse Width (Human – FDA)	600us

Low Current Somatosensory Modality

Transtympanic eABR is performed by providing electrical stimulation to the promontory using needle electrodes. This procedure is usually performed where audiometric or sound stimulation testing is impossible when trying to determine the presence of residual auditory function. Two electrodes are used for stimulation; the electrode that delivers the stimulus is inserted into the ear canal past the tympanic membrane and on the promontory, and a second electrode is placed subcutaneously in the pre-auricle area. Subjects, especially human patients, should be provided with a certain degree of sedation due to the nature of the stimulators. Click stimulation is usually used. A vertex positive array is recommended for testing.

WARNING: This modality is meant for animal testing and research applications at this time, Transtympanic eABR for humans is pending FDA clearance. Consult with your IHS representative for current FDA application status.

WARNING: Patients with an implanted electronic device, such as a pacemaker, should not be subjected to any type of electrical stimulation unless indicated and approved by a specialist.

The Control Panel

Once the software is set to low current somatosensory mode (Click on Stimulus > Modality > Somatosensory > Low Current from the main menu), the control panel will display a different set of options. The modified options are described below.

Intensity: 1000uA	Stim: OFF	Limit: 4500uA	Side: Right	Phase: Rare.
Mod: LowSom	Sweeps:1024	Rate: 5.10/s	Time:100.0us (51.2)	EEG & Amplifier
Acquire	Continue	Clear Page	Load Settings	Save Settings

Figure 37 - Low Current Somatosensory Control Panel

- Intensity is now shown in microamperes. This amount can be incremented and decremented, by the given step size, by left or right clicking the button respectively. The step size is defined under Stimulus in the main menu; minimum step size is 1 microamperes. The intensity field will change color to yellow and red as it gets closer to the defined limit.
- Stimulus ON and OFF determines whether the stimulation is continuously provided or not. This setting does not affect acquisition. If stimulus is set to ON and acquisition is stated, the program will turn stimulation OFF once the system is done acquiring.
- Limit is the maximum current value that can be given to the intensity. This value can be entered by double clicking on the button. This limit should never be set higher than 5000 microamperes for human stimulation; some systems may have that limit built in.
- Side is a button used for file naming purposes; it has no effect on how stimulation is delivered.
- All other buttons and settings are kept the same as previously described (see page 65).

The Somatosensory Stimulus Generation Window

This window is used to generate stimulation files for the somatosensory programs. It is accessible through the stimulus option of the main menu (Stimulus > Select Stimulus). The utility can be used to build stimulus files with up to four continuous segments, each with different stimulation parameters. The following figure shows the Stimulus Generation window with some defined fields:

- Duration: This option defines the duration of the stimulus component; this quantity needs to be entered in microseconds.
- % Amplitude: This option defines the amplitude of the component when related to the Intensity provided. Example: for a component defined at 40%, if the test intensity is set to 2000 microamperes, the intensity of the component will be 800 microamperes.
- Component Type: These radio buttons define whether a Positive pulse, Negative pulse, a Zero stimulation area, or a sinusoidal signal is delivered.
- Frequency: Active only when sinusoidal component type is selected; it defines the sinusoidal frequency of the component.

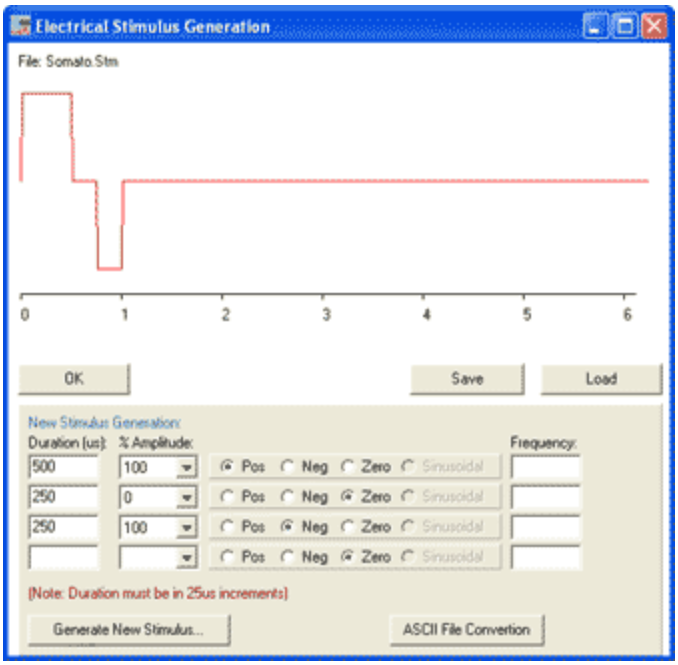


Figure 38 - Somatosensory Stimulus Generation Utility

It is recommended to save the created file before using it. To save the file, click on the Save button to load and/or edit an existing stimulus file, click on the Load button and select the desired file. To close the window and use the selected stimulus as your stimulation file, click on the OK button.

Performing a Test

1. Connect the Electrical Stimulator Box to the USB Box channel F, using the 7 pin Fischer connector.
2. Start the system by turning ON the USB box and starting SmartEP.
3. Choose **Stimulus > Modality > Somatosensory > Low Current** from the SmartEP main menu.
4. On the control panel, make sure the stimulus is OFF (Stim: OFF).
5. Set the limit level for the current on the control panel. NOTE: Even though the system allows for a maximum of 10000uA, the limit for human testing should never fall above 5000uA. Be sure to verify the allowed limit with your institution or region of residence.
6. Insert the Transtympanic needle electrodes in their respective sites. Also, place the surface electrodes as needed.
7. Define your stimulus as necessary using the Somatosensory Stimulus Generation window as explained above. When testing humans, a pulse should not exceed 600microseconds for optimal results. Click the Save button when done, then click OK to close the window.
8. Select the appropriate ear, rate, phase, number of sweeps, and intensity as necessary for the test.
9. Click on the Acquire button on the control panel to begin acquisition.
10. Once done, label, analyze, and organize your recordings as needed.

Software Limits

Due to government regulations and safety considerations, certain limits have been implemented in the software and hardware. Other limitations have been implemented by regulatory entities when it pertains to human subjects. Limitations are given in the following table:

Units	Limits
Voltage (Hardware)	20V
Current (Software)	10000uA
Current (Human – FDA)	5000uA
Stimulation Rate (Software)	100/s
Pulse Width (Human – FDA)	600us

Advanced Research Module

The Advanced Research module allows users to customize every aspect of acquisition and stimulation using an advanced, yet easy to understand, interface. It permits the mixing of stimuli (when the ipsilateral masking hardware is installed), definition of times of acquisition, rejection and other options as follows.

- The Right Ear stimuli options are shown with a light red background in the stimulus setup section and are represented by a light red bar in the Setup Graph. They are activated by checking the ON box.
- The Left Ear stimuli options are shown with a light blue background in the stimulus setup section and are represented by a light blue bar in the Setup Graph. They are activated by checking the ON box.
- In standard systems, it is possible to activate one stimulus file per ear. In systems with ipsilateral masking hardware, it is possible to activate two stimuli to be mixed together.
- Stimuli are chosen from previously generated stimulus files. These files can be generated using the stimulus generation utility or by converting a wav file to the IHS stimulus format. Stimulus files should be calibrated before use and should have a sampling rate of 40 KHz.
- Each stimulus has its own independent intensity value; they can start at any time by modifying the offset value or they can be defined as a continuous presentation. They can also be presented in condensation, rarefaction, or alternating modes.
- The data buffer options include the selection of:
  - Zero time: the point considered as zero in the final time scale. It is labeled as such in the graph by a vertical bar.
  - Start time: the time at which the first buffer will begin acquisition. It is shown on the graph as the starting position of Buffer 1 (Buf1).
  - Sampling rate: determines the size of the acquisition window. It is shown on the graph as the distance between the parallel vertical bars that belong to the buffer.
  - Number of buffers: determines how many recordings will be acquired per set. Each buffer will consist of 1024 data points regardless of size. Multiple buffers are defined as consecutive in the time scale.

- The Main options include:
  - Rate: determines the rate of repetition of the entire set. This value is inversely related to the length of the possible acquisition period; the smaller the rate, the longer the possible acquisition period.
  - EP Type: lets the system know the type of evoked potential that is being attempted. This option does not affect acquisition.
  - Artifact Rejection: these two fields determine the beginning and the end of the artifact rejection window. The rejection value can be selected from the EEG and Amplifier window.

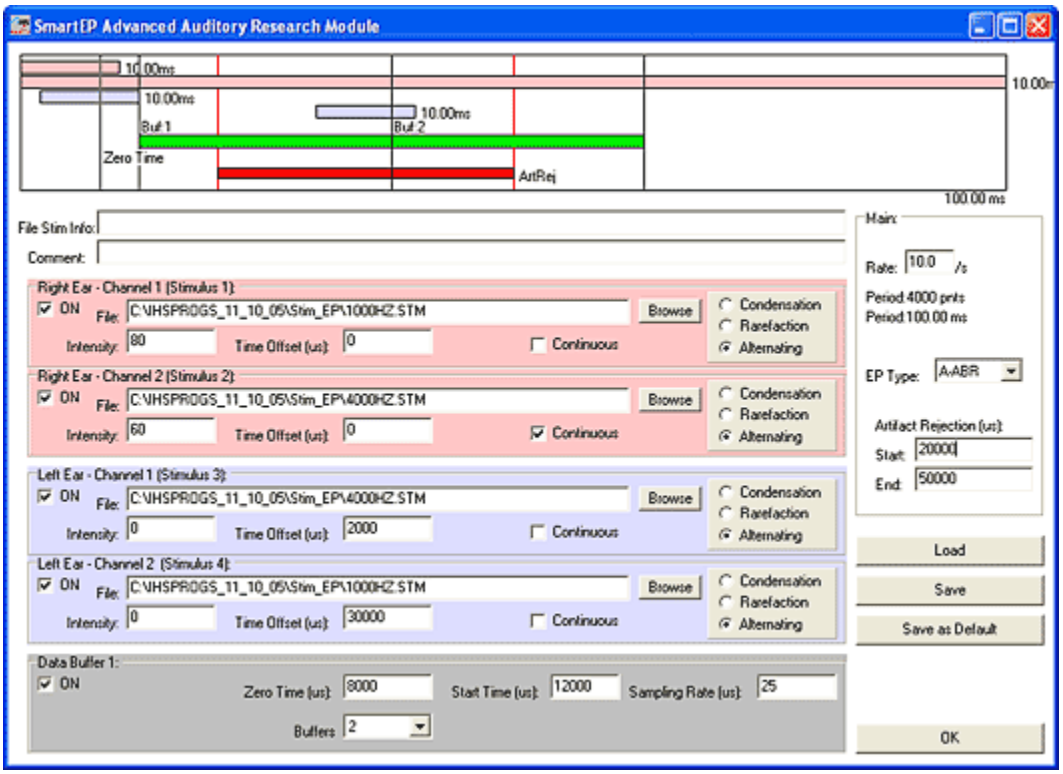


Figure 39 - Advanced Auditory Research Module

Once the test settings have been determined, they can be saved as default or as a file to be loaded later. Acquisition will generate recordings for the selected ears and the number of buffers as selected.

Patient Preparation

8

The quality of an ABR test is contingent on many factors. Aside from properly functioning software and hardware, patient state and preparation are crucial for obtaining reliable and valid test results.

Patient State

In general, it is best to test patients when they are in a state of quiet sleep, except when performing a P300 test. It is advisable that infants be tested midway between feedings, when they are least likely to be engaging in artifact producing muscle activity. It is often convenient to attach the electrodes as described later in this chapter before the infant is put to sleep.

Skin Preparation

The sites at which the electrodes are to be placed must be cleaned and prepared for placement using any of a variety of commercial preparations for reducing skin impedance. If reusable electrode cups are used, the cups should be filled with a conductive electrode paste before placement. Follow the guidelines provided by your hospital or clinic with regard to skin preparation. Detailed instructions on the use of the skin preparation materials supplied by Intelligent Hearing Systems can be found in the Technical Reference Manual.

WARNING:
Follow your hospitals guidelines for skin preparation. When performing skin preparation and/or impedance reduction, only use supplies approved by the FDA (or equivalent regulatory entity for your area).

Electrode Placement

Electrodes should be placed using the following guideline tables:

Single Channel System Configuration	
Red	Right Mastoid
Black	Forehead
Blue	Left Mastoid

Dual Channel System Configuration	
Red, Side A	Right Mastoid
Red, Side B	Left Mastoid
Black	Forehead below blue
Blue	Forehead above black

See page 87 for a visual aid to these tables

Impedance Testing

Impedance should be no more than seven K-Ohms for each electrode. If satisfactory impedance is not achieved, re-taping or re-cleaning of the skin may be necessary. Before performing ASSR testing, electrode impedances must be checked as follows:

1. After applying the electrodes to the patient's skin, connect the electrode leads to the electrodes, and to the Opti-Amp Transmitter or USB Lite Unit.

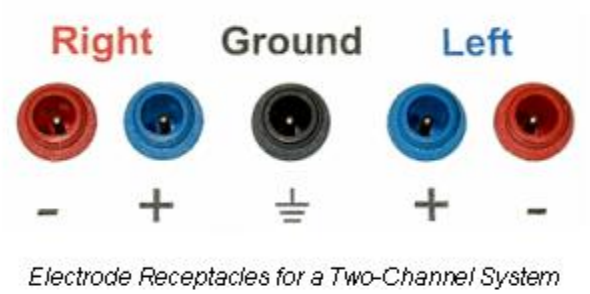


Figure 40 – Opti-Amp Electrode plugs

1. In USB Lite systems, click on the "Check Impedance" button in the EEG and Amplifier window. The impedances will appear on the screen. High electrode impedances will elicit a warning.
2. If using USB or USB Jr. systems, move the rotary switch on the Opti-Amp to each of the electrode impedance checking positions to read the impedance.

Insert Earphone Use

When using insert earphones, check that insert tubes are free of debris and listen to make sure there is sound. If using ear cups, place the cups over the ears and insert the foam tips or insert couplers into the ear cup inlet. Attach the tube onto the nipple end of the insert earphone. Roll the foam tip by pressing with your fingers into the smallest diameter possible and insert it well into the ear cup or ear canal, if using tips directly, according to the manufacturer's instructions. Connect the red transducer's ear-tip into the right side and the blue transducer's ear-tip into the left side of the patient. If using foam, hold the tip in place until the foam is fully expanded. After testing, discard all disposable supplies if used. Further instructions, including those for additional stimulators, are provided in the Technical Reference Manual.

WARNINGS:

- Do not clean ear canal unless you are a certified professional trained in this procedure. Irreversible injury may result to patient if an unqualified person cleans the ear canal.
- Choking Hazard; ear tips, couplers, and other small parts should be kept away from the reach of small children.
- DO NOT reuse ear tips or ear cups; dispose of them properly after use.

Reducing Noise

Before beginning SmartEP testing, check to make sure the EEG activity, in the EEG and Amplifier window, is normal for a relaxed state and that the signal does not contain artifacts or rhythmic waveforms reflecting electrical noise. If noise is present, check that the electrode leads are not in contact with earphone transducers, cables, or electrode leads from other equipment; and that the USB Lite or Opti-Amp are placed as far away as possible from CRT computer monitors, electrical outlets, motors, or other possible sources of ambient noise. Electrode leads may also be braided together to minimize the antenna effect of the cables.

For best results, the ambient noise in the testing environment should be minimized. This includes patient motion, heavy breathing, and other acoustic and electrical noise present in the test area. Highly noisy environments may result in elevated thresholds and/or longer test times.

Electrode Configuration

Right and left ear electrodes may be placed, by the operator, onto preferred sites such as the mastoids, earlobes, and ear canals. Electrodes should only be connected to the Opti-Amp transmitter box or USB Lite. See the following pages for the electrode configurations. Do not connect electrodes to any other plug or alter the circuit to connect electrodes directly to another position.

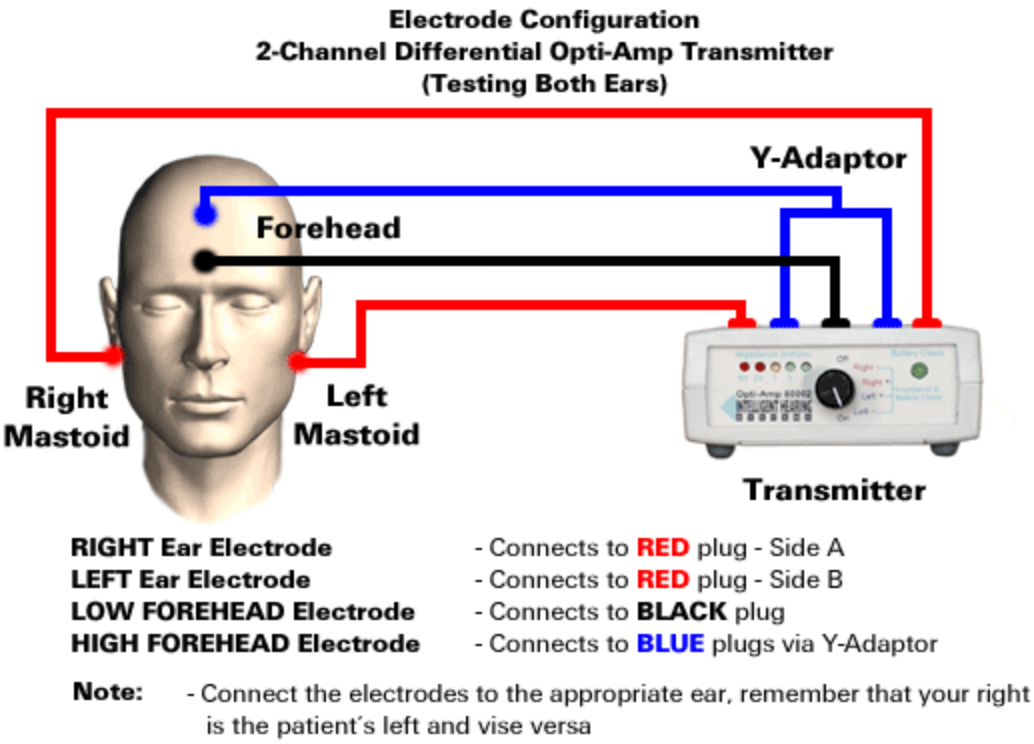


Figure 41 - Dual Channel Both Ears Setup

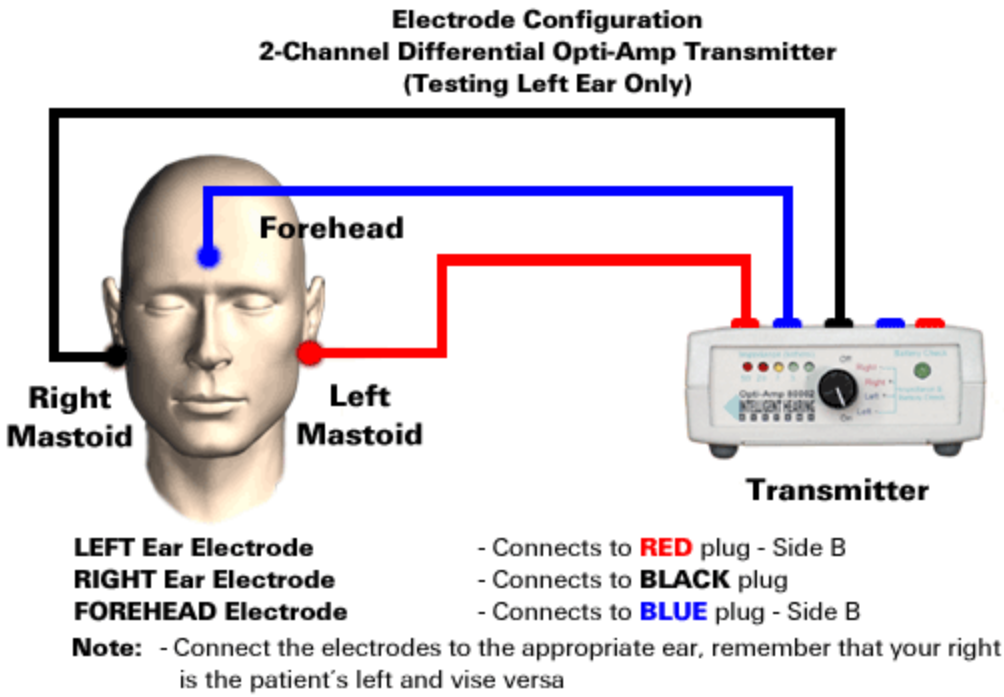


Figure 42 - Dual Channel Left Ear Setup

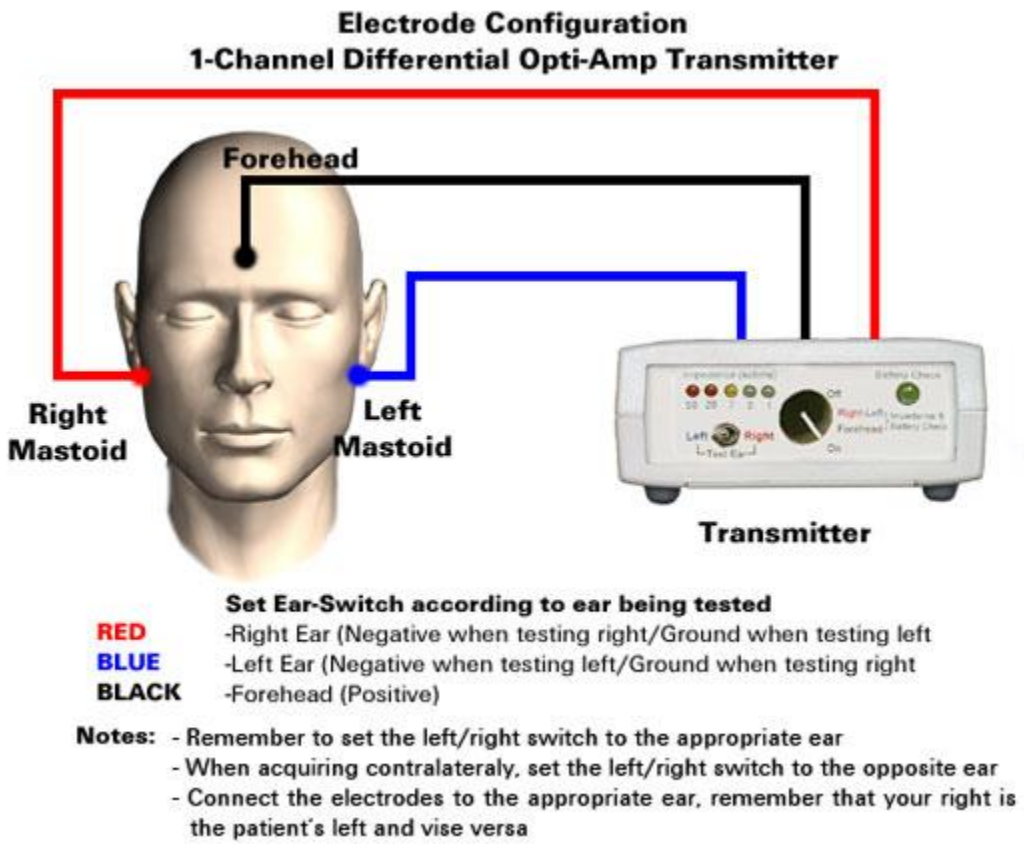
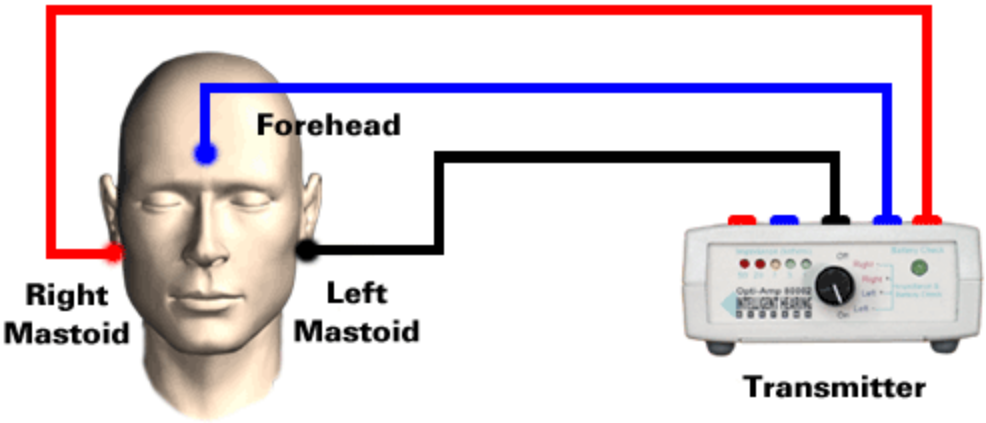


Figure 43 - Single Channel Setup

Electrode Configuration
2-Channel Differential Opti-Amp Transmitter
(Testing Right Ear Only)

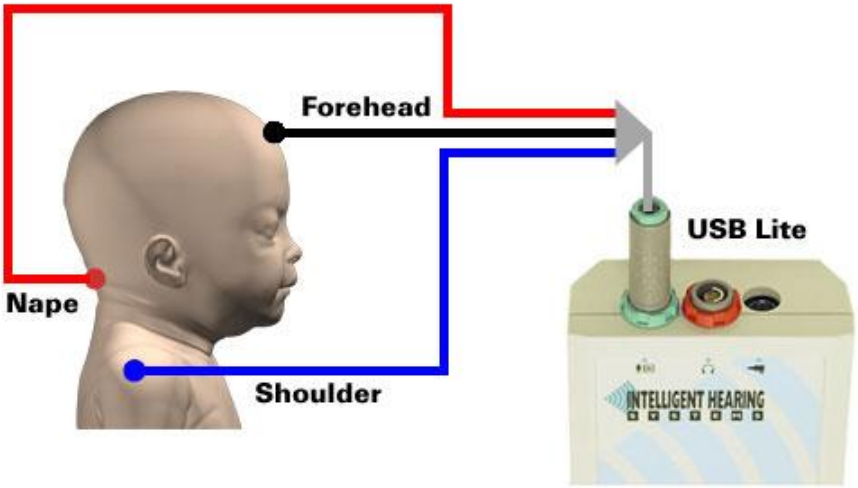


RIGHT Ear Electrode - Connects to **RED** plug - Side A
LEFT Ear Electrode - Connects to **BLACK** plug
FOREHEAD Electrode - Connects to **BLUE** plug - Side A

Note: - Connect the electrodes to the appropriate ear, remember that your right is the patient's left and vise versa

Figure 44 - Dual Channel Right Ear Setup

Electrode Configuration
USB Lite Systems



RED - Nape: Inverting (-)
BLUE - Shoulder: Ground
BLACK - Forehead: Positive (+)

Notes: - ASSRs are acquired for both ears simultaneously
- Connect the electrodes to the appropriate location. Electrodes placed in the wrong postition may cause noisy conditions.

Figure 45 – Setup for USB Lite

Patient Prep.

Calibration

9

Calibration of Pure Tone Signals

It is recommended that you print a copy of the original calibration table before proceeding. Follow this procedure to calibrate a stimulus signal:

1. Select the desired stimulus from the Stimulus menu including frequency and duration.
2. Set the intensity to, preferably, 70 dB nHL.
3. Connect the sound pressure meter to the headphones/insert earphones. Make sure to use the appropriate coupling device as recommended by the manufacturer of the sound pressure meter.
4. Carefully read and follow the sound pressure meter's instructions for the appropriate window, filter, and measurement settings required the signal to be calibrated. Different settings will be required depending on the signal duration and output intensity level.
5. Read the SPL measurement from the sound pressure meter.
6. Make sure to use the correct SPL to nHL correction value for the stimulus signal selected (the click correction value between nHL and SPL is 32 dB).
7. If, after converting the measured SPL value to nHL, that level is not the same as the selected output intensity, enter the proper calibration correction value in the corresponding position of the calibration table in order to make the levels equal.
8. After entering the correction value, re-measure the output signal until the selected intensity matches the adjusted sound pressure reading.

Note: Make sure to select the correct stimulator and side when modifying correction values.

Calibration of Click Stimulus Signals

1. Select SmartEP to output Clicks @ a rate of 10/sec @ 70dB HL
2. Connect the Sound pressure meter with the following settings:

    - Response: Peak
    - Weighting: LIN
    - dB Range: 80-140
    - Mode: SPL

3. Press reset and read the value. The first value should be 102. The SPL to HL conversion for Clicks is approximately 32 dB.

Special calibration values may also be saved for specific stimuli. When a stimulus is generated, the corresponding calibration table value is automatically used. A special calibration value may be used by entering that calibration value in the Auditory Stimulus Generation Dialog Box and saving the stimulus as a file. The program will store the calibration values with the generated stimulus signal. These values will then be used by SmartEP every time the file is loaded. Consult the Launch Pad manual for information about the stimulus calibration utility and the file calibration utility.

Troubleshooting

10

This is a guide to the most frequently asked questions and their solutions. Please be sure to follow this trouble-shooting guide before calling technical support. Make sure to go through all the steps outlined in this guide until the problem is resolved. Even though some of the suggestions may seem to be too elementary to present a solution, they may be the source of the problem. If the procedure does not help you solve the problem, you will need to contact IHS technical support for additional help. Make sure to tell the technical support representative that you followed the steps in this guide.

My USB System is not responding.

- Is the power LED of the USB box illuminated? If not, check that the switch is in the ON position, the power cable is properly connected, the isolation transformer is connected, and all cables are in good condition. If necessary, disconnect the isolation transformer and connect a different electrical appliance to the outlet to verify that the outlet is in good condition and currently providing power. Note that the USB Lite hardware does not have and ON/OFF switch.
- Is the USB cable connected properly, and is the Computer connection and Online LED's illuminated? Check both ends of the cable for proper connection and the cable itself for any possible damage.
- Was the SmartEP program started before the USB box was turned ON? If so, exit the SmartEP program, Turn the USB Box OFF, then back ON and then restart the SmartEP program.
- Was the installation of the software completed properly? Turn the USB system's Switch to the ON position and run System DSP Test from the System menu on the IHS LaunchPad. It is important that you do so in that order. If you can only see the General and Driver tabs, then the computer cannot communicate with the USB box. Check the cables and power connections again; retest the system.
- Otherwise, go through the tabs pressing the buttons to perform tests on the system. If an error arises, contact IHS with that information.

The power LED on the USB box is blinking

The system must be inspected by IHS Immediately. Turn the system OFF and contact Technical Support for details about shipping the system back to IHS for inspection.

I get an error message when starting SmartEP

The problem may be that the USB box was not turned ON or connected to the computer before SmartEP was started. SmartEP needs to verify the connection and download programs to the USB box before it can run. To fix this problem, close SmartEP, turn ON the USB box and restart SmartEP. If this does not solve your problem, there must be something wrong with the USB box or its connections. Follow the steps in the "My USB System is not responding" question to find the problem.

There is no stimulus presented at the Bone Vibrator

- Verify that the USB box is turned ON and working properly by following the steps in the "My USB system is not responding" question.
- Check that the Bone Vibrator is connected to the correct output of the USB box.
- Access the Auditory Stimulus Generation dialog box using the Stim option from the control panel. Verify that the bone vibrator is selected on the right hand panel.

The EEG and Amplifier dialog box shows a flat line signal.

- Is the transmitter switch in the ON position?
- Is the transmitter box receiving power? Move the switch to one of the impedance test positions and verify that the LEDs light up. If not, verify that the power is properly connected.
- Is the fiber optic cable plugged correctly at both ends?
- Unplug the fiber optic cable from the transmitter. Are the sockets where the cable was connected emitting a red light? If not, contact IHS.
- Is the fiber optic cable kinked or bent? Kinks or bends in the fiber optic cable may cause the inside of the cable to break; this will cause the data to stop flowing from the transmitter. Always store the fiber optic cable in a loose circular manner.

To test the internal integrity of the fiber optic cable, reconnect the cable to the transmitter and turn the transmitter on. Disconnect the cable from the receiver and verify that the red light can be seen at the other end of the cable. If you cannot see any changes, the cable may be broken.

- Is the socket where the fiber optic cable is connected to the system broken? This may cause external lighting to enter the transmission receiver, creating too much noise for the signal to be clear.
- Are the electrodes attached correctly?
- Are the transducer silicon tubes or foam tips occluded?

The signal is too noisy

- Check the EEG and Amplifier rejection ratio. Press EEG and Amplifier in the control panel and move the level slider to read between 15 and 30 microvolts for a standard ABR. The rejection level is displayed in the bottom-right information panel. Do the same for all channels that will be used for acquisition.
- Check the high pass and low pass filters in the EEG and Amplifier Dialog Box. Make sure the filters are close to the recommended settings for the type of recording that you wish to acquire.
- Is the transmitter away from other electrical devices? If not, move it away from any electrical machinery that may be causing interference, such as computer monitors or electrical outlets. The EEG and Amplifier dialog box should show the signal as close to a flat line as possible when the patient is not connected.
- Are the fiber optic cables kinked or bent? If they are, they may be broken inside and may need to be replaced. To test the internal integrity of the fiber optic cable, reconnect the cable to the transmitter and turn the transmitter on. Disconnect the cable from the receiver and verify that the red light can be seen at the other end of the cable. If you

cannot see the light, the cable may be broken. Always store the cable in a loose circular manner.

- Is there a 60 Hz (or 50 Hz, depending on your location) signal artifact? Try turning the Notch filter ON.

There is no sound output from the transducer.

- Check to see that the system is turned ON.
- Verify that the stimulator is connected to the appropriate output as indicated in the hardware installation instructions
- Is the intensity of the signal high enough? Check the Intensity option in the control panel. Right click on it to increase its value.
- For the inserts, are the silicon tubes or foam tips occluded with liquid?
- Are the transducer cables near electrical outlets? If so, move them away.
- Are the transducers touching the electrode leads? If so, move them away.
- Are the transducers (insert earphones) on top of the patient? If so, move the transducers to the side of the patient.
- Is the sound presentation set on continuous? Go to the Auditory Stimulus Generation dialog box by clicking on the Stim button in the Control Panel. Verify that the Stimulus Presentation option is set to Continuous.

If you have a voltmeter or other device that can check for continuity follow the next steps if you are using insert earphones. This procedure is intended for experienced users. If a voltmeter is not available, contact IHS or your local representative for help.

1. Disconnect the insert earphones from the system.
2. Pull the plug from the transducer straight out and gently. DO NOT TWIST OR BEND THIS PLUG, doing so may cause it to brake.
3. Notice that the small plugs have two tips, one thicker than the other.
4. Test the small tips against the pins of the Fischer connector and make sure that there is continuity with at least one of the pins for each one of the tips.

If any of the tests fail, the soldering or wiring inside the cable may have been compromised. Contact IHS with this information to get further instructions.

There is a normal EEG signal shown but I cannot acquire an recording

- Can the patient hear the stimulus?
- Are the insert earphones connected properly?
- Is the stimulator cable in good condition?
- Are the tubes from the insert earphones or the ear tips being occluded by any substance?

If the fiber optic cable has not been damaged and you have used new non-expired electrodes from an uncompromised package, this can be due to two reasons:

Pre-Stimulus Artifact: This is an artifact appearing at the start of a recording. This artifact may be invisible depending of the current plot times. To fix this problem:

1. Select the recording that presents the problem by clicking on its signal handle, the small circle at the left of the signal.

2. Click on the PP button in the side menu.
3. Verify that scale is set to normalized
4. Select the Special option under Scaling
5. Change the Scaling start time to a time after the artifact appeared.
6. The rest of the signal should now be scaled to a normalized value. Change the plot start and end times at your convenience to eliminate the clipped artifact from the display.

NOTE: This procedure does not change the signal as was stored in the hard drive of your computer, and only changes the current display.

Post Auricular Muscle Artifact: This is an artifact appearing at the end of a recording, usually around eight milliseconds. To fix this problem:

1. Select the recording that presents the problem by clicking on its signal handle, small circle at the left of the signal.
2. Click on the PP button in the side menu
3. Verify that the scale is set to normalized
4. Select the Special option under Scaling
5. Change the Scaling end time to a time before the artifact appears.
6. The rest of the signal should now be scaled to a normalized value. Change the plot start and end times at your convenience to eliminate the clipped artifact from the display.

NOTE: This procedure does not change the signal as was stored in the hard drive of your computer, only changes the current display.

I cannot find recordings acquired on a previous testing session.

First, you must have had AutoSave ON from the Data option in the main menu. Otherwise, any recordings not saved manually will be lost. The AutoSave option is ON by default, unless changed by the user. The other most probable cause for this problem is that the data was acquired without entering a patient identifier. The computer will automatically save the files under the identifier XX. To place the files to the correct location:

1. Navigate to IHSPROGS > XX from the drive where the program was installed using windows explorer. (Hint: right click on “My Computer” and select explore).
2. Cut the files from the XX folder and paste them in the appropriate folder after locating the patient’s identifier.

NOTE: The recordings in the XX file may belong to more than one patient. To avoid this completely, make sure the patient’s identifier number is always entered into the Patient Demographics dialog box. It is a good practice to load the patient in the Launch Pad program, before starting SmartEP. The file creation date may help you identify which recordings belong to this patient.

My computer freezes or the USB box does not start

Windows XP, by default, turns off USB Root hubs to conserve power when the computer goes into idle state. This may cause the device to stop working properly when resuming operations after sleep, hibernation, or long computer inactivity. To correct this problem:

1. Access the System Properties either by right clicking on the “My Computer” icon and selecting properties or by choosing the System Option from the Windows Control Panel.
2. From System Properties, select the Hardware tab and click on the Device Manager Button.
3. Expand the Universal Serial Bus controllers section by clicking on the [+] sign.
4. Open the properties dialog box for the USB Root Hub item by right clicking on it and selecting properties.
5. From the Properties dialog box, select the Power Management Tab and deselect the option that reads “Allow the computer to turn off this device to save power”, then click OK to save the changes.

Alternatively, you may choose to prevent the computer from going into sleep or hibernation modes. Access the power management options in the Windows control panel to change these options. Consult your computer’s user manual for details about these options.

Some computers may contain a USB chipset that produces a communication signal not supported by the established USB protocols. If this is the case with your computer, the USB Box and other USB devices may not communicate properly. To correct this problem it may be necessary to replace the USB Card on a Desktop computer or attach an external USB Hub to the existing USB port.

References

11

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Appendices

12

Easy Menu Mode

This mode of operation of SmartEP has been designed for ease of use. It is recommended to run the program in this mode when restricted use of the settings is necessary, such as restricting user's access, or when the system is used mainly for screening purposes. The Control Panel and the right side menu disappear and the easy access test buttons appear in place of the Control Panel. All menus, except the help menu, become inactive. In this mode, the operation of the software becomes simplified, allowing the operator to perform tests using pre-made protocol files without worrying about other variables.

Easy Menu Mode Setup:

Create a Shortcut in your desktop to SmartEP, if you do not already have one. An easy way to do this is to look for the SmartEP shortcut in the Start Menu and drag it with the right mouse button into your desktop. When the mouse is released, a menu will appear. Choose the Copy Here option.

1. Right click on the shortcut to SmartEP and select Properties option.
2. Look for a line called "Target" in the properties dialog box.
3. Change the line to read "C:\IHSPROGS\SEPWIN.EXE EASYMENU", where C:\IHSPROGS is the directory where the program was originally installed.

You can also add a password so that the users starting in the Easy Menu Mode are not able to go to the full diagnostic mode unless they know the correct password. To add a password, modify the Target line to read:

"C:\IHSPROGS\SEPWIN.EXE EASYMENU PASSWORD=password"

The password can be any text up to 20 characters long, no spaces. The password is case sensitive. For the password option to work, make sure that SmartEP is started from the shortcut that was modified.

It allows the user to enter a new patient's information in the SmartEP system. Old patient files can also be loaded by using the Load button in the Patient Demographics dialog box. This option is equivalent to Patient Menu > New Patient in the Main menu in expert mode.

Changes the current acquisition ear to left or right. This option is equivalent to the Control Panel's Ear button.

Executes the user-designed protocol saved in SmartEP as SCRN1. File must be saved using the Protocol Menu > Protocol Setup option in the Main Menu in Expert Mode.

Executes the user-designed protocol saved in SmartEP as SCRN2. File must be saved using the Protocol Menu > Protocol Setup option in the Main Menu in Expert Mode.

Opens the Open File dialog box in order to load and execute a previously saved protocol in SmartEP. It is equivalent to the Execute Protocol option under the Protocol Menu.

This option will start execution of chain stimuli test set on the selected testing ear.

It allows the user to load recordings from the hard drive. It is equivalent to the Data Menu > Load File option in the expert mode.

It prints the recordings currently displayed and the Amplitude and Latency of the marked peaks.

This button allows access to the EEG & Amplifier Settings dialog box in SmartEP. All options will be available since they are needed to acquire the recordings correctly.

Quits the easy menu mode and enables the user to have access to all the advanced features of SmartEP. Password access may be implemented .

Time and Rate

The system is programmed to record 1024 data points per stimulus. It acquires 512 points before the stimulus and 512 after the onset of the stimulus. The length of time between points is user-defined, through the Time button (on the Control Panel), and is in the range of 25-800 μ s (microseconds). Therefore, the total time the computer is working per stimulus is equal to 1024*Time, as in the diagram below. For this example, Time is 25 μ s. The figure is not to scale.

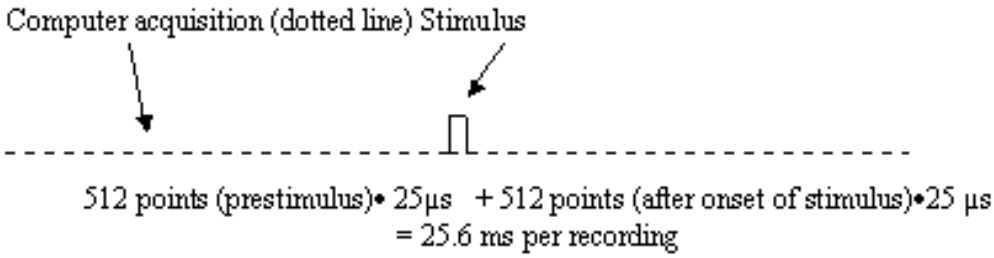

Figure 46

If the Rate is set to 19.3 stimulus presentations per second, then the total time for the stimulus and the computer acquisition before and after the stimulus is 1s divided by 19.3 =1s/19.3 = 51.8 milliseconds. This is more than enough time for the computer to acquire the required

512 points before and after the onset of the stimulus; therefore, there is “dead time” when the computer is not acquiring data. A graph of the acquisition would look like this:

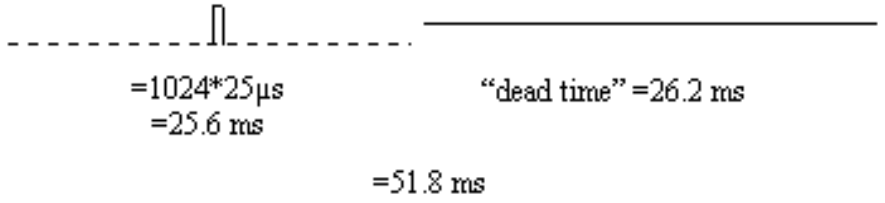

Figure 47

However, if the Rate is increased to 100/s (and Time is not changed) then the total time for the computer acquisition of the 1024 points is only 10 ms, which is less than the 25.6 ms, calculated previously, that are needed for all 1024 points. In fact, this combination is possible in SmartEP, and it is important to be aware of what the computer will do in this situation. Since a Time of 25 µ s is the smallest possible, decreasing the amount of time between points to decrease the whole acquisition window is not possible. Instead, the computer just does not acquire the full 1024 points. It removes all of the 512 points acquired pre-stimulus (before the stimulus) and decreases the number of points acquired after the stimulus accordingly, so that the acquisition is completed by the time the next stimulus is presented. In this case, the maximum possible number of data points is 400, as in the following diagram. In turn, all subsequent stimuli will have the adjusted number of acquisition points.

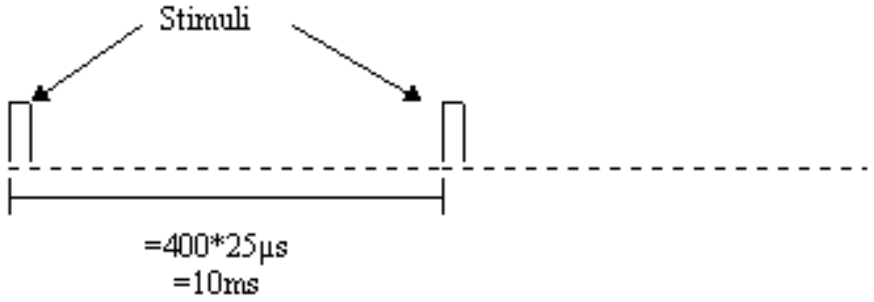

Figure 48

Notes on Acquisition

While acquiring data, the recordings are displayed in a normalized format. After all sweeps have been collected, the recordings will be displayed in the current amplitude scale entered under Scale in the Page Parameter Menu.

A 0.90 ms latency correction is automatically made by the software when using insert earphones to account for the time it takes the sound to go from the transducers to the ear. This is an example of why it is important to set the correct transducer option in the Auditory Stimulus Generation dialog box (see page 38), not only for good record keeping, but also for the software to interpret the data correctly.

Even though SmartEP is capable of multitasking, no new patients may be loaded or created during the acquisition process so that all the recorded data are written to the correct directory. No data recordings may be loaded onto the software module when acquisition is occurring. Only one instance of SmartEP may be running at a time. In addition, SmartEP and other IHS programs cannot be run simultaneously on the same computer.

Filter Characteristics

The degree of low-pass filtering obtained when implementing the off-line smoothing operations depends on which filter is selected. The amount of smoothing is selected by choosing the number of points (filter size or number of filter coefficients) to use or the spectral limits desired.

FIR filters

Finite Impulse Response filters are implemented such that they will not introduce any phase or time shift to the recordings. In order to calculate the cut-off frequency of these filters, you must know the sampling period used to acquire the recordings (depending on the time window used as determined from the first table below) and the appropriate coefficient (resulting from the number of filter points selected as determined from the second table below). Please contact IHS if you need assistance with these calculations.

Time Window (ms)	Sampling Rate (μ sec)
13	25
25	50
51	100
103	200
204	400
409	800

Number of Points	-3 dB Frequency	-6 dB Frequency
7	0.159	0.191
9	0.121	0.145
11	0.098	0.117
13	0.082	0.100
15	0.070	0.0857
17	0.062	0.075
19	0.0565	0.0675
21	0.051	0.061
23	0.047	0.056
25	0.043	0.051

Example Calculation

For example, if the recording sampling rate is 25 μ sec and a 15-point filter is selected, the cut-off frequencies can be calculated by dividing the values given on the table above by the sampling rate in seconds:

$$F - 3dB = 0.070 / 0.000025 \text{ s} = 2800 \text{ Hz}$$

$$F - 6dB = 0.0857 / 0.000025 \text{ s} = 3428 \text{ Hz}$$

Spectral Filters

Spectral filters allow you to specify the desired cutoff frequencies to be applied to the selected recordings. The spectral filters have an infinite falloff, i.e. there is no -3 dB frequency. The minimum high pass frequency is zero. The maximum possible Low pass frequency is given by one-half of the sampling frequency (inverse of the sampling rate). Since the minimum sampling rate is 25 μ sec, the maximum value that the low pass filter can have is 20 kHz. For example, if the recording sampling rate were 50 μ sec, the maximum value of the low pass filter would be:

$$F \text{ max} = 1/2(0.000050 \text{ s}) = 10000 \text{ Hz}$$

Recommended Test Settings

The following tables show the recommended settings for some of the tests that can be executed using the SmartEP system. Always consult with a trained audiologist to determine the best settings to be used. Consult the appropriate literature to obtain detailed information on the required test settings.

MLR Protocol	
Stimulus	0.1 milliseconds Click for Neuro-diagnosis and 2-1-2 Tone Burst for Audiometry
Rate	7.1/sec
Polarity	Alternating
Transducers	Insert Earphones
Intensity	70 dB HL. Lower intensities may be used to determine hearing threshold. Higher intensities may result in post-auricular muscle artifact.
Masking	50dB. Only if Stimulus exceeds 70 dB HL.
Filters	10-1500 Hz or 10-200 Hz
Notch Filter:	OFF
Amplification	75x
Runs	2 or 4
Analysis Time Window	Overall 100 milliseconds
Sweeps	1000
Electrode Montage	Ipsilateral or Contralateral Array

LLR Protocol	
Stimulus	40 milliseconds Tone Burst
Frequency	500 - 8000 Hz as needed
Rate	1.1/sec
Polarity	Alternating

Transducers	Insert Earphones
Intensity	50dB or higher
Filters	1-30 Hz
Notch Filter:	OFF
Amplification	50x
Runs	2 or 4
Analysis Time Window	Overall 600 milliseconds
Sweeps	250
Electrode Montage	Ipsilateral or Contralateral Array

P300 Protocol	
Stimulus	40 milliseconds Tone Burst
Frequent (frequency)	1000 Hz
Rate	1.1/sec
Polarity	Alternating
Transducers	Insert Earphones
Intensity	50dB or higher
Filters	1-30 Hz
Notch Filter:	OFF
Amplification	50x
Runs	2 or 4
Analysis Time Window	Overall 600 milliseconds
Sweeps	250
Electrode Montage	Ipsilateral or Contralateral Array

ECochG Protocol	
Stimulus	0.1 milliseconds Click
Rate	7.1/sec or 99.9/sec. Slow is used to enhance the AP component and the fast rate is used to differentiate the SP (still robust) from the AP (now deteriorated).
Polarity	Alternating
Transducers	Insert Earphones
Intensity	90 to 95 dB HL
Filters	5 or 10 – 1500 Hz
Notch Filter:	OFF. ON, only if there is excessive electrical line noise present.
Amplification	75x or 100x
Runs	3 or 4
Analysis Time Window	5 milliseconds
Sweeps	250 to 500
Electrode Montage	Horizontal Array

ABR Protocol	
Stimulus	0.1 milliseconds Click or Tone bursts
Rate	19.3/sec or 21.1/sec. Slower rates are used to enhance wave I.
Polarity	Rarefaction provides clearest wave I. Use condensation if response is not optimal. Use alternating if there is excessive stimulus artifact.

Transducers	Insert Earphones
Intensity	80 to 95 dB HL, for neuro-diagnosis 90 dB HL down to 0 dB HL for Threshold search.
Filters	30 – 3000 Hz (100 –3000 Hz for Adult ABR and 30-1500 Hz for an infant ABR).
Notch Filter:	OFF. ON, only if there is a lot of electrical line noise present.
Amplification	100x
Runs	2 or 4
Analysis Time Window	12.8 milliseconds
Sweeps	1024 or 2048
Electrode Montage	Ipsilateral or Contralateral Array

Bone Conduction ABR Protocol	
Stimulus	0.1 milliseconds Click
Rate	7.1/sec used to enhance wave I and assure ear-specific response
Polarity	Alternating
Transducers	Bone Vibrator
Intensity	Maximum is 60 dB HL. Please note: Air conduction and Bone conduction latencies should be similar using default calibration values, assuming proper conduction between the bone vibrator and the bone through the skin.
Masking	Always. Masking Level should equal the Intensity Level + 40dB since the effective masking is SPL – 39 dB. If using the Following option the offset should be 40.
Filters	30 – 3000 Hz
Notch Filter:	OFF. ON, only if there is a lot of 60Hz electrical noise present.
Amplification	100x
Runs	2 or 4
Analysis Time Window	12.8 milliseconds
Sweeps	1024 or 2048
Electrode Montage	Ipsilateral or Contralateral Array

Tone-burst ABR Protocol	
Stimulus	Tone burst (Exact Blackman Envelope): 500 Hz 8msec (8000 u sec) in duration 1000 Hz 4msec (4000 u sec) in duration 2000 Hz 4msec (4000 u sec) in duration 4000 Hz 2msec (2000 u sec) in duration 8000 Hz 1msec (1000 u sec) in duration
Rate	27.7/sec
Polarity	Rarefaction or Rarefaction + Condensation
Transducers	Insert Earphones
Intensity	As needed
Filters	30 – 1500 Hz
Notch Filter:	OFF
Amplification	100x
Runs	2 or 4
Analysis Time Window	25.6 msec in order to see peak V at 500 or 1000 Hz

Sweeps	1024 or 2048
Electrode Montage	Ipsilateral or Contralateral Array

Masking Notes:

Air Conduction Masking testing with headphones or insert earphones:

- Masking Level ON option is set to the intensity level.
- Masking Level FOLLOWING option is set to zero (0).

Bone Conduction Masking testing using headphones or insert earphones as masking devices:

- Masking Level ON option is set to the intensity level + 40 dB.
- Masking Level FOLLOWING option is set to forty.

ECochG Testing with Tiptrodes

Follow this procedure to conduct an ECochG test. Ear cleaning must be performed by a trained professional. Remember that the suggested parameters are only a guideline to follow. Consult the appropriate literature for the required parameter settings.

1. Clean the ear canal thoroughly, and then dry it with a cotton swab.
2. Prepare the skin for electrode placement.
3. Follow the guidelines provided by the hospital or clinic with regard to ear cleaning and skin preparation.
4. Coat the Tiptrodes with conductive electrode paste.
5. Complete the patient connections using the appropriate table below.

One Channel Opti-Amp - testing right ear (Remember to set the ear switch to right)

Red (right) tiptrode tube	Connects to RIGHT insert earphone transducer
Red (right) insert tiptrode	Inserts as deep as possible into the RIGHT ear
Red tiptrode lead	Connects to RED plug (-) on Transmitter Box
Left mastoid electrode	Connects to BLACK plug (+) on Transmitter Box
High forehead electrode	Connects to BLUE plug (ground) on Transmitter Box

One Channel Opti-Amp - testing left ear (Remember to set the ear switch to left)

Blue (left) tiptrode tube	Connects to LEFT insert earphone transducer
Blue (left) insert tiptrode	Inserts as deep as possible into the LEFT ear
Blue tiptrode lead	Connects to BLUE plug (-) on Transmitter Box
Right mastoid electrode	Connects to BLACK plug (+) on Transmitter Box
High forehead electrode	Connects to RED plug (ground) on Transmitter Box

Two Channel Opti-Amp - testing right ear

Red (right) tiptrode tube	Connects to RIGHT insert earphone transducer
Red (right) insert tiptrode	Inserts as deep as possible into the RIGHT ear
Red tiptrode lead	Connects to RED plug (-) on Transmitter Box side A
Left mastoid electrode	Connects to BLUE plug (+) on Transmitter Box side A
High forehead electrode	Connects to BLACK plug (ground) on Transmitter Box

Two Channel Opti-Amp - testing left ear

Blue (left) tiptrode tube	Connects to LEFT insert earphone transducer
Blue (left) insert tiptrode	Inserts as deep as possible into the LEFT ear
Blue tiptrode lead	Connects to RED plug (-) on Transmitter Box side B
Right mastoid electrode	Connects to BLUE plug (+) on Transmitter Box side B
High forehead electrode	Connects to BLACK plug (ground) on Transmitter Box

Suggested acquisition parameters

Mode:	Ipsi
Stimulus	Click
Phase:	Alternating
Sweeps:	650 or higher
Rate:	7.1/sec (can be modified to 5.1 to 11.1/sec)
Filters:	3-1500 Hz or 10-1500 Hz
Gain:	100k or as needed
Scale Region:	0-6 msec (in the System Menu)
EP Type:	E-ECochG (to obtain SP and AP markers and amplitude calculations)

Bone Vibrator Testing

Follow this procedure to complete a bone vibrator test.

1. Make sure the system is turned OFF and the patient is disconnected from all stimulators
2. Connect the Bone Vibrator and earphones as shown in the Hardware Installation Instructions manual.
3. Turn the computer ON and execute the SmartEP program.
4. Access the Auditory Stimulus Generation dialog box using the Stim button from the control panel and select Stimulator to be Bone Vibrator.
5. Turn Masking ON. At the Mask Intensity option in the dialog box, type the masking intensity in dB SPL.
6. Place electrodes on Right Midline montage.
7. Place the Bone Vibrator stimulator at the Right or Left Mastoid. If using an ipsilateral or contralateral array electrode montage, make sure that the Bone Vibrator stimulator does not touch the electrodes.
8. Set the masking switch to the ear you wish to mask.
9. The user must select the ear in the Control Panel as usual in order to acquire data from the appropriate electrodes.

MMN Area Calculations

SmartEP performs the calculation of the MMN area automatically, after marking the MMN peak. The peak can be placed by right clicking over the desired position on the selected recording and selecting MMN from the “mark other peak” option. Dragging the bottom marker to the appropriate location will calculate the area of the valley contained between the top and bottom markers in microvolt-microsecond (uVus) units. The calculated area will be shaded in yellow.

Loop Back Testing

Loop-Back testing ensures that the whole system is working properly. Performing this test, verifies the operation of sound output, signal receiving, amplification, and filters and data communications. The system contains a utility to perform this test automatically. Please refer to the Launch Pad Manual section about the System DSP Test utility for further details.

SmartEP 1 Channel Loop-Back Test Procedure

LOOK BACK

Before performing a Loop-Back test, make sure sound is present in the sound transducer. Set the intensity to 80 dB HL and select the appropriate ear for sound output. You may have to click on the Acquire button in the Control Panel. If no sound is being produced in the transducer, then that is the source of the problem; contact IHS for assistance.

1. Connect the Loop-Back cable to the 300-Ohm earphone connector of the USB Box.
2. Connect the test leads of the Loop-Back cable as follows to the Opti-Amp:
    - Black lead to black plug
    - Blue lead to blue plug
    - Red lead to red plug
3. Run the Smart-EP program.
4. Set the Gain in Smart-EP to 100k, the high-pass filter to 50 Hz, and the low-pass filter to 5000 Hz on the EEG and Amplifier window.
5. Open the artifact rejection to at least 75% in the EEG & Amplifier window.
6. Turn Opti-Amp ON.
7. Set the test Ear to Right and the Mode to Ipsi in the Control Panel. Set the ear switch on the Opti-Amp to RIGHT.
8. In the Protocol Menu, select the Execute Protocol option. Then select the Loopback.pse protocol file and open it.
9. Make sure Amp scale is set to N (Normalized) in the Page Parameter Menu (PP Button) and Show all waveforms by Intensity.
10. Print Results.

The values obtained in the printed table should be close to the ones shown in the following table. There is an allowable error margin of 15% to the shown values. If you have any large discrepancies to the values shown, fax the printed result to IHS for evaluation.

Intensity (HL)	Peak-to-Peak Voltage (Vp-p)
90 dB	9.28 microvolts
80 dB	2.89 microvolts
70 dB	0.92 microvolts
60 dB	0.34 microvolts
50 dB	0.17 microvolts

Smart-EP 2-Channel Loop Back Test Procedure

Before performing a Loop-Back test, make sure sound is present in the sound transducer. Set the intensity to 80 dB HL and select the appropriate ear for sound output. You may have to click on the Acquire button in the Control Panel. If no sound is being produced in the transducer, then that is the source of the problem; contact IHS for assistance.

1. Connect the Loop-Back cable into the 300-Ohm earphone connector of the USB Box.
2. Run the Smart-EP program. Set the Gain in Smart-EP to 100k, the high-pass filter to 50 Hz, and the low-pass filter to 5000 Hz.
3. Open the artifact rejection (boundary lines) completely (from the EEG window).
4. Connect the feedback test cable as follows to Opti-Amp:
    - Black lead to black plug
    - Two blue leads to blue plugs
    - Two red leads to red plugs
5. Turn Opti-Amp ON.
6. Set Ear to Right and Mode to Ipsi in the Control Panel.
7. In the Protocol Menu, select the Execute Protocol option. Then select the Loopback.pse protocol file and open it.
8. Make sure Amp Scale is set to N (Normalized) in the Page Parameter Menu (PP Button) and Show all waveforms by intensity. Print results.
9. Clear the page.
10. Set Ear to Left and Mode to Ipsi in the Control Panel.
11. In the Protocol Menu, select the Execute Protocol option. Then select the Loopback.pse protocol file and open it.
12. Make sure Amp Scale is set to N (Normalized) in the Page Parameter Menu and Show all waveforms by intensity.
13. Print the results.

The values obtained in the printed table should be close to the ones shown in the following table. There is an allowable error margin of 15% to the shown values. If you have any large discrepancies to the values shown, fax the printed result to IHS for evaluation.

Intensity (HL)	Peak-to-Peak Voltage (Vp-p)	
90 dB	9.4 – 12.6	microvolts
80 dB	3.4 – 4.0	microvolts
70 dB	0.94 – 1.26	microvolts
60 dB	0.3 – 0.5	microvolts
50 dB	0.1 – 0.3	microvolts

Troubleshooting

When performing the Loopback.pse protocol, several 1000Hz pure-tones will be displayed on the screen, which correspond to several stimulus intensities. The optimum peak-to-peak voltage levels are displayed above for a total calibration (SPL+SPL-to-HL) of –19 dB. This particular total calibration value corresponds to 300-Ohm insert earphones. For other types of transducers, the total calibration value may be different. The total calibration value is located in the Auditory Stimulus Generation Dialog Box.

If a different total calibration value is desired, the peak-to-peak voltages can be calculated using the following equation:

$$\text{Inverse Log } (\beta / 20) = \text{peak-to-peak Voltage (Vp-p)}$$

$$\text{Where } \beta = \text{Total Calibration (SPL+SPL-to-HL)} + \text{Intensity (HL)-90}$$

Example:

- Pure-tone = 1000Hz (default for Loopback.pse protocol)
- Total Calibration = -19 dB
- Intensity = 80dB HL

$$\text{Vp-p} = \text{Inverse Log}\left(\frac{-19+80-90}{20}\right)$$

Vp-p = 3.54 microvolts

Note that the above equation is only applicable when using a 1000Hz pure-tone stimulus.

If both channels are not working: make sure that the optical cable is well connected and not damaged in any way. Make sure you are using alkaline batteries that are charged. Repeat the loop-back test. If you are unable to acquire the test signal, contact IHS for assistance.

If only one of the channels is not working: Switch optical cables at the transmitter end and receiver end. If there was a problem with only one channel and the problem is now occurring in the other channel, the problem was caused by a faulty optical cable. Replacing the optical cable should fix the problem.

If the problem persists in the same channel, perform the following: Switch optical cables at the transmitter side only. This will make the Channel A receiver collect data from the Channel B transmitter. If the problem is resolved in the problem channel and now appears in the other channel, a faulty transmitter channel caused the problem. Contact IHS for further instructions.

If the problem persists, the problem could be caused by a component inside your computer. Please contact IHS for assistance. Please remember to inform your technical support representative of the troubleshooting steps that were completed.

Filename Conventions

File Extensions

The SmartEP Software uses a variety of file types to store the acquired data, preferences, protocols, and other preferences that may be saved for later use. Since these files will be saved to your hard drive, it is important that you know how the software uses these files. The following symbols are wildcard characters, which are filled in by the user or are dependent on the patient demographics and other parameters chosen.

- '*' Is a string of letters or numbers of indeterminate length entered by the user,
- '?' Represents a single alphanumeric character,
- '#' Represents a single numeric character

File	Contents	Directory
*.dmg	Demographics File (patient information)	Dmg
*.rpt	Report: All pages of data recordings	Patient
*.set	Control Panel acquisition settings	SEPWin

*.stm	Stimulus Settings	SEPWin
*.pse	Testing Protocol	SEPWin
????##?.###	One data recording for a patient	Patient's Folder
??proto.###	Patient recordings from one protocol acquisition	Patient's Folder

File naming

Data recordings have the following automatic naming convention:

Example: XXLD90A.###

- XX represents the first and last name initials of the patient
- L represents the ear being stimulated
  - L for the left ear
  - R for the right ear
  - B for both ears
- A represents the recording type
  - D for ASSR
  - A for ABR
  - M for MLR
  - L for LLR
  - E for ECochG
  - P for P300/MMN
  - O for TEOAE
  - B for VEMP
  - X for User Defined
  - Other: any letter or number
- 90 represents intensity (dB SPL) of the stimulus
- A represents the channel from which recorded
- ### Represents the automatically assigned recording number with the exact same ear, recording type, intensity and channel settings (range: 0-999).

SNR and RN

The SNR and RN peak-to-peak values are calculated using the split-sweep acquisition characteristic of SmartEP. Their values are given by the following formulas, where A is the peak-to-peak amplitude of the first buffer and B is the peak-to-peak amplitude of the second buffer.

$$RN = A - B$$

$$SNR = (A + B) / (A - B)$$

Saving ASCII Files

ASCII Files are simple text files that contain any type of data. In SmartEP, it is possible to export acquired recordings to ASCII files in order to obtain the numeric data as comma-separated values that can be imported into other programs, such as Microsoft Excel®, for further manipulation and study. The files produced by saving an ASCII file from the Data Menu will contain the following fields in the order outlined:

- Identifying Information: Patient Identifier, File Name
- Acquisition Parameters: Ear, Channel, Intensity, Rate, Mode, Sweeps, Artifacts, Sample Period, Amplifier Gain, Low Pass Filter, High Pass Filter, Notch Filter, Type of Stimulus, Stimulus Frequency, Stimulus Duration, Masking, and Stim. Point (point at which the stimulus occurred).
- Peak Information: Peak Label (Label), Peak Top Marker (Peak (+)) position in microvolts, Peak Top Marker (Peak (+) (ms)) time position in milliseconds, Peak Bottom Marker (Peak (-)) position in microvolts, and Peak Bottom Marker (Peak (-) (ms)) position in milliseconds.
- Data points, 1024 as comma-separated values: Data Point number, Data Point position in milliseconds, Average Data point Position, Average equivalent microvolt value, Buffer1 point position, Buffer1 equivalent microvolt value, Buffer2 point position, and Buffer2 equivalent microvolt value.

Note that all recordings use a maximum of 1024 data points, however, some older recordings may not use all point and show 0s in some of the fields. In addition, all recordings contain two buffers, one for holding data of condensation mode and the other for rarefaction mode (where the average is the alternating mode value); if the recording was acquired using only one of these modes, the opposite buffer will contain no data.

ASCII files may also be generated from the Power Spectrum window, accessible through the Process option in the main menu. Saving a file from this window will yield a different file that includes different data for the data points. While the Identifying information, Acquisition Parameters and Peak information will be shown the same way, the Data along with Spectral components will contain the following fields' equivalent to the time domain:

- Data: data point of acquisition.
- Average: Average amplitude of the response (e.g. rarefaction + condensation) both buffers added.
- Buffers: Average amplitude of the individual buffers as previously explained.

These additional fields contain information reflecting the frequency domain and independent from the previously described data points:

- Freq: Frequency data point from the spectrum.
- Pre-Spectra: This is the power spectrum of the pre-stimulus region at each of the frequency points outlined in picovolts squared.
- Pre-Phase: This is the phase of the frequency in the pre-stimulus area.
- Post-Spectra: This is the power spectrum of the post-stimulus region at each of the frequency points outlined in picovolts squared.
- Post-Phase: This is the phase of the frequency in the post-stimulus area

It may be able to understand the fields better keeping in mind the following XY plot possibilities with the provided data:

- Plotting Data vs. Average yields the original recording in the time domain.

- Plotting Data vs. Buffer yields the corresponding rarefaction or condensation; in case of P300, odd vs. normal.
- Plotting Frequency vs. Pre-Spectra yields the pre-stimulation frequency components, e.g. the red part of the graph in the Spectral Analysis.
- Plotting Frequency vs. Post-Spectra yields the post-stimulation frequency components, e.g. the blue part of the graph in the Spectral Analysis window.

The Single Channel Transmitter

The use of a single channel transmitter can be difficult to understand. Here we provide you with reference materials and some guidelines to understand the connections, as they are set inside the unit. The transmitter has a left-right flip switch and a rotary switch that help to test the impedance for all available electrode leads combinations.

Impedances are measured between the active and the ground electrode. The relative positions of these electrodes change internally, as described, when the switches are set in the position specified by the following table. This table may also be helpful to determine the source of a problem when a particular electrode or electrode lead is defective. You must consider these when testing electrode impedances, keeping in mind which electrodes are used together.

Rotary Switch	Left-Right Switch	Red Plug	Black Plug	Blue Plug
Left-Right	Right	Active	-	Ground
Forehead	Right	-	Active	Ground
Left-Right	Left	Ground	-	Active
Forehead	Left	Ground	Active	-

In the following figure, there is a graphical representation of how the channels are connected at a specific time. This set of rules applies to the impedance testing and to the actual setup during a normal test. Notice that the Black lead, which is connected to the high forehead, remains all the time as the non-inverting electrode. The ground and the inverting electrode will interchange places as the switch is toggled between the left and right positions, with the inverting electrode positioned at the side specified by the switch.

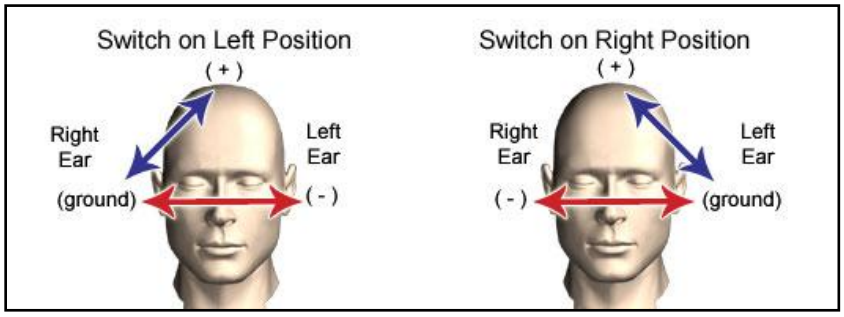

Figure 49 - Single Channel Transmitter Connections